

LUMA

A little light. A lot of personality.

Printable robotic desk companion



Illustrated assembly manual + engineering drawings

Raspberry Pi 4 · four moving joints · round animated LCD · five distance sensors · white light + RGB halo

Revision A / September 2026 / Digital prototype release

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Meet LUMA

LUMA is a printable Raspberry Pi 4 desk companion with an articulated arm, a circular animated face and two independently controlled light rings. It is an original prototype inspired by the posture and personality of the Autonomous Lamp. It is not a conversion kit for the commercial lamp, and its parts do not claim compatibility with that product.

The head contains a Waveshare ESP32-S3-Touch-LCD-1.85 display, a small white-light ring and a larger RGB halo. The Raspberry Pi makes the behavior decisions. The ESP32-S3 built into the display serves as a graphics peripheral over USB. This preserves the requested Pi 4 control while using the actual QSPI display interface supported by the Waveshare board.

The selected 1.85-inch screen is a 360 × 360 IPS TFT LCD, not an OLED. Its visible circular area is approximately 45.68 mm across. Do not substitute a similarly named Waveshare B, C, AMOLED or bare display module without checking both the mechanical envelope and firmware pin mapping.

What the prototype does

- Wink: one eye closes briefly, the head makes a small gesture and the halo responds.
- Say Hi: a friendly greeting appears on the face, with an audible greeting through the specified audio hardware.
- Be happy: smiling eyes, a bright expression and a gentle arm gesture.
- Sense nearby objects using exactly five Adafruit VL53L1X STEMMA QT boards, connected through separate I2C multiplexer channels.
- Illuminate the face area with the white channel of a 24-pixel RGBW ring and express color with a separate 60-pixel RGB ring.

The provided behavior software runs locally. A conversational language model, camera, microphone, speech recognition and cloud account are not required by this design.

What is in the project

The printable STL files and editable CAD define the mechanical prototype. The drawings in the back of this manual document those parts. The bill of materials identifies purchased hardware separately from printed parts. Wiring instructions and software source accompany the manual, along with three rendered promotional videos and their editable production scripts.

The project is a digital prototype release. Mesh and software checks can verify files and logic, but they cannot establish physical fit, holding torque, optical performance or print strength. The first assembled unit is the fit-and-function validation build. Record component revision, printer settings and any measured corrections before duplicating it.

Design choices that matter during assembly

The 55 mm square display board cannot fit through the small light ring's roughly 52.2 mm center. It is installed behind that ring in a separate plane. Follow the head stack order instead of trying to place all boards side by side.

The large RGB ring consists of four purchased 15-pixel quarter boards. Its printed support carries the mechanical load; the solder bridges only connect the electronics. All four arcs are needed to create the complete 60-pixel ring.

Frosted material belongs over the light rings. Keep the LCD viewing area open and the five time-of-flight optical apertures clear. A diffuser over a ranging sensor can create reflection and crosstalk.

The two NeoPixel rings use different pixel formats. The large ring is RGB and the small ring is RGBW. They use separate data outputs and separate driver objects. The normal brightness limits are part of the power and thermal design.

Reading the drawings

All mechanical dimensions are in millimeters. STL files do not carry a unit declaration: select millimeters in the slicer and confirm the imported bounding dimensions before printing. Use the drawing's stated scale; page fitting in a PDF viewer changes paper scale. Numerical dimensions and the editable model control geometry.

Purchased circuit boards, bearings, fasteners and servo horns are not printable substitutes. Match the exact part numbers in the bill of materials. Minor manufacturing differences are handled by the stated clearances and the first fit coupon, not by forcing a board into a printed pocket.

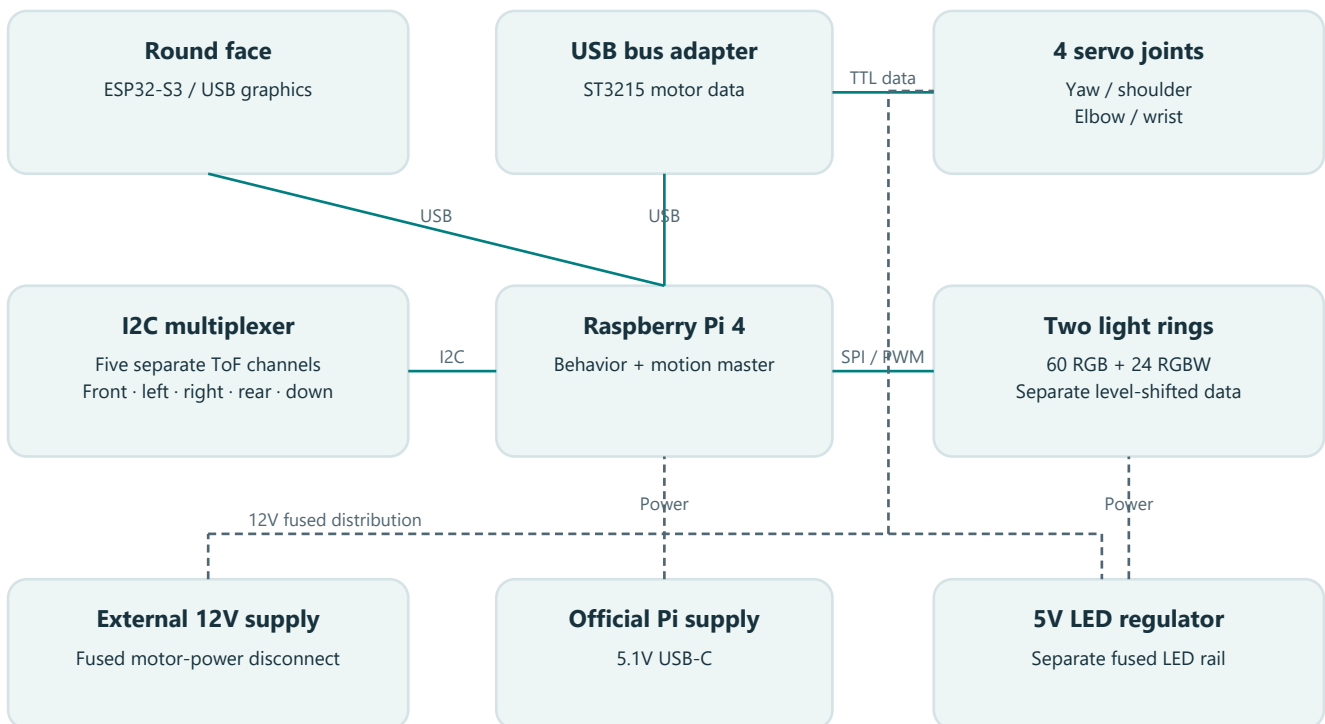
Build sequence

1. Check the purchased component revisions and inspect the dimensioned interfaces.
2. Print and measure the fit coupon, then slice the structural parts and diffusers.
3. Bench-test each electrical subsystem before fitting it into the arm.
4. Assemble the base, joint supports and arm links with the motors unpowered.
5. Assemble the layered head, display, light rings and sensor mounts.
6. Route and restrain the wiring, leaving service loops at every moving joint.
7. Set unique motor IDs, calibrate the neutral pose and enable a restricted first movement.
8. Validate sensing, fault response, balance, temperature and the three expressions.

Reference and attribution

The appearance reference is the [Autonomous Lamp](#). The LUMA geometry, manual layout and animations are newly authored for this project. Vendor specifications and documentation are identified in the component source register and bill of materials. Promotional files show the digital prototype, not a filmed or physically tested product.

System at a glance



Solid: data. Dashed: power. Wiring chapter defines grounds, fuses and the motor feed.

The Pi supplies control commands. Dedicated motor and LED power paths keep the actuator load off the Pi's supply. The detailed wiring chapter is authoritative for the power-disconnect circuit and common grounds.

Parts and purchasing

Planning allowance for the purchased items below: **US \$778.78** before shipping and tax. Figures are budgeting allowances, not checked-out prices. Filament and any workshop tools are included only where explicitly listed. Reuse of existing supplies can change the total substantially.

Order the exact display and voltage variant specified. Generic fasteners, connector harnesses, relay contacts and wire must meet the listed dimensions and electrical ratings. Buy the servo horns and their screws with the servos; verify their actual thread and engagement before ordering extras.

Controller, sensing, lights and power

Ref / qty	Item	Required specification	Budget / each
E01 / 1	Raspberry Pi Raspberry Pi 4 Model B 4GB	Master controller. 40-pin GPIO and four USB ports	\$60.00
E02 / 1	Raspberry Pi 15W USB-C Power Supply	Separate Pi supply. 5.1V 3A USB-C	\$12.00
E03 / 1	Reputable flash vendor 32GB or larger microSD A1	Operating system card. High endurance preferred	\$12.00
E04 / 1	Waveshare ESP32-S3-Touch-LCD-1.85 SKU28514	Central face display. 55x55mm;360x360 ST77916 QSPI;USB-C;no B/C suffix	\$35.00
E05 / 5	Adafruit 3967 VL53L1X STEMMMA QT	Five interaction ranging sensors. 25.5x17.5x4.6mm;l2C0x29	\$14.95
E06 / 1	Adafruit 5626 PCA9548 STEMMMA QT	Mux for five same-address sensors. TCA9548A-compatible;40.6x20.2x4.8mm;0x70	\$6.95
E07 / 5	Adafruit STEMMMA QT JST-SH4 cable	One separate cable per sensor. Length selected to assembled routing;100-300mm nominal	\$2.00
E08 / 1	Adafruit STEMMMA QT to female header cable	Pi to mux upstream connector. 3.3V logic only	\$2.00
E09 / 4	Adafruit 1768 quarter NeoPixel60 ring	Outer RGB emotion halo. 15RGB per quadrant;four form OD157.5mm ID144.8mm ring	\$9.95
E10 / 1	Adafruit 2862 NeoPixel24 RGBW natural white	Inner white lamp ring. 24RGBW;OD65.5mm;only W channel used;4500K nominal	\$24.95
E11 / 1	Adafruit 1787 74AHCT125 DIP14	Two data level shifters. 5V VCC;3.3V logic accepted;AHCT required	\$1.50
E12 / 2	Generic 330ohm 0.25W resistor	LED input series resistors. One per independent data line	\$0.10
E13 / 2	Generic 1000uF 10V electrolytic	LED ring supply reservoir. Observe polarity;one each ring	\$1.00
E14 / 1	Generic 100nF ceramic	74AHCT125 decoupling. Across DIP14 and DIP7 at socket	\$0.10
E15 / 1	Pololu D24V50F5 item2851	LED5V buck converter. 5V5A typical thermal-limited;17.8x20.3x8.8mm	\$32.95
E16 / 4	Waveshare ST3215 30kg.cm@12V	Four joint bus servos. 12V version;4096ticks/rev;2.7A stall each;1Mbps default	\$22.00
E17 / 1	Waveshare Bus Servo Adapter A SKU25514	USB to TTL half-duplex bus adapter. ST selected;USB mode;separate low-current12V branch	\$10.00
E18 / 1	Mean Well GST220A12-R7B	External enclosed motor/LED supply. 12V15A model;R7B powerDIN;keep outside printed enclosure	\$95.00
E19 / 1	Generic Matching R7B powerDIN harness	DC supply breakout. Correct polarity;all specified parallel contacts;15A aggregate rated	\$15.00
E20 / 1	Generic IEC C13 mains cord	External supply cord. Local approved cord;earth conductor intact	\$7.00
E21 / 1	Generic Latching mushroom stop two NC contacts	Motor stop plus isolated Pi sense. NC1 interrupts12V relay coil;NC2 connects BCM27 to GND	\$15.00

Ref / qty	Item	Required specification	Budget / each
E22 / 1	Generic 12VDC coil relay with 30A DC NO contact and socket	Hardware motor supply disconnect. Coil $\leq$ 200mA; NO contacts $\geq$ 15A at 12VDC; socket 16AWG	\$12.00
E23 / 1	Generic 1N4007 diode	Relay coil flyback. Stripe/cathode to coil positive	\$0.10
E24 / 1	Generic 15A DC fuse and holder	Main 12V input branch. Holder and connector rated $\geq$ 15A	\$3.00
E25 / 4	Generic 3A time-delay DC fuse and holder	Individual servo V+ feeds. 18AWG star feeds; one fuse per servo	\$3.00
E26 / 1	Generic 2A DC fuse and holder	Buck 12V input branch. LED firmware capped 20 percent	\$3.00
E27 / 1	Generic 1A DC fuse and holder	Bus adapter power branch. Do not run servo group current through adapter jack	\$3.00
E28 / 1	Generic 0.5A DC fuse and holder	Relay coil branch. NC1 switches coil only	\$3.00
E29 / 1	Generic 3A DC fuse and holder	5V LED output branch. Protects LED 18-22AWG harness	\$3.00
E30 / 1	Adafruit 3369 mini USB speaker	Spoken Hi. USB audio; 84x43x32mm; 73.6g; keep volume modest	\$12.50
E31 / 1	Generic USB-A to right-angle USB-C data cable	Pi to display. Flexible cable; strain relief; 15mm rear plug envelope assumed	\$7.00
E32 / 1	Generic USB data cable matching adapter	Pi to bus adapter. Use actual adapter USB connector type	\$5.00
E33 / 1	Generic 16AWG / 18AWG silicone wire and rated distribution terminals	Power harness. 16AWG trunk; 18AWG motor branches; 22AWG LED branches	\$20.00
E34 / 1	Generic Small perfboard and DIP14 socket	Level shifter assembly. Insulate underside; strain relief all wires	\$5.00
E35 / 1	Generic Pi4 heatsink and ventilation hardware	Thermal management. Check CPU temperature after assembled operation	\$7.00
E36 / 1	Generic 10kohm resistor	External E-stop sense pullup. BCM27 to 3.3V; open NC2 means asserted	\$0.10

Mechanical hardware and consumables

Ref / qty	Item	Required specification	Budget / each
M01 / 1	Local metal supplier Custom ballast disc	Steel ballast plate. Mild steel diameter180 thickness6 mm; 4 holes diameter4.4 at XY(+/-45,+/-35); 4 holes diameter3.4 at XY(+/-26,-36/-8); deburr; approx1.19kg	\$25.00
M02 / 1	Generic 6808-2RS	Yaw support bearing. 40mm bore x52mm OD x7mm width; sealed deep groove	\$12.00
M03 / 4	Generic M4x25	Base and tray bolts. M4 x25 socket head; steel plate6 + floor3 + spacer5 + tray3	\$0.25
M04 / 4	Generic M4x5 spacer	Tray standoffs. Metal spacer5mm length; 4.3mm bore; OD8 minimum	\$0.40
M05 / 4	Generic M4 nut	M4 lock nuts. ISO metric nyloc	\$0.12
M06 / 8	Generic M4 washer	M4 load spreading washers. OD12 minimum	\$0.06
M07 / 4	Generic M3x16	Yaw cradle through bolts. Through printed cradle3 steel6 and base3; nuts below base; tray has7mm head access	\$0.16
M08 / 2	Generic M3x45	Yaw rear body clamp bolts. Through bottom3 + body35 + keeper2.6; do not overtighten case	\$0.30
M09 / 2	Generic M3x50	Shoulder clamp bolts. Clamp through44mm tower width plus nuts	\$0.30
M10 / 4	Generic M3x16	Shoulder foot bolts. 6mm tower foot plus6mm platform plus nuts	\$0.16
M11 / 8	Generic M3x60	Arm rail cross bolts. 5+44+5mm stack plus nut; 4 per arm; include4 cassette ear positions	\$0.40
M12 / 24	Generic M3x10 nominal	Pitch horn attachment screws. Nominal M3; 5mm rail +3.3mm spacer leaves1.7mm horn engagement; confirm vendor horn thread	\$0.15
M13 / 4	Generic M3x6 nominal	Yaw horn attachment screws. 3mm web plus1mm washer leaves2mm engagement; confirm vendor horn thread	\$0.15
M14 / 6	Generic M3x12	Base lid screws. Into2.8mm plastic-thread-forming printed pilots	\$0.15
M15 / 4	Generic M3x12	Head yoke screws. 6mm fork mounting face +3mm shell floor; nuts on inside	\$0.15
M16 / 8	Generic M3x10	Outer bezel screws. 4mm bezel +1mm shim;2.8mm plastic-thread-forming pilot	\$0.15
M17 / 4	Generic M3x12	Central face screws. 4mm plate +1mm shim;2.8mm plastic-thread-forming pilot	\$0.15
M18 / 4	Generic M3x10	White ring carrier screws. 2mm carrier on dedicated radius45 bosses at z32.5	\$0.15
M19 / 4	Generic M3x25	LCD cradle and retaining plate screws. 3mm retainer +15mm cradle; thread into head pillar at z14	\$0.20

Ref / qty	Item	Required specification	Budget / each
M20 / 16	Generic M3x1 shim	Precision axial shims. M3 bore; thickness1mm; 12 underface/bezel plus4 yawhorn washers	\$0.12
M21 / 10	Generic M3x10	Sensor pod attachment bolts. 2 per pod through shell or frontplate	\$0.15
M22 / 10	Generic M2x14	Sensor edge frame screws. 2 per pod; pilot2.2 through holes; nuts	\$0.15
M23 / 10	Generic M2 nut	Sensor retainer nuts. ISO metric M2	\$0.08
M24 / 30	Generic M3 nut	General M3 nuts. Include spare quantities for clamps sensorpods and platform	\$0.10
M25 / 50	Generic M3 washer	General M3 flat washers. Nominal0.5mm; not replacement for1mm axial shims	\$0.05
M26 / 4	Generic M2.5x6	Raspberry Pi fasteners. Pi4 four mounting holes; printed standoffs may be tappedM2.5	\$0.12
M27 / 4	Generic Rubber feet	Adhesive anti slip feet. 20mm diameter x5mm high minimum; place radius90	\$0.40
M28 / 1	Generic Translucent PETG	Frosted optical printing filament. Natural/translucent; approximately60g including trials; solid1.2mm optical skin	\$20.00
M29 / 1	Generic PETG	Structural filament. Approx1kg roll; actual slicer mass varies;6 walls on arm rails	\$22.00
M30 / 1	Generic Neutral cure silicone	Small adhesive tube. 3 small removable dots for inner diffuser and edgefoam; electronics compatible	\$6.00
M31 / 1	Generic Thin PCB foam	Closed cell foam tape. 0.5 and1mm thickness; LCD edge preload only	\$4.00
M32 / 1	Generic LED retaining tape	Electronics compatible thin adhesive strips. Attach LED ring PCB on support posts without covering LED optics	\$3.00
M33 / 1	Generic Desk clamp	Optional commercial padded desk clamp. Use only if assembly reach/mass exceeds validated stable envelope; not a printed load bearing clamp	\$12.00
M34 / 12	Generic M2.5x12	Optional harness clip screws. For6 printed cable saddles; use selftapped pilots or cableties	\$0.12
M35 / 10	Generic Soft closed-cell foam strip	Required ToF edge preload strips. 18x2x6mm each; compress gently to5.4mm between PCB edge and retaining frame;2 per sensor	\$0.20

Printable part schedule

27 unique STL files; 52 printed pieces including the fit coupon and optional cable clips. Quantities control the print run. The drawing index uses the same part order.

Drawing / qty	STL identifier	Material / print orientation
P01 / 1	base_tub	PETG; floor down

Drawing / qty	STL identifier	Material / print orientation
P02 / 1	base_lid	PETG; flat annulus down
P03 / 1	electronics_tray	PETG; plate down
P04 / 1	yaw_mount	PETG; base down
P05 / 1	yaw_cap	PETG; flat down
P06 / 1	turntable	PETG; neck down
P07 / 1	shoulder_tower	PETG; foot down
P08 / 1	shoulder_cap	PETG; flat down
P09 / 2	upper_rail	PETG; broad face down
P10 / 2	forearm_rail	PETG; broad face down
P11 / 2	servo_cassette	PETG; closed side down
P12 / 2	cassette_cap	PETG; flat down
P13 / 4	cross_spacer	PETG; cylinder upright
P14 / 6	horn_spacer	PETG; flat down
P15 / 1	head_yoke	PETG; rear mounting plate on bed; rotate180X
P16 / 1	head_shell	PETG; rear disc down
P17 / 1	outer_bezel	PETG; front lip down
P18 / 1	face_center	black PETG; flat front down
P19 / 1	white_carrier	PETG; flat down
P20 / 1	lcd_cradle	PETG; flat base down
P21 / 1	lcd_retainer	black PETG; flat down
P22 / 1	outer_diffuser	natural/translucent PETG; smooth optical face on bed
P23 / 1	inner_diffuser	natural/translucent PETG; smooth optical face on bed
P24 / 5	sensor_pod	PETG; rear flange down
P25 / 5	sensor_retainer	PETG; flat down
P26 / 6	cable_clip	PETG; flat down
P27 / 1	fit_coupon	PETG; flat down

Printing and finishing

Prepare one interface before a full print run

Use a 0.4 mm nozzle as the baseline. Print the fit coupon supplied with the CAD, measure its holes and pockets with calipers and test the actual fasteners and bearings. This is especially useful for horizontal holes, which can print smaller than their modeled diameters. Keep the slicer at 100% scale; compensate holes locally or revise the relevant CAD clearance rather than scaling the complete machine.

The part manifest gives quantities, material and orientation. The STL orientation is the manufacturing orientation when the manifest identifies it as such. The assembled model is for viewing and should not be sliced as a single object. Check that each sliced part has continuous perimeters and that screw holes remain open.

Structural material

Use PETG for the first functional build. For arm links and load-carrying brackets, start with 0.20 mm layers, five perimeters, six top/bottom layers and 40% gyroid infill. Increase local solid infill around fasteners if your slicer supports modifier volumes. Print link plates flat so their length lies in the XY plane.

For non-load-bearing covers, start with four perimeters, 0.20 mm layers and 20% infill. Use the orientation and support notes specific to each part. Remove supports without gouging bearing seats or the surfaces that locate circuit boards.

PLA can be useful for quick dimension checks, but warm motors, LED boards and sustained joint loads make it a poor default for the final load-carrying structure. A material change also changes shrinkage and fit; repeat the coupon.

Frosted light covers

Print both annular diffusers in natural translucent PETG. The small ring covers the white LEDs and the large ring covers the RGB LEDs. Use 0.12–0.16 mm layers and a continuous solid optical face. Preserve the modeled thin wall; do not replace it with sparse infill that leaves a visible lattice in the illuminated area.

Make a small test of the intended wall thickness before committing both rings. Translucent filaments vary greatly in pigment and light transmission. A nominal 1.2 mm optical face is a starting point, not a measured optical specification.

Wet-sand the outside very lightly with 600–1000 grit abrasive to obtain a uniform matte surface. Wash and dry the part before installation. Do not sand the locating lip so far that the diffuser becomes loose. Check the result at the firmware's normal brightness limit with the head closed: rings that look uniform in room light can reveal hotspots when lit.

Keep the opaque separator between the two light paths. A black or charcoal separator helps prevent RGB light washing across the face. Leave an air gap between LEDs and diffuser as defined by the head stack. Avoid adhesives that touch the LED emitters or the LCD glass.

Fasteners and tolerance

Use the specified metal nuts, washers and servo horns. Deburr holes before insertion. A bolt should enter a clearance hole by hand without driving threads into the plastic. Seat the head and washer firmly, then stop before the printed layers visibly deform. Tighten opposing fasteners alternately to prevent distorting a PCB cradle or joint bracket.

If a heat-set insert is specified, test the insert type and soldering-iron temperature on scrap from the same filament. Support the boss while inserting it, allow it to cool completely, and confirm the insert is perpendicular. Do not add heat-set inserts to a design location intended for a through-bolt and captive nut.

Press bearing outer races squarely into their stated seats. Apply force to the race being fitted. A bearing that becomes rough after installation is a sign of an undersized or distorted seat; correct the seat before assembling the joint.

Harness preparation

Make a service loop at each joint, and move the unpowered mechanism through its intended travel while watching the loop. The wire must not become taut, rub a sharp printed edge, enter a gear/horn gap or prevent the joint from reaching its limit. Fit the provided cable guides and add soft sleeving where the harness moves against a surface.

Use flexible stranded wire for moving runs. Keep the I2C wiring short, route power and motor wiring away from it where practical, and retain the sensor cables so their small connectors do not take bending loads. A STEMMA QT connector is a signal interconnect, not a high-current power distribution connector.

Record the first print

Record printer, nozzle, material brand, layer height, wall count, infill, measured coupon hole sizes and any CAD clearance change. Weigh the completed head with its display, sensors, screws and wiring. Use the measured head mass when reviewing the arm torque calculation.

Estimated print duration and filament mass depend on the slicer and printer. The CAD validation report includes geometric volume; the slicer's estimate after applying the actual walls and infill is the useful production estimate.

LUMA mechanical assembly, revision A

This is an original four-axis prototype design. The complete editable geometry is `cad/luma.scad`; `cad/build.py` exports the printable parts and their exact assembly transforms. Open `cad/assembly_preview.scad` to view the entire assembled STL model, or use `cad/assembly.json` for its transforms. The master OpenSCAD default view is only a base subassembly preview. For an individual part set part to its manifest ID. The illustration meshes in `cad/visual_only` are purchased-component envelopes, not printable replacements.

The indexed neutral position has a 115-degree upper arm and a 45-degree forearm measured counterclockwise from the forward horizontal (+Y), with the head facing +Y. Both pitch axes run along X. The base yaw axis is Z. The elbow relative angle is -70 degrees. Dimensions: base diameter 216mm, upper pivot pitch 140mm, forearm pivot pitch 120mm, shoulder axis 115mm above the base floor datum, head diameter 176mm and 42mm optical depth. The center of the head's rear mounting plane is approximately Y25.7+21=46.7mm, Z326.7mm; this is not the center of mass. The housing top is 414.7mm. Rubber feet add 5mm to tabletop height. The LCD active face is recessed approximately 10.8mm behind the diffuser's front surface.

Vendor geometry and fit datums

The Waveshare ST3215 vendor drawing gives a 45.22 x 24.72mm side envelope, 35mm body axis width, 37.25mm across the two horns, 19.2mm metal horn diameter and 14mm diameter four-hole pattern. The shaft axis is 10.11mm from the nearest body end. Files downloaded from Waveshare are in `cad/references/ST3215-2D` and `cad/references/ST3215-3D`. Source: https://www.waveshare.net/wiki/ST3215_Servo and <https://www.waveshare.net/w/upload/0/08/ST3215-2D.zip>. The supplied drawing does not specify the horn thread; the BOM's M3 horn screws are a purchasing assumption that must be checked on one actual servo before ordering the full set. Never force a mismatched screw. Vendor guidance explicitly warns that a protruding horn screw can contact the servo housing and overload the motor.

The printed cassettes grip the rear portion of the servo case, avoiding assumptions about threaded case holes. Both original metal horns remain installed. The far-side factory horn/idler supports the opposite link plate; the design does not place the arm on one unsupported output screw. One purchased 6808-2RS bearing supports the yaw platform independently of the motor shaft. The 52.15mm bearing pocket and 39.75mm neck are intentionally clearance fits. Tune the fit coupon for the actual printer before printing the load-bearing parts.

The selected display is the standard Waveshare ESP32-S3-Touch-LCD-1.85 with a 55 x 55mm board/body outline. It is retained at its edges in a 55.8mm square pocket. No guessed PCB mounting holes are used. The design allows a 12mm board/body stack and uses foam edge shims to take up sample variation. Inspect the purchased screen's actual glass projection and connector position before tightening; the assembly illustration assumes a 31.2mm active-face datum. The USB relief is 19mm wide; use a low-profile right-angle plug and check its actual bend clearance in the open head before closing it.

Print settings

Print all units in millimetres at 100% scale. Largest XY footprint is 216 x 216mm. On a 220mm bed this leaves only 2mm per edge: disable an external brim on the base, or use a larger bed if bed clips reduce usable area. Structural parts use PETG, 0.20mm layers, 4 walls generally, 6 walls and 40% infill on the arm rails. Use solid local infill around the horn and cassette screws. Print the side rails flat so the layers follow the arm's length. Print the head fork with its broad mounting plate against the bed, rotated 180 degrees about X from its STL coordinates; enable supports beneath its fork ears. Remove support material from all mating surfaces.

Print the two optical diffusers in natural/translucent PETG at 0.15mm layers, 3 perimeter walls and 100% infill. Their optical face is 1.2mm thick: specify at least 8 solid layers at 0.15mm. Smooth the face with 600–1000 grit paper for frosting, wash and dry it, and test its diffusion with the real rings. The outer diffuser is open behind the LED annulus; it is not a sealed hollow torus. Do not apply frosting, paint or a diffuser over any time-of-flight aperture.

Clear through-holes with the matching drill by hand. The lightly loaded cover bosses have 2.8mm plastic-thread-forming pilots for M3 screws; these are not standard metal tapping drills. The 2.1mm Pi standoff pilots are tapped M2.5. For frequently serviced covers, enlarge the bosses as necessary in the parametric model and use suitable inserts. Do not use the

motor to force a tight printed joint. General designed sliding clearance is 0.3–0.4mm. Check flatness and remove elephant's-foot at the bearing, horn and LCD fits. Filament quantity and print duration should be taken from the actual slicer, not the solid CAD volume.

Prepare purchased metal parts

1. Cut a mild-steel disc 180mm diameter and 6mm thick, approximately 1.19kg before drilling. `cad/ballast_plate.dxf` provides the cut/drill geometry; `cad/ballast_plate.svg` is a dimensioned drill template. With disc center as XY origin, drill four 4.4mm holes at ($\pm 45, \pm 35$) and four 3.4mm holes at ($\pm 26, -36$) and ($\pm 26, -8$). Deburr and protect the surface from corrosion. The same eight holes exist in the base floor; orient the disc so the asymmetric yaw hole pattern points rearward (-Y). There is no central hole in this version: the passive yaw horn has approximately 1.9mm clearance above the steel at the indexed assembly height.
2. Confirm the bearing is a 6808 measuring 40 x 52 x 7mm, not a 608 skateboard bearing. Check it on a fit coupon or sacrificial test print. Keep the bearing clean; do not hammer it into plastic.
3. Check the servo's original metal horns against the 14mm pattern. For the three pitch joints, 5mm printed plate + 3.3mm printed spacer leaves approximately 1.7mm engagement with a 10mm screw. For yaw, use a nominal 6mm screw through the 3mm web plus 1mm washer, leaving 2mm engagement. Adjust length and washers after checking the real thread depth. These are not torque specifications.

Assemble the weighted base

4. Put the steel disc on the 3mm-thick base floor. Install the yaw cradle on the disc at Z9, aligned with the four asymmetric M3 holes. Use four M3x16 bolts with washers and nuts below the enclosure. Confirm screw tips and nuts remain above the eventual rubber-foot contact plane.
5. Place servo ID1 in the yaw cradle with its output axis vertical and body extending toward -Y. Its body envelope occupies Z12–47; the output-horn face is approximately Z48.1. Fit the 2.6mm keeper above the rear of the case at Z47 and use two M3x45 bolts through the cradle ears. Tighten evenly only until the case is securely held. The keeper ends below Z50 and clears the lid.
6. Install the electronics tray at Z14 on four 5mm metal spacers seated on the steel disc. Use four M4x25 bolts through base, steel, spacer and tray, with load-spreading washers and lock nuts. The tray and base share the 90 x 70mm bolt pattern. The large central opening clears the lower yaw case and cradle; four 7mm access openings clear the yaw mounting screw heads. Mount the Pi4 on the four integral standoffs with M2.5x6 screws. Mount the small boards using the universal slots, insulated standoffs and cable ties. Keep the external AC adapter and any high-current distribution/fuse enclosure outside this compact base. Put the optional USB speaker outside the base if its 84 x 43 x 32mm case does not fit the chosen cable layout.
7. Route the harness through the 13mm rear cable port with a fitted grommet and strain relief. The rectangular secondary port accepts USB/control wiring. Fit four 5mm-high rubber feet near radius 90mm. Leave a service loop for ± 25 -degree yaw; this assembly has no slip ring and must not rotate continuously.
8. Seat the 6808 bearing in the lid's 52.15mm pocket. Place the turntable neck through its 40mm bore. Attach the internal 3mm horn web to the yaw metal horn using four nominal M3x6 screws and 1mm washers. A long hex key reaches the screws through the platform's 22mm center aperture. Neck start is Z47.8 and the platform top remains Z68. Check free movement through the limited yaw travel before fastening the lid with six M3x12 screws.

Assemble shoulder and links

9. Bolt the shoulder tower to the turntable using four M3x16 bolts on the 52 x 36mm pattern. Install servo ID2 vertically with its shaft axis at Z115 and its case extending downward. Fit the separate keeper at the open side using two M3x50 bolts. The original two metal horns remain accessible above the clamp.
10. Assemble one upper link from two `upper_rail` pieces, two 44mm cross spacers, one rear-body cassette and its cap. Plate inner faces are 44mm apart. Two M3x60 bolts pass through the cross spacers at local X35 and 88; two more pass through the cassette ears at local X115, $Y \pm 18.5$. Servo ID3 sits in the cassette at the 140mm distal pivot. Fit the cap before tightening the cassette bolts. Keep both connectors accessible and preserve the servo's factory horn retaining screws.

11. With ID2 electronically centered at 2048, index the upper link at 115 degrees above the forward horizontal. Put a 3.3mm horn spacer between each metal horn and its printed rail. Secure both sides with four nominal M3x10 screws per horn, eight total. Verify engagement depth manually. The passive-side connection must turn without binding.
12. Assemble the forearm similarly using two forearm_rail pieces. Cross spacers are at local X35 and 68; cassette bolts are at local X95, Y±18.5. Servo ID4 sits at the 120mm distal pivot. Index the forearm at 45 degrees above the forward horizontal while ID3 is centered; its angle relative to the upper arm is -70 degrees. Attach the proximal rails to both ID3 horns with spacers and eight nominal M3x10 screws.
13. Fit the head fork to both ID4 horns with the remaining two 3.3mm spacers and eight nominal M3x10 screws. At indexed neutral, the fork's broad mounting plate faces forward (+Y) and lies 21mm in front of the wrist axis. Leave the head off until the unloaded link motion check succeeds. Route cables along the inner side of the rails with loose loops at each joint. Use the optional printed saddles or cable ties without clamping a servo connector or a moving horn.

Assemble the optical head

All following Z values use the head's rear face as 0, increasing toward the viewer; they are not global heights. The head floor occupies 0–3, and the shell rim reaches 36. The LCD is behind the white ring, allowing its square corners to occupy space that would otherwise intersect the LED annulus.

14. Attach the rear shell to the fork with four M3x12 bolts on its 54 x 30mm pattern, washers and nuts inside the shell. Install the front, rear, left, right and downward sensor pods using two M3x10 bolts per pod. The front pod is above the display. The rear pod faces away from it. The side pods' long dimension follows the rim tangentially so both mounting bolts land in the shell wall.
15. Put each 25.4 x 17.78mm VL53L1X board into its 26.1 x 18.1mm pod recess. The PCB rear sits at pod Z3; a 1.6mm PCB has its top at 4.6, while the retaining-frame underside is 10. Fill the 5.4mm edge gap with two required soft closed-cell foam strips 18 x 2 x 6mm per sensor, gently compressed to 5.4mm. Place them along the upper and lower PCB edges, centered near Y±8.3; do not place them over the center optical package or the side connectors. Fit the retaining frame with two M2x14 screws and nuts and verify the board cannot rock. If the purchased PCB thickness differs, measure and adjust the edge shims to provide light preload. Both ±X pod walls have 8mm-wide reliefs for the horizontal STEMMA QT plugs. The remaining aperture must leave the emitter and receiver unobscured; the sensor sees through air with no printed optical window. Connect all five sensors before closing the shell.
16. Join the four Adafruit 1768 RGB quarter-rings into one 60-pixel circle. Dry-fit its 158mm OD and 145mm ID around the radius 76 support posts. Use thin electronics-compatible adhesive strips on the posts to retain the PCB at head Z24. Leave solder-joint and wire clearance at the quadrant joins; check for sharp leads before fitting the diffuser.
17. Set the LCD cradle on the four corner pillars at head Z14, with its USB relief toward the intended cable route. Place the actual LCD in the 55.8mm pocket on thin edge foam. The rigid body envelope is 17–29; the front glass datum is assumed 31.2. Place the corner retaining plate at 29–32. Four M3x25 screws go through retainer and cradle into the lower shell pillars. Tighten only until foam is just compressed. Confirm no load is applied to the active glass, and test display/USB operation while it is accessible.
18. Install the white-ring carrier on its separate four cardinal bosses at radius 45, head Z32.5, using four M3x10 screws. It is 2mm thick and does not intersect the LCD corner posts or retainer. Retain the Adafruit 2862 RGBW ring on it with thin adhesive; the intended PCB underside is 34.5, PCB top 36.1 and LED top approximately 38.1. Check the actual board and LED heights; the optical skin begins at 40.8, leaving about 2.7mm above the assumed LED top.
19. Place 1mm axial shims on each of the four radius 60 face bosses and eight radius 83 bezel bosses. Fit the outer diffuser with its smooth face outward: its front is 42 and its rear flange is 35.8–37. The front bezel and center baffle capture the flanges without squeezing the optical skin. The outer flange has screw-clearance notches. The 142.6mm center baffle leaves 0.2mm radial clearance to the diffuser's 143mm inner wall. Secure the center plate with four M3x12 screws and the outer bezel with eight M3x10 screws.
20. Place the inner translucent diffuser over the RGBW ring and secure it with three tiny removable dots of neutral-cure silicone at its outer edge. Keep its 47.8mm opening centered over the active 45.68mm screen. Its outer diameter 68mm clears the center plate's 68.6mm opening. Let adhesive cure fully before operating the rings. The central visible LCD remains recessed; verify the intended viewing angle before making final foam adjustments.

Bring-up and mechanical limits

21. Set each servo ID separately before daisy-chaining its data lines. Index all four neutral counts to 2048 with the physical pose above. Start with the head supported, torque disabled, and inspect the complete manually moved limited envelope for cable tension and contact. Then operate the empty structure slowly. Install the head only after the unloaded checks. Initial commanded ranges are yaw ± 25 degrees and shoulder/elbow/wrist ± 10 degrees about neutral; these are commissioning limits, not a verified full workspace.
22. Recheck fasteners and case-clamp grip after the first short motion run and after the first thermal cycle. Read servo load/current/temperature while stationary. Reduce ring brightness during initial enclosed thermal checks. Physical proof-load, drop, fatigue, diffusion, thermal and cable-flex tests remain to be performed on a built sample; mesh checks cannot establish those properties.

Load and stability calculations

The 30kg.cm vendor torque is a stall value, approximately 2.94N.m, not a continuous design rating. For preliminary sizing, assume a 0.40kg complete head including fork, a 0.13kg forearm including wrist servo, a 0.12kg upper link including elbow servo, and low acceleration. With a 0.06m effective head offset forward from the wrist, the folded neutral head center is about 0.086m ahead of the shoulder. Its gravitational shoulder moment is $0.40 \times 9.81 \times 0.086 = 0.34\text{N.m}$. The forearm's center of mass is near $Y = 0.017\text{m}$ and the upper link near $Y = 0.030\text{m}$, so those masses partially counterbalance the head. Use the absolute sum rather than crediting counterbalance for a conservative neutral estimate: $0.34 + 0.13 \times 9.81 \times 0.017 + 0.12 \times 9.81 \times 0.030 \approx 0.40\text{N.m}$.

If the same assembly is extended horizontally, a 0.32m head moment arm alone produces 1.26N.m, and the links add about 0.30N.m. This approaches the stall rating too closely for prolonged operation without a measured thermal/load assessment. The default limited folded pose deliberately avoids claiming that extended posture as continuous-duty capability. Weigh the built head; if it exceeds 0.40kg, reduce reach and motion and repeat the calculations using measured mass.

The 1.19kg steel disc alone supplies a nominal restoring moment $1.19 \times 9.81 \times 0.108 \approx 1.26\text{N.m}$ about the base edge before considering the rest of the base mass. The calculation assumes a flat rigid support and does not include dynamic forces, cable pulls or foot friction. It supports low-acceleration folded commissioning, not a stability certification. Use a commercial padded desk clamp if the measured mass/reach or intended motion requires additional restraint. Do not use the printed lamp as a lifting arm.

Drawing conventions and verification

Each printable STL has its own local coordinates. The manifest identifies orientation, dimensions, quantities and hardware. Assembly transforms are row-major 4x4 matrices with millimetre translations; apply them directly to the STL vertices. Purchased metal horns, fasteners, wiring and the actual display geometry are not generated by the print exporter. Mechanical projection sheets should be derived from the exact final STLs. `validation/assembly_check.json` records static indexed-neutral printed-part collision results; its scope does not include a proven continuous motion envelope. Keep the final report with the manual.

LUMA electrical design, revision A

The Raspberry Pi 4 is the only behavior and motion master. It reads exactly five Adafruit 3967 VL53L1X STEMMMA QT sensors, commands four ST3215 position servos, selects face expressions, produces local speech, and drives both light rings. The Waveshare board is a wired display coprocessor; its Wi-Fi, touch, microphone and battery circuitry are unused.

The selected display is the **Waveshare ESP32-S3-Touch-LCD-1.85, SKU 28514, without B or C suffix**. This is a 360 × 360 TFT LCD, not OLED. Its 55 × 55 mm circuit assembly and ST77916 QSPI interface are vendor documented. USB serial avoids presenting the bare QSPI panel as a directly compatible Pi display. Do not substitute a visually similar 1.85B/1.85C: their pin assignments and revisions differ. Reserve a 15 mm rear right-angle plug/cable clearance as a design assumption, then measure the purchased cable and board before final printing. See [Waveshare documentation](#).

```
Pi4 USB —┬─ Waveshare ESP32-S3 1.85 LCD / face firmware
          │├─ Bus Servo Adapter A — DATA — ID1 yaw — ID2 shoulder — ID3 elbow — ID4 wrist
          └─ USB speaker / local espeak-ng speech
Pi I2C — PCA9548/TCA9548A-compatible mux — ports0..4 — five VL53L1X
Pi BCM10 SPI MOSI — 74AHCT125 +330Ω — 60RGB outer ring
Pi BCM18 PWM0 — 74AHCT125 +330Ω — 24RGBW inner ring (white only)
Pi BCM27 — isolated NC stop contact — GND
```

Power and physical wiring

Use the official Pi 5.1 V/3 A USB-C supply. Use an external enclosed Mean Well GST220A12-R7B 12 V/15 A supply for the actuators and the LED buck converter. All grounds share one DC star point; the Pi 5 V rail and LED 5 V rail are never joined. The enclosure contains DC only. The chosen supply uses a locking R7B power DIN, not a standard small barrel plug. Order a correctly rated mating harness and wire all parallel contacts exactly as the supply drawing specifies. [Mean Well specification](#).

Each 12 V ST3215 can draw 2.7 A stalled. Four together can demand 10.8 A; that is an abnormal transient design load, not a permissible continuous duty point. A 15 A main DC fuse protects a 16 AWG input trunk. A normally-open 12 V coil relay with a contact rating of at least 15 A at 12 VDC disconnects the four motor feeds. A latching stop's first NC contact interrupts the fused relay coil supply; its second, electrically isolated NC contact grounds BCM27 in the normal state. Fit a coil flyback diode with stripe to coil positive. **The stop must work with the Pi disconnected.** Match actual connector, relay and wire ratings; the generic accessories in the BOM are procurement specifications, not falsely verified exact products. [Servo electrical specifications](#).

Distribute motor V+ and GND as four separate 18 AWG feeds with 3 A time-delay fuses on V+. Do not route all motor current through one servo lead or the USB adapter's barrel jack. The adapter gets its own 1 A fused supply; its DATA and GND join the servo signal harness while its V+ is disconnected from the motor harness. Observe the markings on the actual servo connector; do not rely only on red/black/white wire conventions. The stop disconnects motor V+, while Pi, face and lights remain powered to report the stop.

The Pololu D24V50F5 converts 12 V to LED 5 V through a 2 A input fuse and 3 A output fuse. Its advertised 5 A output is thermally limited. Software caps the outer RGB ring at 20%. The outer RGB worst-case allowance at that cap is approximately $60 \times 60 \text{ mA} \times 0.2 = 0.72 \text{ A}$. The inner ring uses only white with RGB forced to zero and W limited to 192/255; approximately $24 \times 20 \text{ mA} \times 192/255 = 0.36 \text{ A}$, or 1.8W electrical LED input. Exact LED revisions differ, so measure assembled current. Uncapped all-channel full white would be about 5.52 A across the rings and is not an allowed operating mode. This first prototype is an expressive accent/task-area lamp; lux, CRI and lumen output are not validated and are not equivalent to the reference commercial lamp. [Pololu specification](#).

Use two channels of a **74AHCT125**, powered at LED 5 V, to shift BCM10 and BCM18. HC125 alone is not the same input threshold. Pin-by-pin connections are in `wiring.csv`. Put a 330 Ω resistor at each LED data input, a 1000 μF/10 V polarized capacitor across each ring supply, and a 100 nF ceramic across DIP pins14/7. The four outer quadrants require a structural backing, not just soldered joints. Solder DATA through only three joints; leave the last DOUT open. The RGB and RGBW streams remain separate. [NeoPixel ring documentation](#), [RGBW ring](#).

Five sensors and bus allocation

Mux address is 0x70; each sensor remains at its default 0x29 on a separate channel. Port0 is front, port1 left, port2 right, port3 rear, port4 down, using the lamp's own left/right. A rear hand gesture requests a wink. The mux is Adafruit5626 PCA9548, electrically compatible with TCA9548A and the same Python driver. Upstream I2C uses Pi physical3/5 (BCM2/3); logic power is Pi3.3V. Do not power the upstream pull-ups at 5 V. [Mux documentation](#), [sensor pinout](#).

Place the mux in the head to keep the five branch cables short. Keep the Pi-to-head trunk about 500 mm or shorter at 100 kHz and separate it from motor power. A longer bus requires a measured signal integrity check or a differential I2C extender. The mux isolates addresses, not optical beams: software ranges one sensor at a time at a 33 ms timing budget in short mode. A normal sweep is roughly 0.25–0.4 seconds including overhead. Out-of-range/no-return measurements are invalid, never assumed clear. Short mode is intentionally conservative for indoor interactions; it does not promise the device's headline4m maximum.

Remove shipping films. Each ranging aperture must remain open or use a specifically characterized IR window. **Do not put the frosted print over a ToF sensor.** The down sensor must see a nominal desk distance above100mm throughout the allowed pose range; if the support surface is closer, reposition the sensor before commissioning. An object below100mm, missing sensor, invalid range, or reading older than650ms latches a hold. These are interaction interlocks, not a safety-rated collision detector, and five narrow cones do not cover all pinch points.

Motion indexing and feedback

Set servo IDs individually with the vendor tool before making the bus:1 yaw,2 shoulder,3 elbow,4 wrist. Set position mode and1Mbps. Mount horns at encoder2048 with the mechanical neutral pose: upper link115° measured from the+Y horizontal direction, forearm45° from+Y, and head face vertical facing+Y. The relative elbow angle is–70°. The shoulder pivot remains atZ115mm and the wrist is approximatelyY25.69mm/Z326.74mm in the CAD reference. Initial software limits are yaw±25°, shoulder±10°, elbow±10°, wrist±10°. `joint_signs` must be measured against the mechanical axes; the shipped all+1 values are calibration defaults, not verified motor handedness.

At startup hardware motion stays disabled unless both `calibrated=true` and `--arm` are supplied. Before torque enable, all feedback must be valid and within3° of indexed neutral. Runtime checks encoder range,9.0–12.6V feedback, temperature below65°C and serial checksums; targets are limited to12°/s. A detected interaction fault sends the most recently measured hold position once and latches; it does not resume after a hand moves away. Restart requires clear sensors and an explicit arm command. Servo status feedback does not establish joint structural strength or torque capacity.

On normal exit the servos keep holding their final position; lights turn off. A power cut releases holding torque. Support the arm before pressing the physical motor stop or removing power. Fit the mechanical support/prop during calibration. If the Pi hangs, feedback checks cannot execute; the independent physical motor stop is the available intervention. No watchdog or safety-rated stop controller is claimed.

Bench commissioning sequence

1. Keep the arm supported and all motor V+ feeds disconnected. Check DC polarity and continuity; verify3.3V logic,5VLED and12V motor rails independently.
2. Flash and test the exact display with PING/PONG, then FACE wink, FACE hi, FACE happy. Confirm the canvas is centered and colors are correct before fitting the bezel.
3. Install the Pi software and enable I2C/SPI. With motors supported and stopped, check that channels0–4 each show one0x29 sensor. Cover each aperture individually and confirm the intended channel changes.
4. Test outerRGB at20% or lower and innerW at192/255 or lower. Verify that the inner ring emits white only, and the outer ring has60 independently drivenRGB pixels. Check the diffuser for hot spots and measure surface/LED/CPU temperatures during a30-minute run.
5. Program one servo at a time to IDs1–4, position mode,1Mbps. With the power stop in reach, read neutral positions while supported. Index horns and signs to the CAD reference. Do not substitute uncalibrated encoder2048 for a mechanically checked pose.

6. Verify that the physical stop removes motor voltage even if the Pi is unplugged. Check BCM27 reports stop when either NC2 wire is removed. Release the stop with the arm supported.
7. Set actual persistent USB device paths and measured joint signs/neutral ticks in `config.json`. Set `calibrated=true` only after the previous checks. Use hardware mode without `--arm` first, then run one supported6-second scene with `--arm`.
8. Check full allowed motion for cable clearance, head/arm/base collisions, stability and servo temperature/current. Measure actual load; reduce travel or redesign counterbalance if any servo cannot sustain the pose. Only remove the support after those checks pass.

Evidence and limits

The host behavior/packet tests are executable and were run in a desktop simulator. Electrical continuity, servo torque, USB power budget, optical ranging and thermal behavior require the physical prototype. Vendor dimensions and electrical ratings are cited above; mounting clearances, wire lengths, fuse coordination and unknown board revisions require fit checks. Budget figures in the CSV are planning allowances, not quotations or stock promises.

Point-to-point wiring schedule

Use these connection names together with the electrical chapter. Physical Raspberry Pi header numbers and BCM GPIO numbers are explicitly distinguished. Disconnect the supplies while changing wiring.

Net	From → to	Cable / connection detail
3V3_LOGIC	Raspberry Pi4: physical1 3V3 → Mux5626: VIN	STEMMAQT upstream red. All I2C pullups stay 3.3V
GND_LOGIC	Raspberry Pi4: physical6 GND → Mux5626: GND	STEMMAQT upstream black. Connect DCSTARground also
I2C_SDA	Raspberry Pi4: physical3 BCM2 → Mux5626: SDA	STEMMAQT upstream blue. I2C bus1 at 100kHz
I2C_SCL	Raspberry Pi4: physical5 BCM3 → Mux5626: SCL	STEMMAQT upstream yellow. I2C bus1 at 100kHz
TOF_FRONT	Mux5626: port0 → VL53L1X front: STEMMAQT4pin	4conductor JSTSH. Address 0x29 isolated on channel 0
TOF_LEFT	Mux5626: port1 → VL53L1X left: STEMMAQT4pin	4conductor JSTSH. Left as seen by lamp facing user
TOF_RIGHT	Mux5626: port2 → VL53L1X right: STEMMAQT4pin	4conductor JSTSH. No sensor shares mux channel
TOF_REAR	Mux5626: port3 → VL53L1X rear: STEMMAQT4pin	4conductor JSTSH. Leave sensor optical aperture clear
TOF_DOWN	Mux5626: port4 → VL53L1X down: STEMMAQT4pin	4conductor JSTSH. Nominal desk range must remain above 100mm
RGB_DATA3V3	Raspberry Pi4: physical19 BCM10 MOSI → 74AHCT125: DIP2 1A	26AWG. Do not connect SPI clock to ring
RGB_DATA5V	74AHCT125: DIP3 1Y → RGB ring quadrant1: DIN via 330ohm	26AWG. Resistor at first ring input
RGB_CHAIN1	RGB quadrant1: DOUT → RGB quadrant2: DIN	solder jumper. Connect common 5V and GND at all quadrants
RGB_CHAIN2	RGB quadrant2: DOUT → RGB quadrant3: DIN	solder jumper. Mechanical backing supports solder joints
RGB_CHAIN3	RGB quadrant3: DOUT → RGB quadrant4: DIN	solder jumper. Do not connect last DOUT back to first DIN
WHITE_DATA3V3	Raspberry Pi4: physical12 BCM18 PWM0 → 74AHCT125: DIP5 2A	26AWG. Disable onboard analog audio
WHITE_DATA5V	74AHCT125: DIP6 2Y → 24RGBW inner ring: DIN via 330ohm	26AWG. Separate GRBW data stream
SHIFT_ENABLE	DCstar: GND → 74AHCT125: DIP1 /1OE and DIP4 /2OE	26AWG. Active outputs; tie both low
SHIFT_UNUSED	5VLED: +5V → 74AHCT125: DIP10 /3OE and DIP13 /4OE	26AWG. Unused outputs disabled
SHIFT_UNUSED_INPUT	DCstar: GND → 74AHCT125: DIP9 3A and DIP12 4A	26AWG. Unused DIP8 and DIP11 outputs unconnected

Net	From → to	Cable / connection detail
SHIFT_SUPPLY	5VLED: +5V → 74AHCT125: DIP14 VCC	24AWG. 100nF betweenDIP14 andDIP7
SHIFT_GROUND	DCstar: GND → 74AHCT125: DIP7 GND	24AWG. Shared reference toPi andboth rings
PI_GROUND	Raspberry Pi4: physical9 GND → DCstar: GND	22AWG. Common ground only; never tie5VLED toPi5V
PI_POWER	Pi official supply: USB-C5.1V3A → Raspberry Pi4: USB-Cpower	factory cable. Independent from motor12V
FACE_USB	Raspberry Pi4: USB-Aport1 → DisplaySKU28514: USB-C	data cable. Do not add a second5V power feed
SERVO_USB	Raspberry Pi4: USB-Aport2 → BusAdapterA: USB	data cable. ST mode;USB bridge mode
AUDIO_USB	Raspberry Pi4: USB-Aport3 → USB speaker3369: USB	captive cable. LimitUSB total current toPi specification
DC_INPUT	MeanWellGST220A12: R7B+12V → Main15Afuse: input	16AWG. Use correctparallel contacts from supply drawing
DC_MAIN	Main15Afuse: output → DC12Vdistribution: +12V	16AWG. Supply remains outside printed enclosure
DC_RETURN	MeanWellGST220A12: R7Breturn → DCstar: GND	16AWG. All high-current returns star here
STOP_COIL_FEED	DC12Vdistribution: +12V → 0.5Afuse: input	22AWG. Coil branch
STOP_NC1	0.5Afuse: output → StopNC1: COM	22AWG. NoPiGPIO in this branch
STOP_COIL_PLUS	StopNC1: NC → Relay: coilpositive	22AWG. Diodecathode stripe here
STOP_COIL_MINUS	Relay: coilnegative → DCstar: GND	22AWG. Diodeanode here
MOTOR_RELAY_IN	DC12Vdistribution: +12V → Relay: COM highcurrent	16AWG. NOcontact>=15A12VDC
MOTOR_RELAY_OUT	Relay: NO highcurrent → Servo4way fusedstar: input	16AWG. StopopensallmotorV+independentofsoftware
SERVO_POWER1	Servo3Afuse1: output → Yawservoid1: V+	18AWG. Separatefeed not powered through adapter
SERVO_POWER2	Servo3Afuse2: output → Shoulderservoid2: V+	18AWG. Separatefeed
SERVO_POWER3	Servo3Afuse3: output → Elbowservoid3: V+	18AWG. Separatefeed
SERVO_POWER4	Servo3Afuse4: output → Wristservoid4: V+	18AWG. Separatefeed
SERVO_GROUNDS	Allfourservos: GND → DCstar: GND	18AWG. One return per servo
ADAPTER_POWER	DC12Vdistribution: +12V via1Afuse → BusAdapterA: DC5.5x2.1mm positive	22AWG. Adapterpoweronly;detachV+ fromdataharness

Net	From → to	Cable / connection detail
ADAPTER_GROUND	BusAdapterA: DCnegative/GND → DCstar: GND	22AWG. ReferenceforDATA
SERVO_DATA	BusAdapterA: STDATA → Servos1-4: DATA daisy-chain	26AWG. Preservegroundreference;do not bridge separateV+ feeds
STOP_SENSE	Raspberry Pi4: physical13 BCM27 → StopNC2: COM	26AWG. External10kohm toPi3.3V
STOP_SENSE_RETURN	StopNC2: NC → Raspberry Pi4: physical14 GND	26AWG. Opencontact orbrokenwire readsstop
BUCK_INPUT	DC12Vdistribution: +12V via2Afuse → PololuD24V50F5: VIN	22AWG. ENleftunconnected
BUCK_GND	PololuD24V50F5: GND → DCstar: GND	22AWG. Shortreturn
LED_FUSED	PololuD24V50F5: VOUT5V → 3Afuse: input	22AWG. Verify5VbeforeconnectingLEDs
LED_RGB_POWER	3Afuse: output5V → Outer60RGB: 5V atoppositequadrants	22AWG. 1000uF10Vcap at ring
LED_WHITE_POWER	3Afuse: output5V → Inner24RGBW: 5V	22AWG. 1000uF10Vcap at ring
LED_RETURNS	Bothrings: GND → DCstar: GND	22AWG. KeepLEDreturnseparatefromI2Creturnuntilstar

LUMA Pi4 controller and face firmware

The project includes a deterministic simulator and Raspberry Pi adapters for five ranging sensors, four ST3215 bus servos, two independent LED rings, USB face display and local spoken Hi. The simulation needs only Python3.11 or later.

```
cd software
python -m luma.app --scene wink
python -m luma.app --scene hi
python -m luma.app --scene happy
python -m unittest discover -s tests -v
```

Simulation prints timestamped pose/light/face events as JSON. It advances deterministically without sleeping. `--seconds` sets duration. Scenes last 6 seconds, then return to idle. The three promotional scenes remain within the initial CAD travel envelope; they are repeatable face/motion scripts and do not require a cloud service.

Raspberry Pi installation

Use Raspberry Pi OS Bookworm64bit on Pi4. The following setup commands affect only a Pi used for this project; do not execute them on the development Windows computer.

```
sudo apt update
sudo apt install python3-venv python3-pip python3-dev i2c-tools espeak-ng alsa-utils
sudo raspi-config
```

In Interface Options enable I2C and SPI. In `/boot/firmware/config.txt`, confirm `dtoverlay=i2c_arm=on`, `dtoverlay=i2c_arm_baudrate=100000`, and `dtoverlay=spi=on`. Set `dtoverlay=audio=off` because BCM18's PWM peripheral drives the RGBW ring and analog audio conflicts. USB audio is separate. Reboot after these settings.

```
cd software
python3 -m venv .venv
.venv/bin/pip install -e '[hardware]'
ls -l /dev/serial/by-id/
aplay -l
```

Enter the actual servo/display `/dev/serial/by-id/` paths in `config.json`. Select the USB speaker as the system default ALSA output, or set `audio_device` to the ALSA identifier listed by `aplay -l` (for example `plughw:CARD=Device,DEV=0`; actual name varies). Test speech at low volume. Three USB devices share Pi4's downstream USB power budget; verify measured total current, and use a separately powered hub if the selected speaker/display exceed it. The LED rings never take power from Pi USB or GPIO.

Before connecting motor power, test the sensor chain and flashed display independently. These commands never instantiate the servo bus or LED drivers and do not require a calibrated configuration:

```
.venv/bin/python -m luma.bench sensors --seconds 10
.venv/bin/python -m luma.bench display --port /dev/serial/by-id/YOUR_DISPLAY_DEVICE --scene all
--seconds 10
```

The sensor check reports millimeters, sample age and physical mux channel for all five sensors. Present a matte target within range to each optical port; a missing/out-of-range return is not treated as clear space. `near` means a valid reading below 100mm: the communication check can pass, but armed motion would fault. The display check cycles `idle/wink/Hi/happy` and checks real firmware replies; its pass result verifies communication, not correct pixels. Inspect the screen yourself. Exit status 0 means the final check passed, 1 means failed, 130 means interrupted. Sensor tests require the configured I2C permissions (normally Pi OS's `i2c` group); USB serial requires `dia1out` access. Diagnose an access error before using `sudo` indiscriminately.

The PWM-backed NeoPixel library requires access normally provided by root on Pi4, so the explicit hardware command uses the virtual environment's interpreter with `sudo`. Leave `calibrated=false` while wiring/testing. Hardware mode without `--arm` still requires powered servo feedback but does not send movement commands. Use the separate bench checks above when motors are unpowered.

```
sudo .venv/bin/python -m luma.app --hardware --scene hi --seconds 6
```

Complete the commissioning procedure in `../electronics/architecture.md`, including supported neutral indexing, polarity/sign checks, physical stop tests and five valid ranges. Only then set `calibrated=true`. Start each run from within 3° of the indexed neutral pose:

```
sudo .venv/bin/python -m luma.app --hardware --arm --scene happy --seconds 6
```

Any fault latches for that process. Inspect the logged reason, support the arm, remove the cause, return to the indexed pose and restart explicitly. Normal exit leaves servos holding. Motor supply interruption releases them. The software has no unattended boot/auto-arm service.

Display firmware

`display_firmware/` targets **ESP32-S3-Touch-LCD-1.85 SKU28514 only**. It uses USB serial115200, an ST77916 QSPI controller on CS21/SCK40/D046/D145/D242/D341, and a TCA9554 at0x20 over SDA11/SCL10 for LCD resetEXIO2/P1. Backlight is GPIO5. This pin mapping is from the [Waveshare board documentation](#) and schematic. Firmware reads panel register0x04 and chooses one of the exact two vendor initialization tables: ID00-7F-7F-7F or00-02-7F-7F. Unknown IDs produce an error and do not advertise READY. Register command/data/delay values were converted from the vendor's April2025 demo to [Arduino GFX](#) operations. A successful compile does not prove physical display operation; verify the purchased panel on the bench.

Install PlatformIO and build/flash the connected board:

```
python -m pip install platformio
cd display_firmware
python -m platformio run
python -m platformio run --target upload --upload-port YOUR_SERIAL_PORT
python -m platformio device monitor --port YOUR_SERIAL_PORT --baud 115200
```

Commands are newline-delimited ASCII. PING replies PONG. FACE idle, FACE wink, FACE hi, FACE happy, and FACE fault reply OK <name>. Startup announces READY LUMA1. Invalid or overlong commands reply ERR. After1.5seconds without a host command/heartbeat the display shows WAITING. The host sends heartbeat every250ms and latches a motion hold if replies stop. No touch/audio/SD/battery functions are enabled on the display board; speech is generated on the Pi and played by the USB speaker.

Source/verification notes

`luma/servo_bus.py` implements only checked position reads and volatile sync writes; no runtime EEPROM changes. Register addresses and little-endian packet arrangement were compared with the [Waveshare ST/SC Python SDK/STServo_Python.zip](#)), `scservo_sdk/sms_sts.py` and `protocol_packet_handler.py`. Controller tests cover all scene travel/speed envelopes, five-sensor requirements, stale/NaN/no-return data, obstruction hold, stop/display/servo faults, speech event debouncing and malformed servo packets. Hardware paths have not been exercised on a physical Pi/arm.

The pinned firmware toolchain is self-contained in `platformio.ini`; downloaded SDK/toolchain caches are development artifacts and do not need redistribution. Python hardware dependency versions are bounded for installation flexibility, not a physical acceptance-tested lockfile. Record actual installed versions with `pip freeze` when commissioning the Pi.

First-build verification and maintenance

Verify before enclosing

Power and exercise the display, rings and each sensor on the bench. Confirm that the large ring shows the intended colors, the small ring uses only its white channel during lamp operation and every sensor is associated with its labeled multiplexer port. This makes a reversed ring or swapped cable easy to locate while it is still accessible.

Connect each motor individually to set its unique ID. Confirm the servo voltage variant and the expected operating mode before attaching the horn. Mark the calibrated neutral pose on the fixed and moving brackets. Save the calibration with the assembled unit rather than assuming a replacement motor will have the same zero offset.

Mechanical acceptance record

Check	Acceptance and record
Printed interfaces	Actual purchased boards and horns seat without forced deflection; record component revision.
Fasteners	Required washers/nuts present, rotating surfaces free and wires not under screw heads.
Bearings and joints	Free motion when unpowered; no binding, visible cracks or loose axial play.
Cable routing	Slack remains throughout allowed travel; no pinched insulation or tight connector bends.
Stability	Base secured as specified; verify measured mass and center of gravity at the maximum allowed reach.
Head mass	Weigh the completed head and replace assumed mass in the torque calculation.
Horns	Metal horns secured with the servo's correct center screw; no printed spline substitutes.

Electrical and motion acceptance

Start with the arm supported and a single joint enabled at reduced speed. Move a few degrees about the calibrated neutral position, verify the sign of rotation, then return to neutral. Repeat for each joint before a coordinated expression. A wrong sign or zero offset must be corrected before enlarging the motion range.

Confirm that the hardware motor-power disconnect removes motor power independently of software. Support the head when testing a power cut: these joints are not self-locking. Verify the software fault response for a missing sensor, invalid range, lost motor connection and lost display connection using the documented tests and a controlled bench check.

Run the normal white-light and RGB settings with the head assembled. Record temperatures at 5, 15 and 30 minutes at the LED support, near the screen and at the most heavily loaded servo. For the first PETG build, use 45 °C on an accessible printed enclosure surface as a conservative investigation threshold; it is a project test criterion, not a material or product certification. If temperature continues to rise, reduce brightness or load and improve the relevant airflow before extending the test.

Verify the five optical directions with a matte target at several distances. Record ranges and reject invalid readings rather than treating them as open space. Check for reflections from the diffuser, desk and nearby arm components. These five sensors do not cover every direction around the mechanism, and the behavior must stay inside the commissioned motion envelope.

Verification worksheet

Measurement	First-build result
Build date / builder	_____
Printer and filament	_____
Component revisions	_____
Completed head mass	_____
Base / ballast mass	_____
Motor supply at motion peak	_____
LED supply at normal brightness	_____
Pi undervoltage check	_____
Maximum observed enclosure temperature	_____
Mechanical travel limits and zero offsets	_____
Sensor directions and valid range test	_____
Hardware power-cut test	_____

Care and service

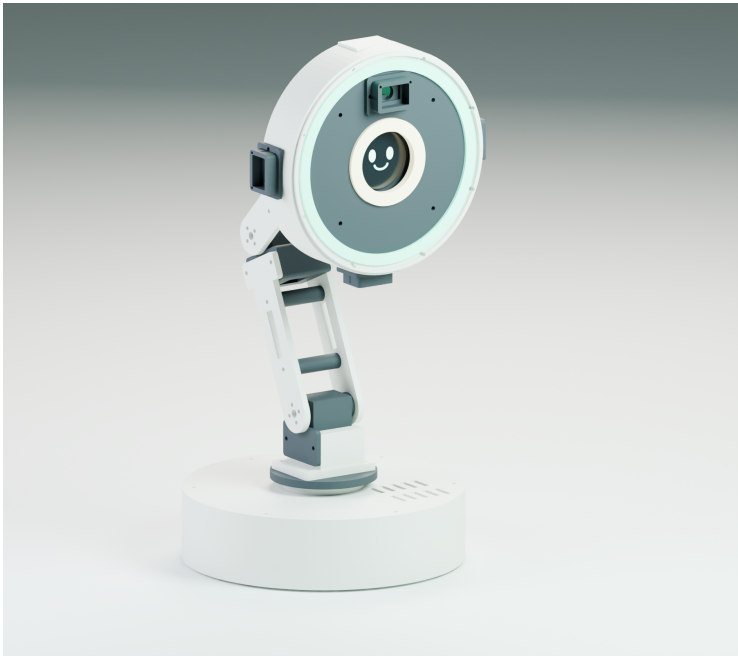
Inspect the horn screws, bearing mounts and cable strain relief after the first hour of motion and again after the first week. Stop using a printed part if a layer crack, whitening around a fastener or increasing joint play appears. Correct the cause before replacing the part.

Clean the diffuser with a soft damp cloth. Keep solvents away from printed surfaces and the display. Dust near a time-of-flight aperture can change measurements; remove it without pushing debris onto the optical package.

For service, shut down the Pi, disconnect the low-voltage supplies and support the arm before opening a joint or loosening the head. Store the final calibration, BOM revision and any dimensional changes with the project so the next printed replacement matches the machine.

Finished design and promotional films

The promotional films show the LUMA digital prototype built from the supplied CAD geometry. The lighting, camera exposure, screen graphics and sound are presentation choices; the films do not establish the physical lamp's brightness, motor sound or production finish.



Finished LUMA digital prototype, rendered from the project geometry.

The three films

Film	Included file	Action
The wink	media/videos/01_wink.mp4	A brief one-eye wink with a coordinated head gesture and colored halo.
Hi, there	media/videos/02_hi.mp4	A friendly face greeting, arm gesture and an audible "Hi!"
Happy to see you	media/videos/03_happy.mp4	Cheerful eyes and a small happy movement, with warm halo color.

Open `START_HERE.html` in a browser after extracting the archive to play all three clips. The videos are also individual files for editing or presentation use. The project includes editable animation sources under `media/sources/`.



Frames from the three included promotional animations.

Assembly visualization



Exploded view of the prototype. Explode distances expose components and do not prescribe insertion travel.

The engineering sheets at the back of the manual are the dimensional references. The rendered images explain appearance and assembly relationships; use the numbered assembly sequence for fasteners, wiring and the optical stack.

Digital release checks

The included validation files record what was checked on the development computer. They should travel with the source and manual when the project is revised. Passing these checks does not replace the first-build measurements in the preceding chapter.

Executed host software checks

The controller, bench diagnostic and servo packet test suite passed 18 tests. The checks exercise all three expressions over repeated control frames, joint angle and slew limits, the requirement for five sensor readings, stale and invalid measurements, a close-object hold, latched faults, physical stop polarity, lost display or servo feedback, and the one-shot spoken greeting.

Additional tests verify that the inner ring receives only white-channel data at its cap, that servo status packets have the right ID and checksum, that uncalibrated motion is refused, and that a stop uses measured position with a defined fallback when feedback cannot be refreshed. The simulator was exercised without activating any hardware.

Meshes and assembly

`cad/validation.json` records STL properties from the exporter. `validation/assembly_check.json` records an independent mesh check and static boolean intersection check on the assembled printed solids. It lists watertightness, positive volume, dimensions, print quantity and whether each XY bounding box fits a 220 mm bed.

The intersection check is limited to the indexed neutral pose and the printed components represented in the assembly. It does not establish moving-cable clearance, exact purchased-component fit or a fully collision-free operating envelope. Follow the manual's supported movement check with the actual hardware before using the programmed expressions.

PDF and media

`validation/final_report.json` records the final PDF page/bookmark count, text bounds and full decoding of each promotional MP4. The verification script also generates a contact sheet of the manual for visual review. Video metadata and frame counts verify the encoded files; the animation is a visualization of the digital design.

Display firmware build status

The firmware source includes the exact-board pin mapping and the vendor's two panel initialization variants with attribution. **Its cross-compilation was not completed in this environment:** the required PlatformIO compiler download did not finish, and an alternate download was truncated. No firmware binary or successful cross-compile is claimed. Run the supplied PlatformIO build on a connected development computer before flashing, and test the purchased display on the bench.

Physical work still required

Actual Raspberry Pi GPIO and USB operation, screen operation, screw engagement, motor holding ability, electrical continuity, stop-relay behavior, sensor optics, light diffusion, stability and enclosed temperatures have not been tested on a physical unit. The specification-based parts and the prototype clearances are intended for a first fit-and-function build. Record the measured results and any revisions on the verification worksheet.

Component source register

Vendor information checked for this digital release. Order by exact part number and verify the revision supplied. Prices in the BOM are budgeting references, not quotations.

S01 · [Autonomous Lamp — appearance reference](#)

Inspiration for an articulated desk companion; no commercial-lamp geometry reused.

S02 · [Raspberry Pi 4 Model B](#)

Master computer and USB-C supply selection.

S03 · [Waveshare ESP32-S3-Touch-LCD-1.85 documentation](#)

Exact integrated LCD board, display resolution, interface and vendor firmware reference.

S04 · [Adafruit VL53L1X STEMMA QT — product 3967](#)

Five sensor boards, nominal dimensions and default I2C address.

S05 · [Adafruit VL53L1X guide](#)

Sensor software and wiring reference.

S06 · [Adafruit RGBW 24-pixel ring — product 2862](#)

Inner white-light ring, RGBW format and mechanical envelope.

S07 · [Adafruit 60-pixel RGB ring quarter — product 1768](#)

Four supported arcs form the outer RGB halo.

S08 · [Adafruit NeoPixel rings guide](#)

Ring diameter reference.

S09 · [Waveshare ST3215 servo documentation](#)

12V servo dimensions, bus control and mechanical drawing.

S10 · [Waveshare ST3215 servo product specification](#)

30 kg-cm stall rating at 12V, current and encoder resolution.

S11 · [Waveshare Bus Servo Adapter \(A\)](#)

USB connection from Pi to serial servo bus.

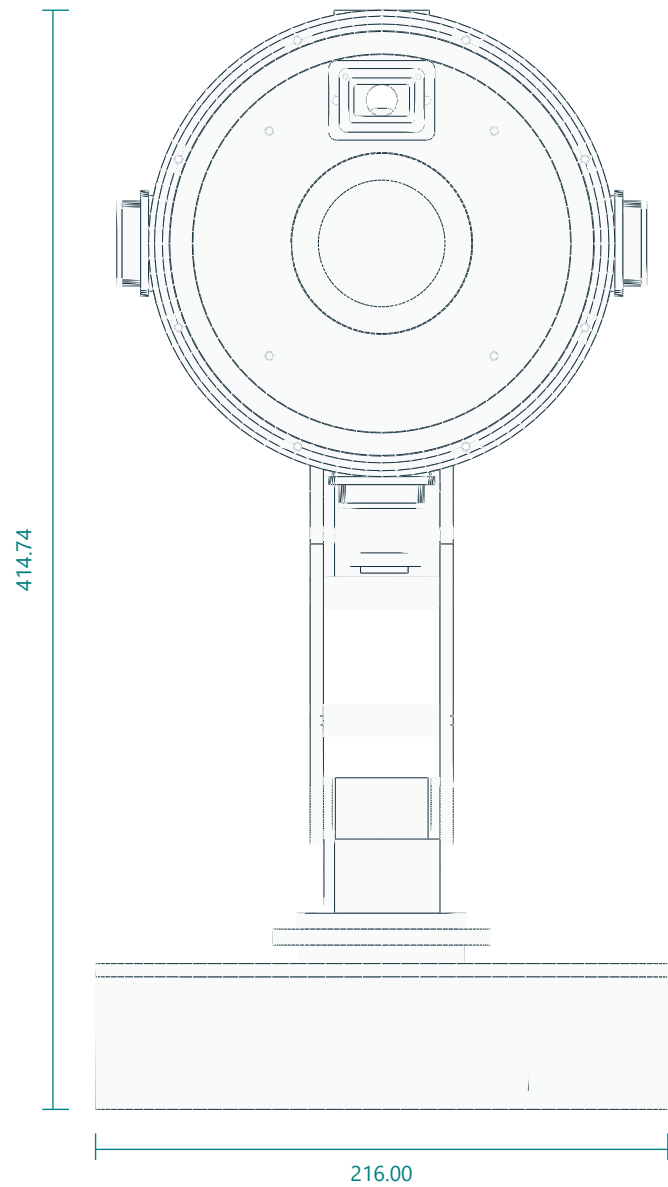
S12 · [Adafruit PCA9548 STEMMA QT multiplexer — product 5626](#)

Independent channels for five sensors with identical addresses.

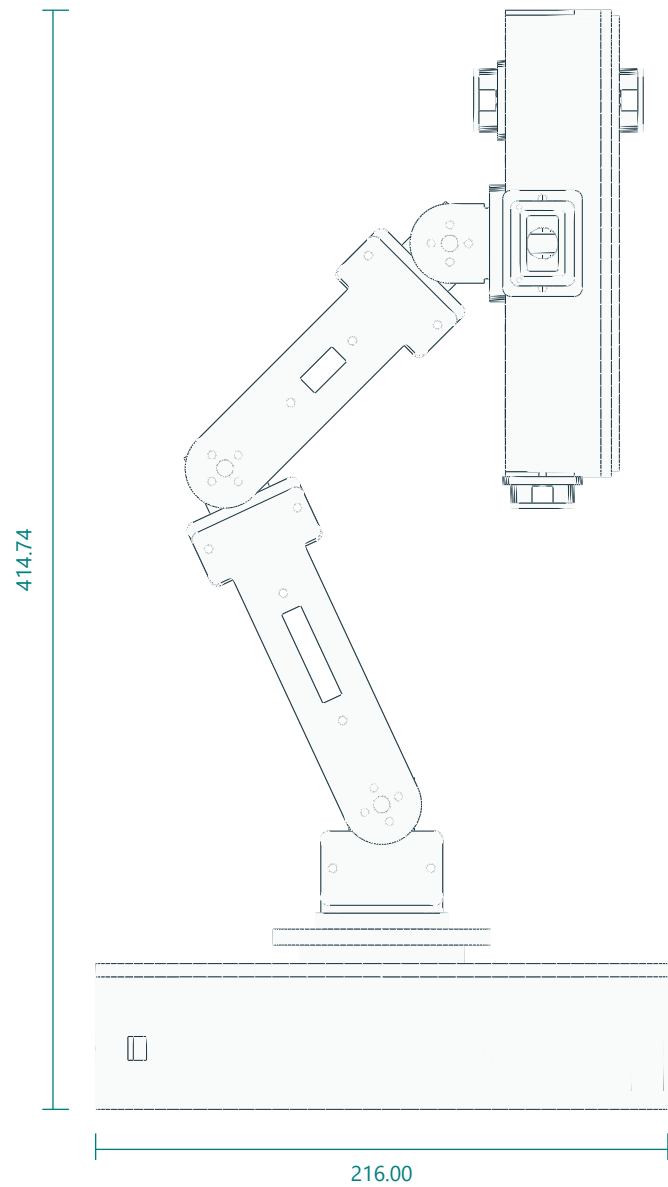
Finished assembly / indexed neutral pose

Assembled geometry from the same STL files supplied for printing. Purchased parts are simplified envelopes.

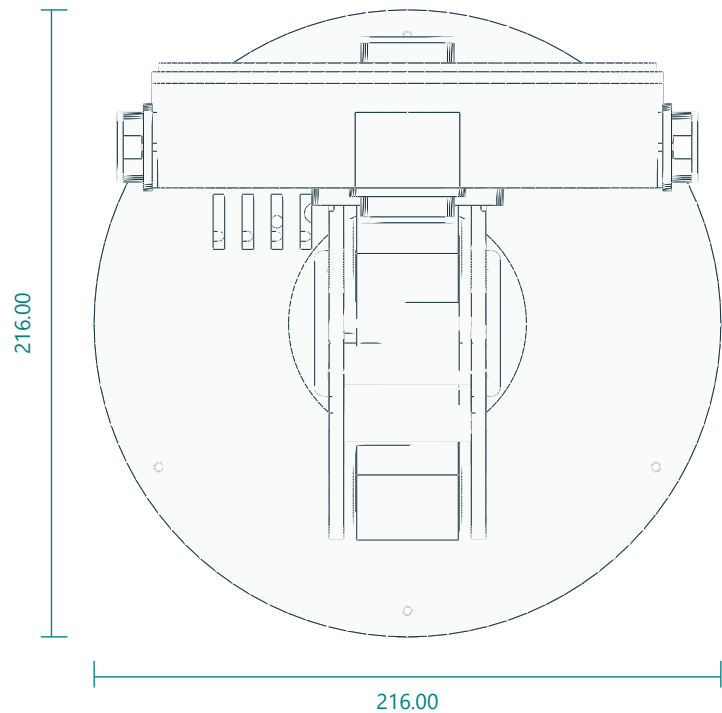
FRONT / LCD FACING VIEW



RIGHT SIDE / NEUTRAL POSE



TOP / FOOTPRINT

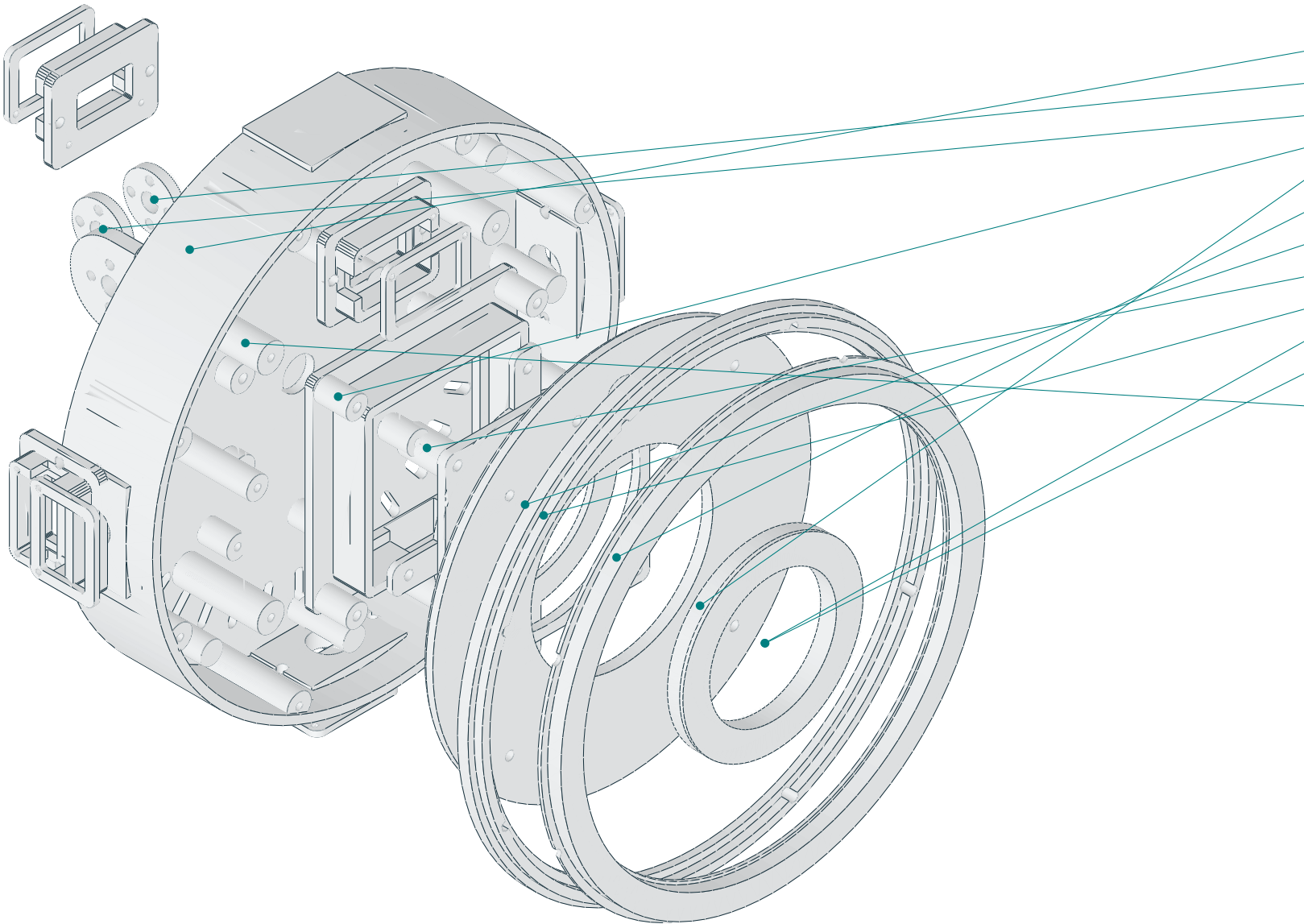


Four axes: yaw about Z; shoulder, elbow and wrist about local X. Datum is the base underside at Z=0.
Nominal linkage pitches: shoulder–elbow 140 mm; elbow–wrist 120 mm. The illustrated height is a pose dimension, not a motion envelope.
Neutral: shoulder link 115° above horizontal; forearm 45° above horizontal; face directed toward +Y. Follow indexed calibration in the manual.
Do not infer screw lengths from the illustration. Fasteners and service loops are specified in the assembly chapter.

Head / layered assembly study

Exploded positions expose the optical parts; displacements are illustrative and are not insertion paths.

EXPLODED HEAD / MODEL-DERIVED VIEW



LAYER REFERENCES

- 01 Head tilt fork
- 02 head left horn spacer
- 03 head right horn spacer
- 04 Head Shell
- 05 Outer Bezel
- 06 Face Center
- 07 White Carrier
- 08 Lcd Cradle
- 09 Lcd Retainer
- 10 Outer Diffuser
- 11 Inner Diffuser
- 12 Purchased LCD envelope

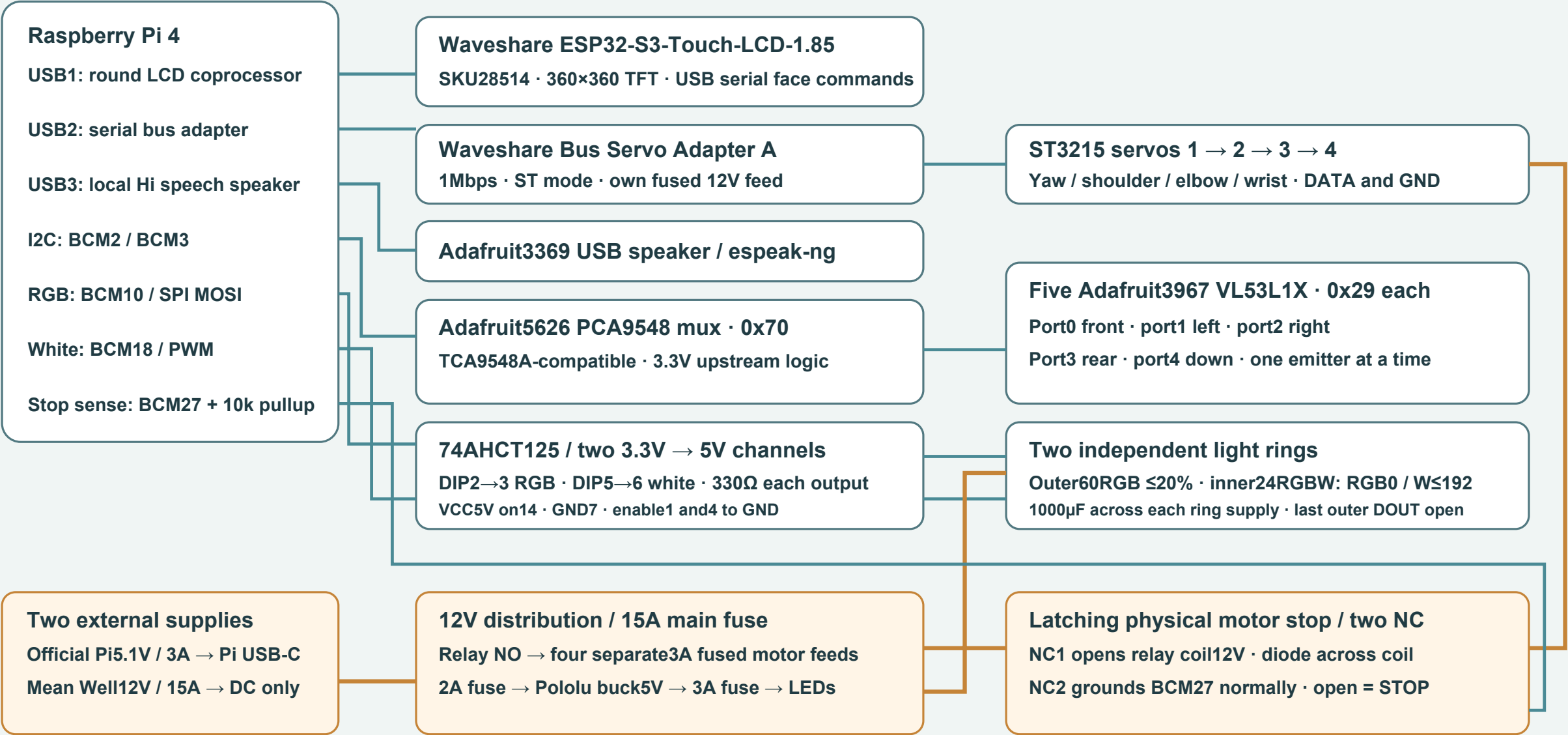
Central LCD board is behind the inner white ring. Keep all optical components at their documented axial heights.
Use natural PETG for both light diffusers and opaque PETG for the optical baffle. Leave LCD glass and all ToF apertures clear.
The four outer ring PCB quarters require continuous mechanical support. Solder joints are electrical connections only.
Refer to the head assembly chapter for screw order, spacers, wire relief and board retention. Purchased LEDs are not printable parts.

Electrical connections / system wiring

Follow the point-to-point wiring schedule for connector pins, polarity and conductor ratings.

LUMA / Electrical connections

Revision A · Raspberry Pi 4 master · four bus servos · exactly five ranging sensors



Read wiring.csv for exact connections, fuses, DIP pins and cable notes.

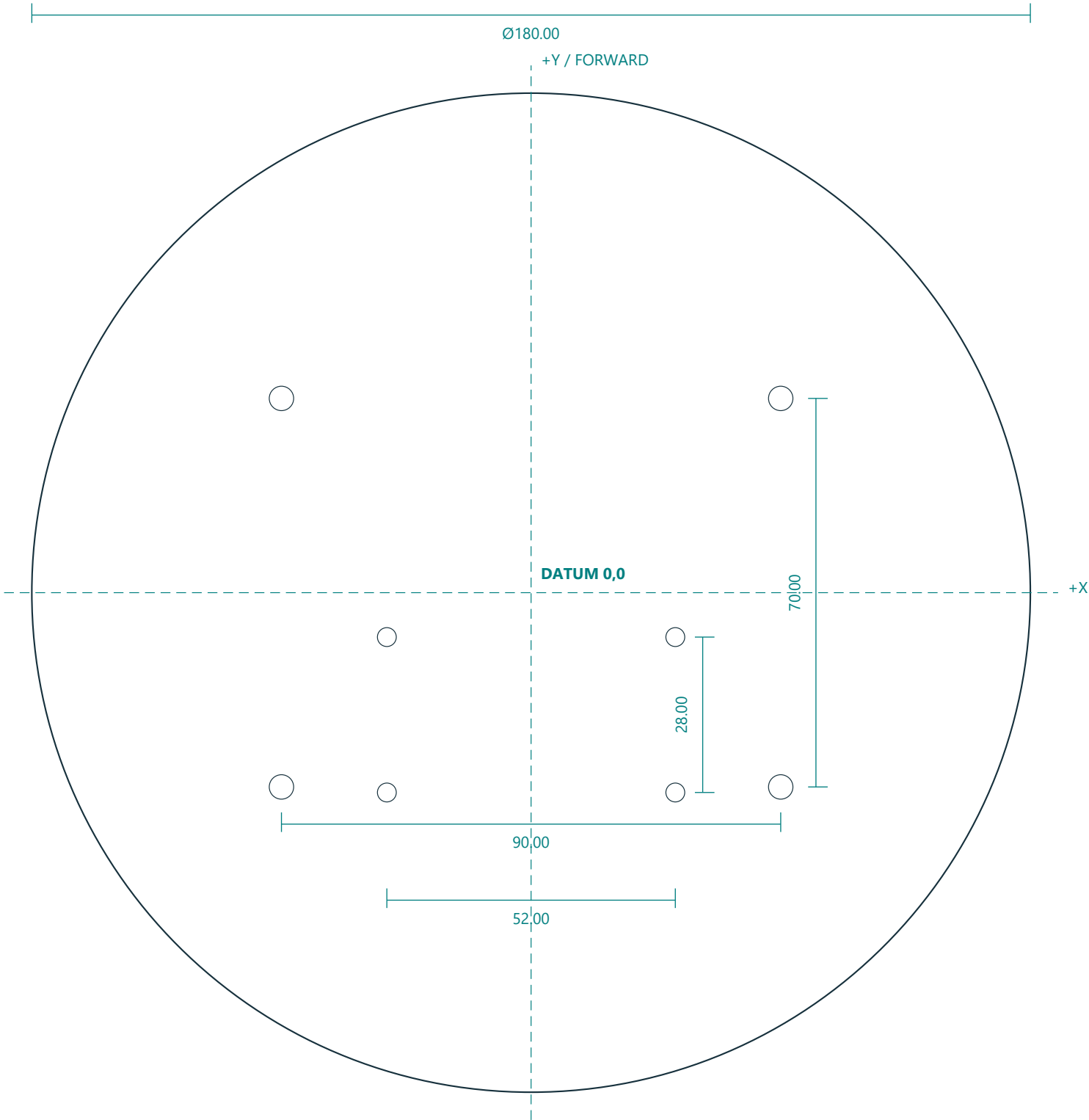
All DC grounds meet at one star point. Never join Pi5V and LED5V. Never carry all motor current through the adapter jack.

Motor stop cuts torque: support the arm. Five narrow ToF cones are interaction inputs and do not cover every pinch point.

Power wires shown in amber; signal routes in blue. Schematic overview: routing lengths and connector positions are not to scale.

Steel ballast plate / fabrication drawing

Purchased/custom-fabricated metal part M01. This sheet is not a printable-plastic substitute.



MATERIAL + INTERFACES

Material: mild steel; finished thickness 6.00mm .
Outer diameter 180.00mm ; deburr all edges and holes.
Datum origin is the disc center. All hole axes normal to plate.
 $4 \times \varnothing 4.40$ through: $X = \pm 45.00$, $Y = \pm 35.00$.
 $4 \times \varnothing 3.40$ through: $X = \pm 26.00$, $Y = -36.00$ and -8.00 .
Do not add a center hole. Rear yaw bolt pattern points-Y.
Nominal mass before drilling: 1.19kg at steel density 7.85g/cm^3 .
Fit to printed base floor before final coating. Inspect burrs and screw-head clearance.
Suggested workshop positional allowance $\pm 0.20\text{mm}$; confirm actual printed mating pattern before drilling.

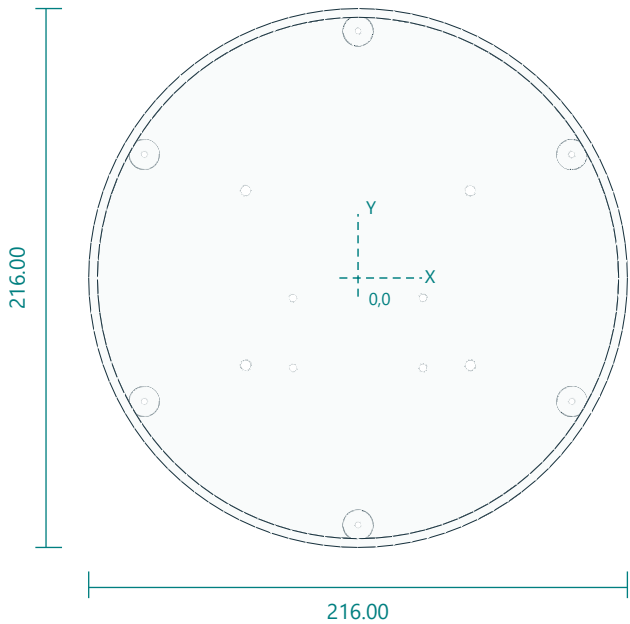
SIDE VIEW / THICKNESS



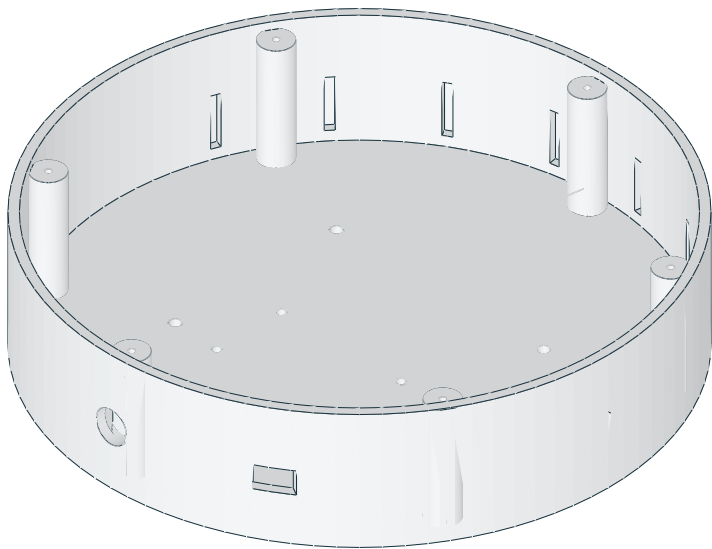
Weighted base enclosure

base_tub.stl | PRINT QUANTITY 1 | PETG

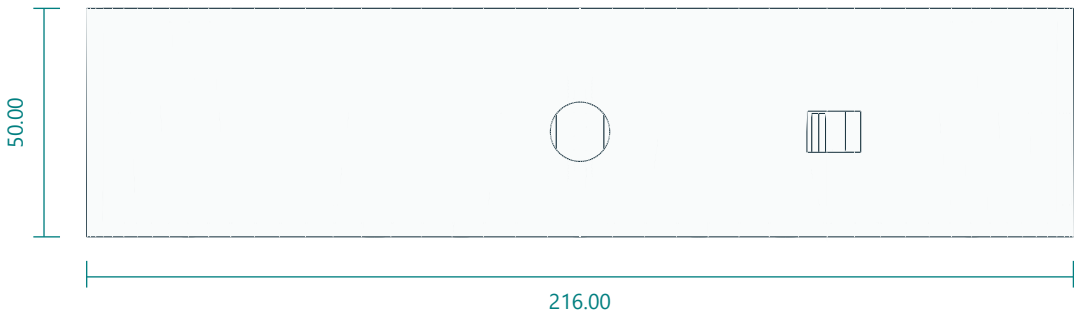
TOP / XY



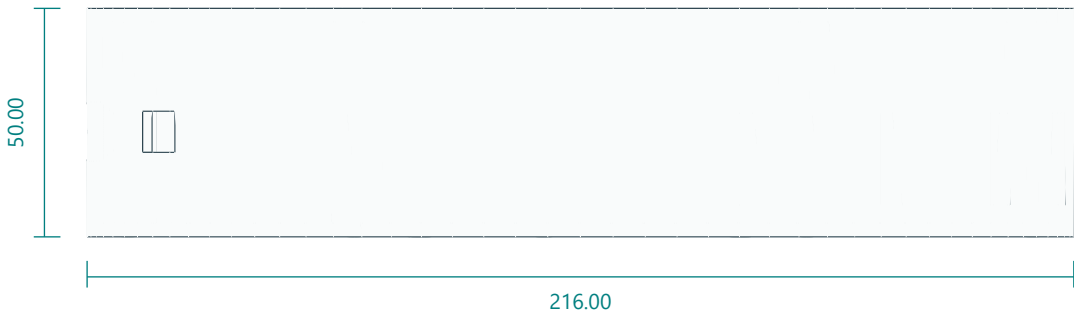
ISOMETRIC / REFERENCE



FRONT / XZ



RIGHT / YZ

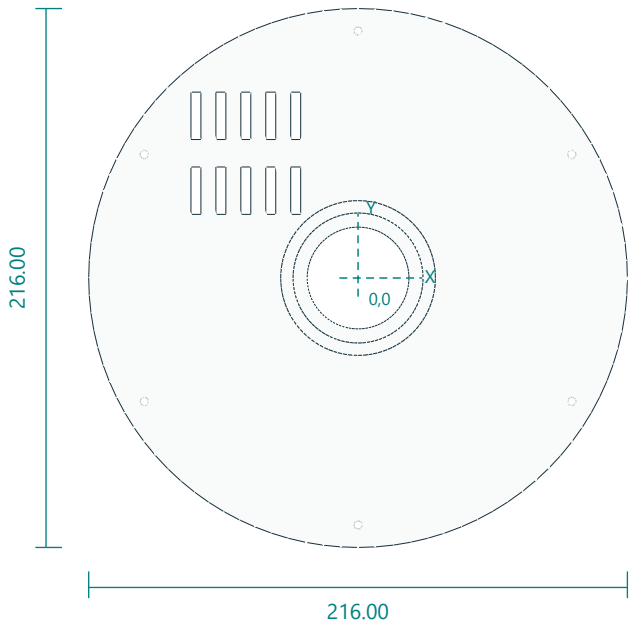


Interfaces: 216 diameter; 50 high; 3 floor; 3.5 wall.
Fasteners / retention: 6 M3x12 lid screws; 4 M4 ballast retainers.
Print orientation: floor down. Layers 0.2 mm; 4 walls; 25% infill baseline.
Feature locations: M4 clearance4.4 at (+/-45,+/-35). Yaw bolts3.4 at (+/-26,-36) and (+/-26,-8). Six lid pilots2.8 at radius99 angles30+60n deg.
General fit: inspect and deburr holes; check the fit coupon and actual hardware. No unlisted machining tolerances are implied.

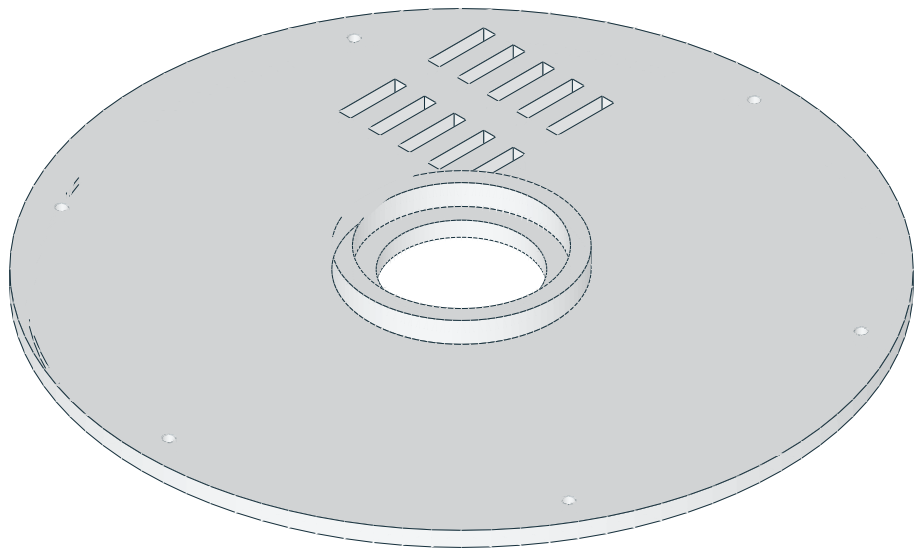
Vented base lid and yaw bearing seat

base_lid.stl | PRINT QUANTITY 1 | PETG

TOP / XY



ISOMETRIC / REFERENCE



FRONT / XZ



RIGHT / YZ

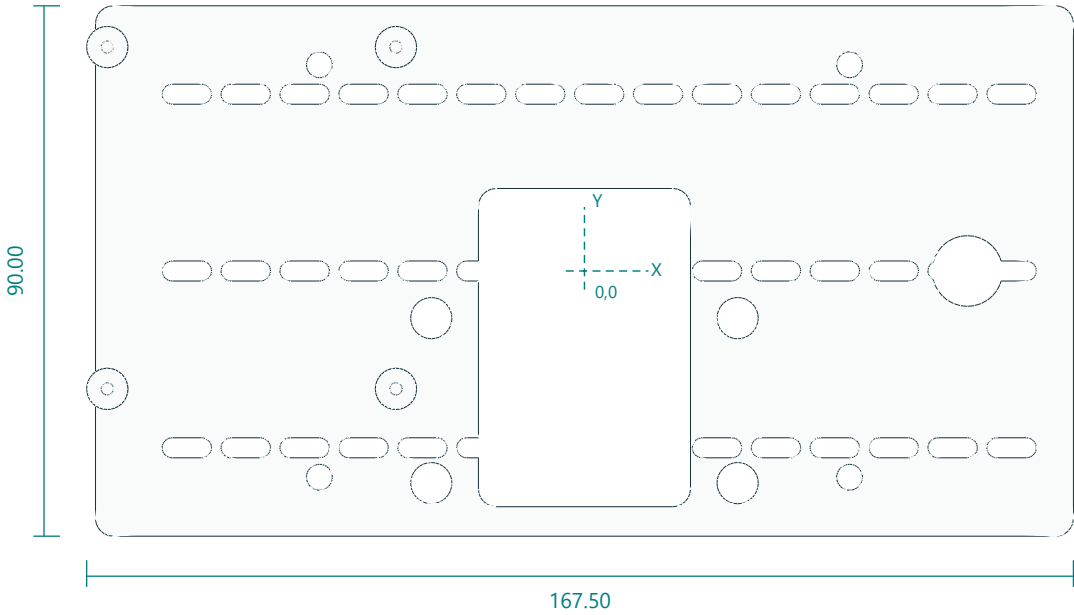


Interfaces: 216 diameter; bearing pocket 52.15 x 7; 40.8 cable/hub bore.
Fasteners / retention: 6 M3x12.
Print orientation: flat annulus down. Layers 0.2 mm; 4 walls; 25% infill baseline.
Feature locations: Six 3.4 holes radius 99 angles 30+60n deg. Bearing 52.15 bore by 7 depth. STL bottom datum is 2mm below principal lid underside.
General fit: inspect and deburr holes; check the fit coupon and actual hardware. No unlisted machining tolerances are implied.

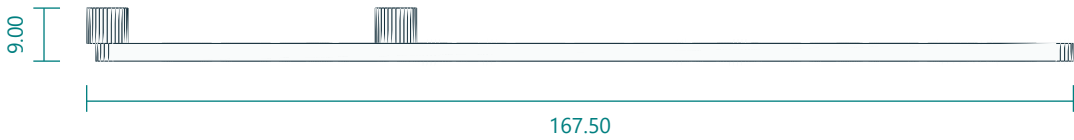
Universal electronics tray with Pi4 standoffs

electronics_tray.stl | PRINT QUANTITY 1 | PETG

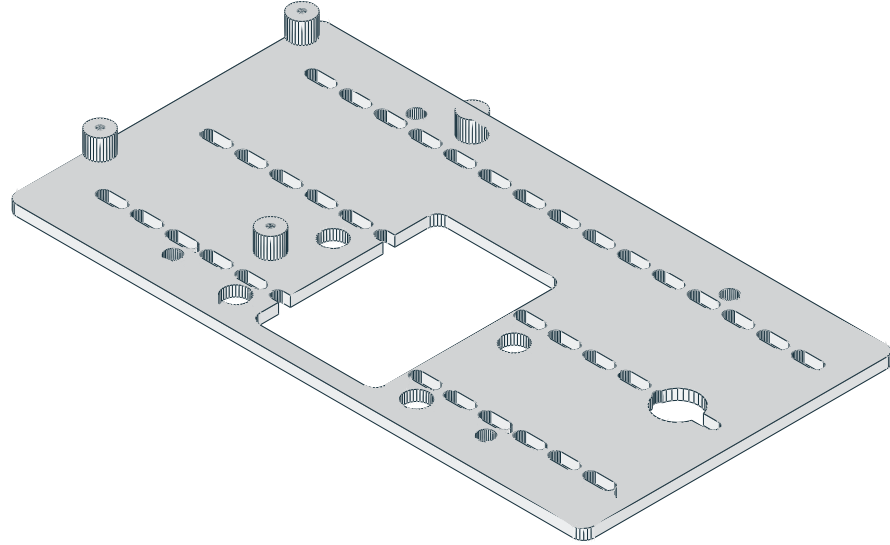
TOP / XY



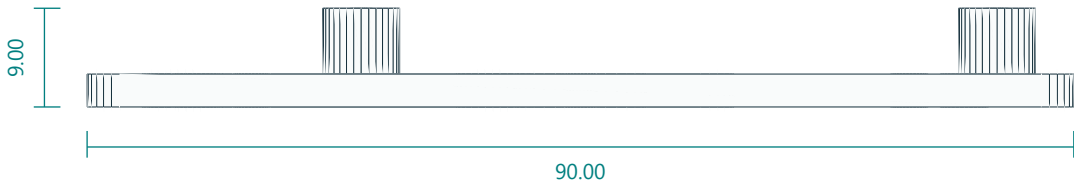
FRONT / XZ



ISOMETRIC / REFERENCE



RIGHT / YZ

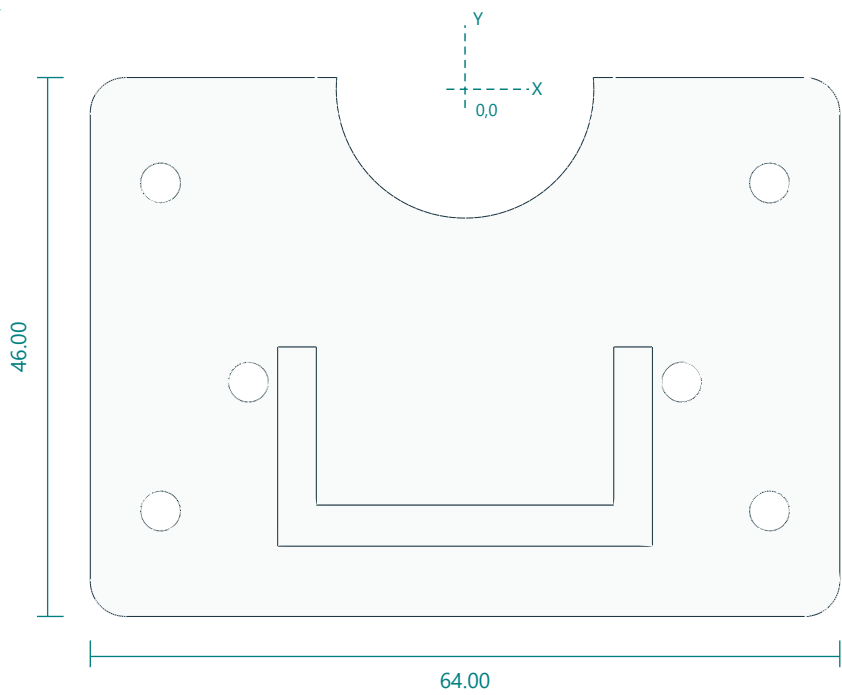


Interfaces: 166 x90; Pi4 hole pitch49 x58; 2.1mm M2.5 tapping pilots.
Fasteners / retention: 4 M2.5x6 Pi; 4 M4x25 with metal spacers.
Print orientation: plate down. Layers 0.2 mm; 4 walls; 25% infill baseline.
Feature locations: M4 holes (+/-45,+/-35). Pi M2.5 pilots2.1 at X=-81,-32 and Y=-20,38. Central rounded36x54 clearance centered(0,-13). Four7mm tool apertures at yaw bolts.
General fit: inspect and deburr holes; check the fit coupon and actual hardware. No unlisted machining tolerances are implied.

Yaw servo rear body cradle

yaw_mount.stl | PRINT QUANTITY 1 | PETG

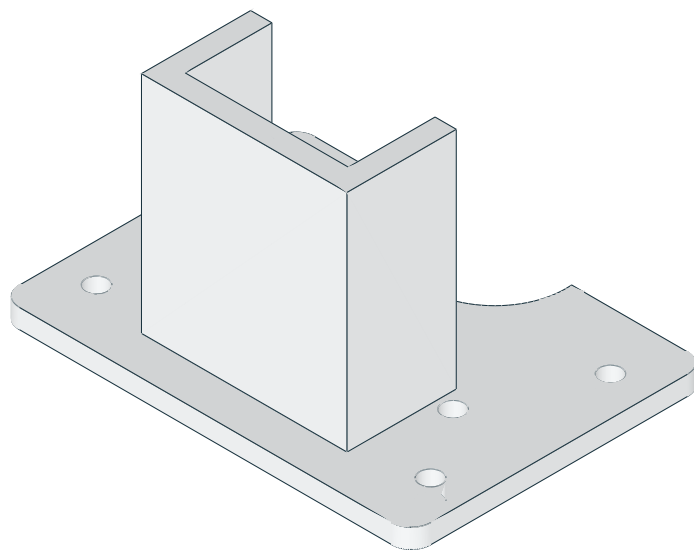
TOP / XY



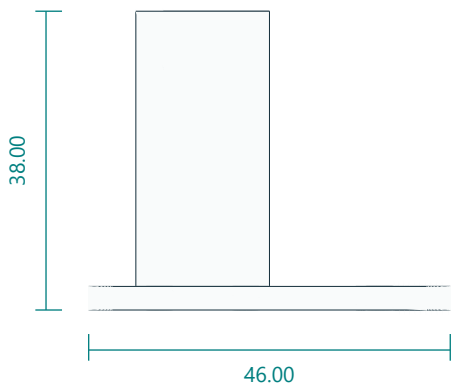
FRONT / XZ



ISOMETRIC / REFERENCE



RIGHT / YZ

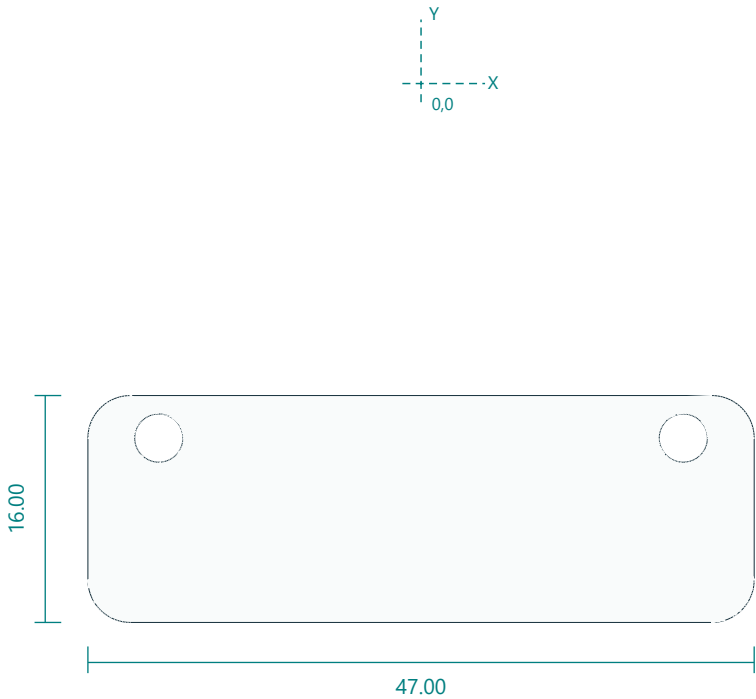


Interfaces: 64 x46; pocket25.4; rear body clamp.
Fasteners / retention: 4 M3x12 mount; 2 M3x45 clamp.
Print orientation: base down. Layers 0.2 mm; 4 walls; 25% infill baseline.
Feature locations: Four3.4 floor holes (+/-26,-36/-8). Keeper bolts at (+/-18.5,-25). Servo shaft atXY(0,0).
General fit: inspect and deburr holes; check the fit coupon and actual hardware. No unlisted machining tolerances are implied.

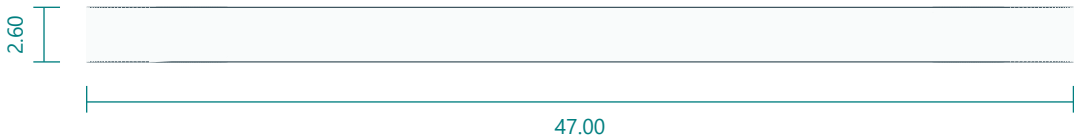
Yaw clamp keeper

yaw_cap.stl | PRINT QUANTITY 1 | PETG

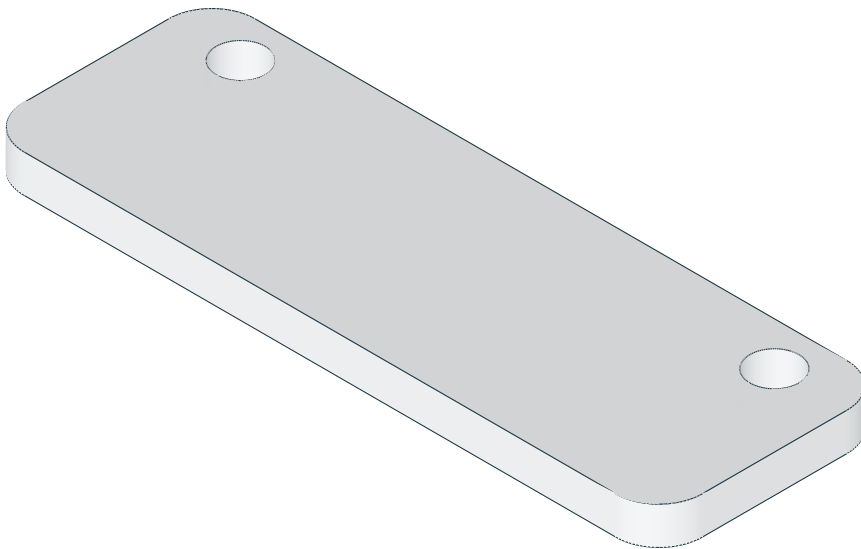
TOP / XY



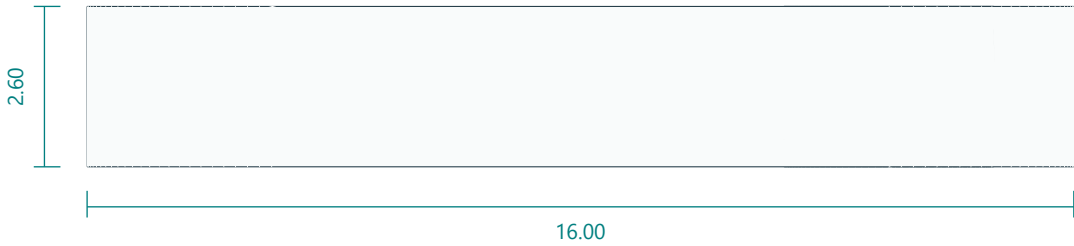
FRONT / XZ



ISOMETRIC / REFERENCE



RIGHT / YZ

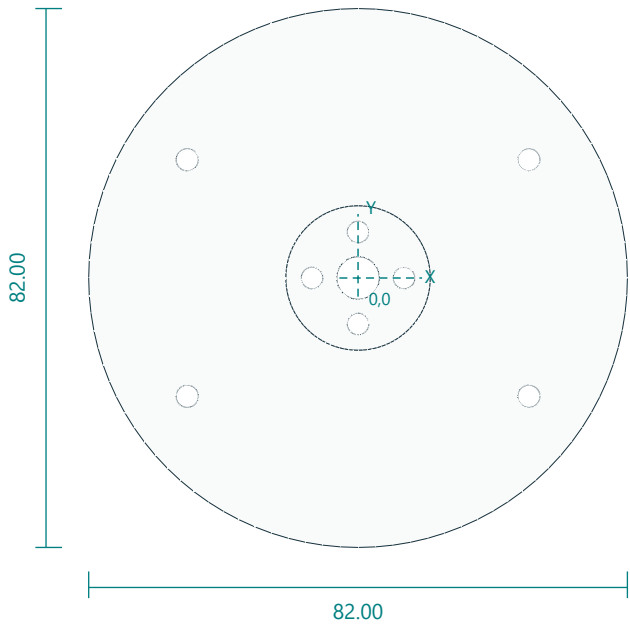


Interfaces: 47 x16 x2.6; clamp hole pitch37.
Fasteners / retention: 2 M3x45.
Print orientation: flat down. Layers 0.2 mm; 4 walls; 25% infill baseline.
Feature locations: Two3.4 holes (+/-18.5,-25). Keeper center(0,-30); thickness2.6.
General fit: inspect and deburr holes; check the fit coupon and actual hardware. No unlisted machining tolerances are implied.

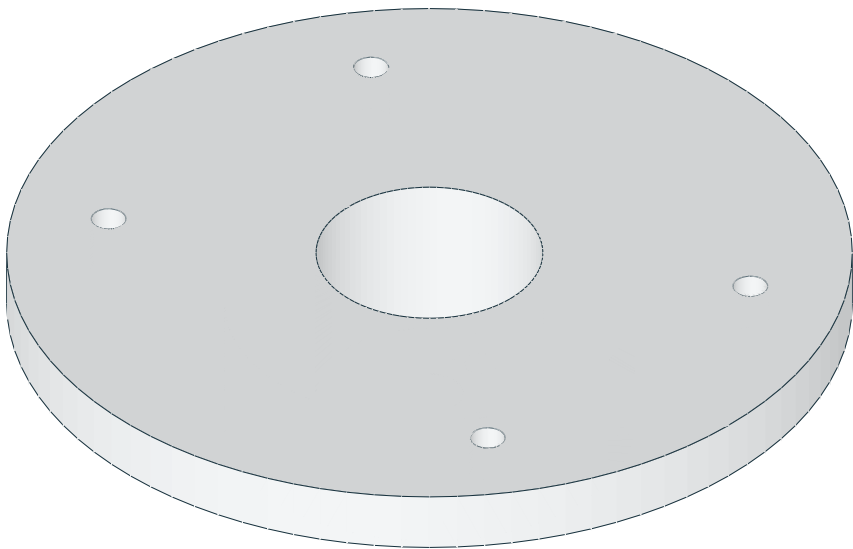
Bearing supported yaw turntable

turntable.stl | PRINT QUANTITY 1 | PETG

TOP / XY



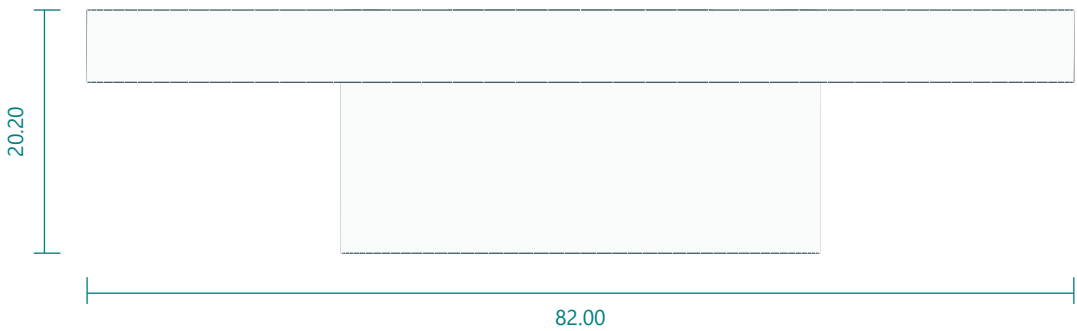
ISOMETRIC / REFERENCE



FRONT / XZ



RIGHT / YZ

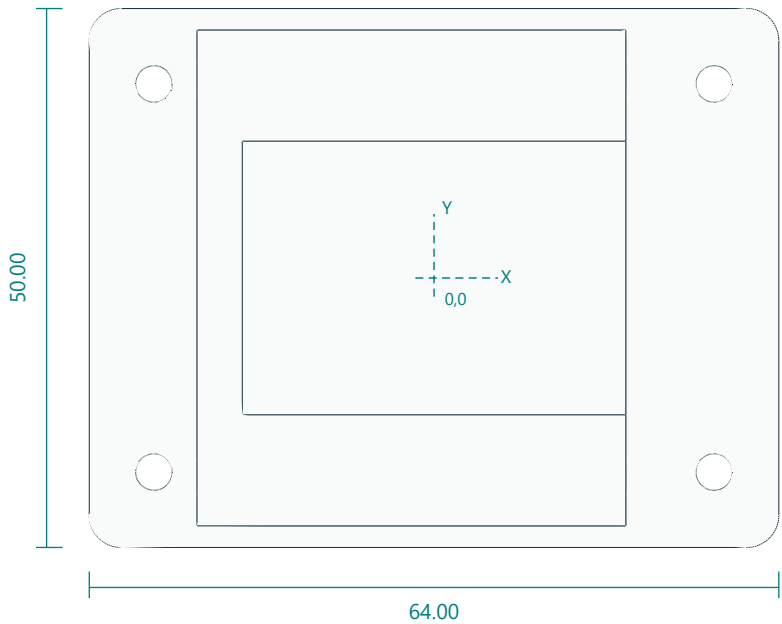


Interfaces: 82 diameter; neck39.75 x14.2; horn PCD14; tower52 x36.
Fasteners / retention: 4 nominal M3x6 horn screws; 4 M3x16 tower.
Print orientation: neck down. Layers 0.2 mm; 4 walls; 25% infill baseline.
Feature locations: Horn4x3.3 at (0,+/-7),(+/-7,0); center6.5 in3mm web. Tower4x3.4 at(+/-26,+/-18). Top central tool aperture22.
General fit: inspect and deburr holes; check the fit coupon and actual hardware. No unlisted machining tolerances are implied.

Shoulder servo body tower

shoulder_tower.stl | PRINT QUANTITY 1 | PETG

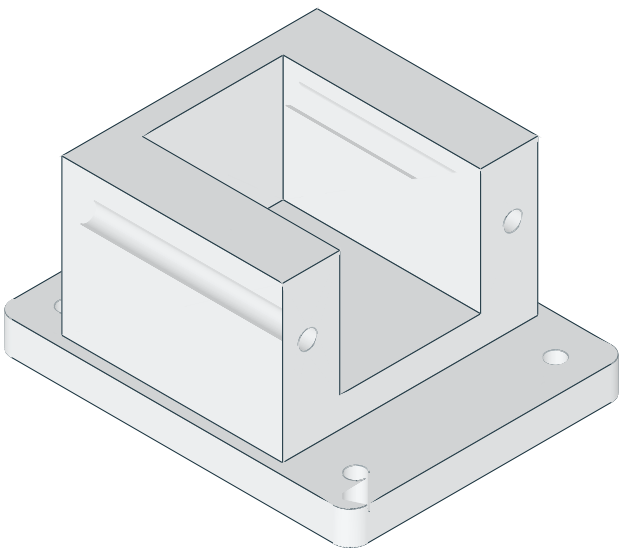
TOP / XY



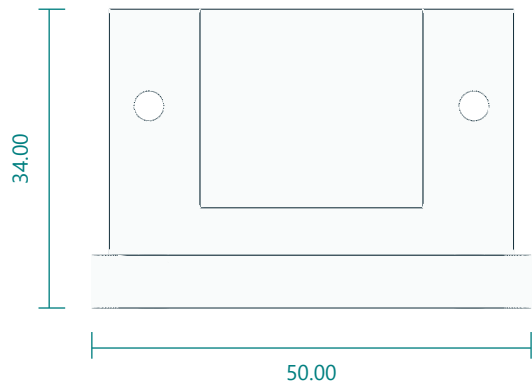
FRONT / XZ



ISOMETRIC / REFERENCE



RIGHT / YZ

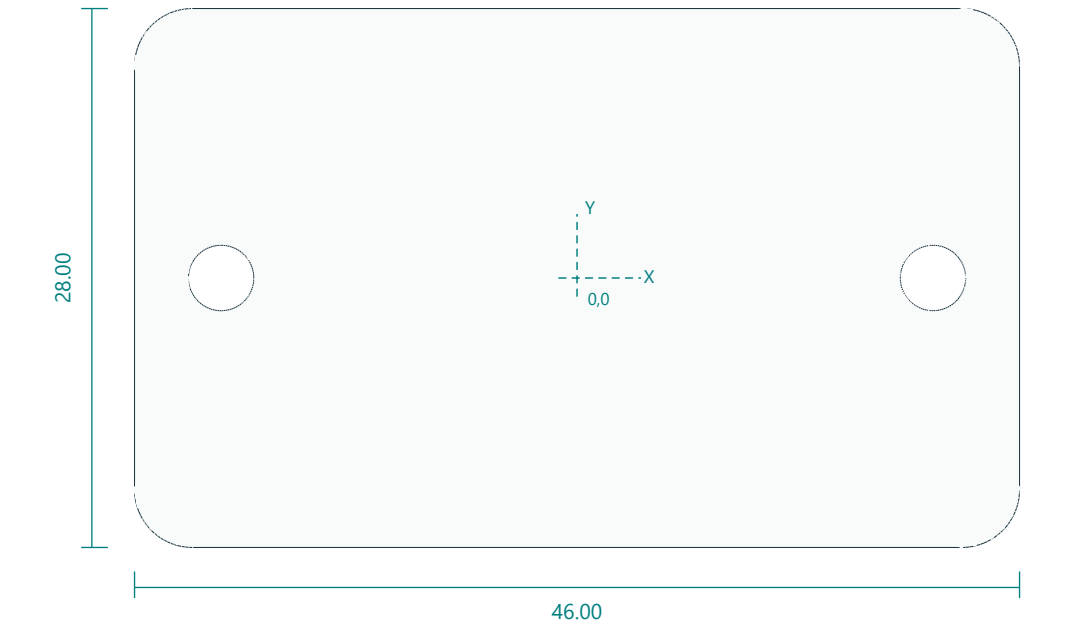


Interfaces: 64 x50; pitch axis47 above foot; body35.6 x25.4.
Fasteners / retention: 4 M3x16; 2 M3x50 clamp.
Print orientation: foot down. Layers 0.2 mm; 4 walls; 25% infill baseline.
Feature locations: Foot4x3.4 at(+/-26,+/-18). Clamp holes alongX atY+/-18.5,Z23. Pitch axis47 abovefoot.
General fit: inspect and deburr holes; check the fit coupon and actual hardware. No unlisted machining tolerances are implied.

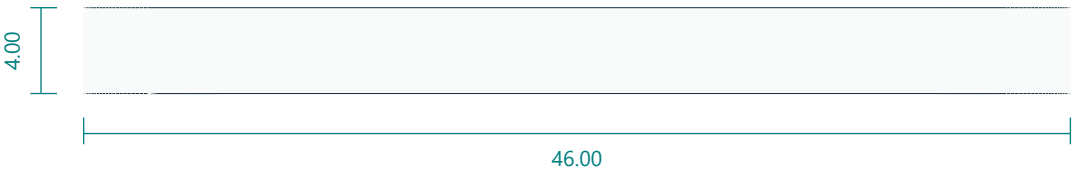
Shoulder clamp keeper

shoulder_cap.stl | PRINT QUANTITY 1 | PETG

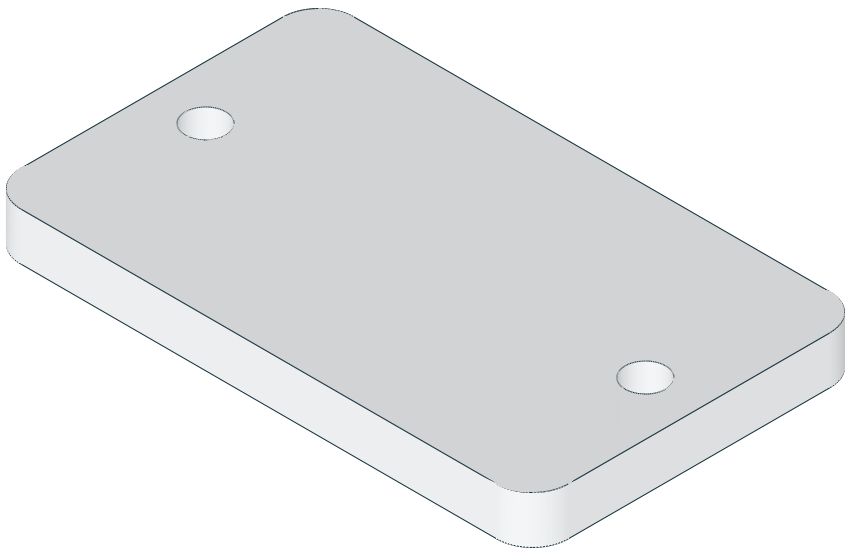
TOP / XY



FRONT / XZ



ISOMETRIC / REFERENCE



RIGHT / YZ

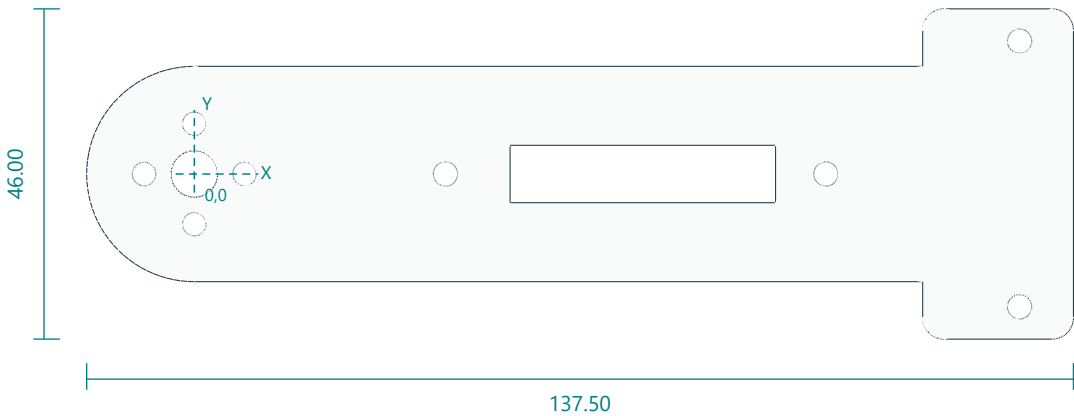


Interfaces: 46 x28 x4; clamp pitch37.
Fasteners / retention: 2 M3x50.
Print orientation: flat down. Layers 0.2 mm; 4 walls; 25% infill baseline.
Feature locations: Two3.4 holes (+/-18.5,0). Printedface46x28.
General fit: inspect and deburr holes; check the fit coupon and actual hardware. No unlisted machining tolerances are implied.

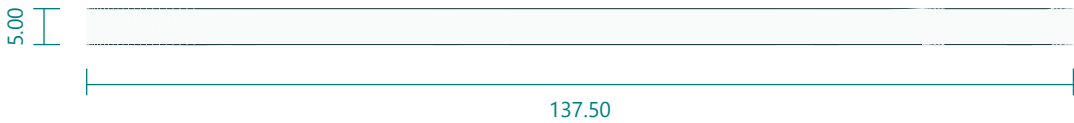
140 mm upper arm side rail

upper_rail.stl | PRINT QUANTITY 2 | PETG

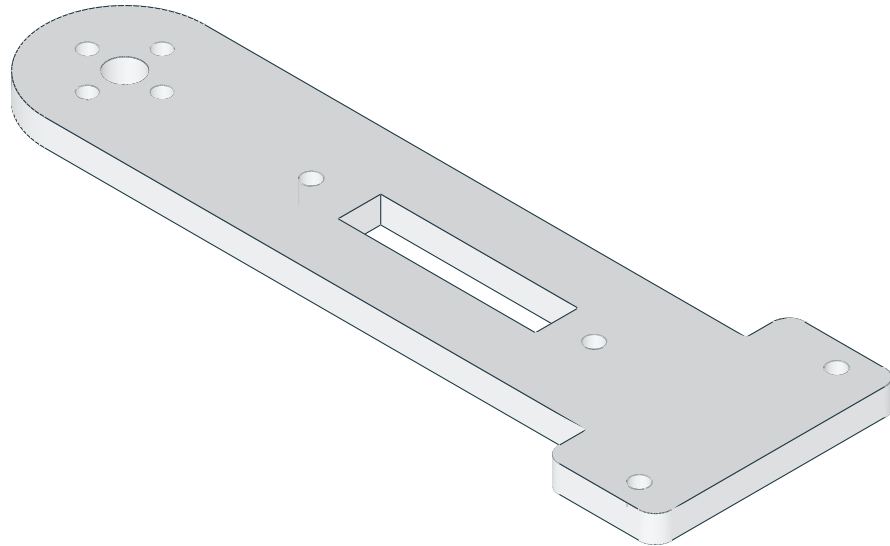
TOP / XY



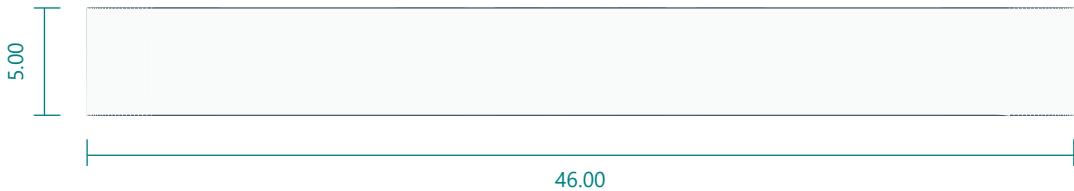
FRONT / XZ



ISOMETRIC / REFERENCE



RIGHT / YZ



Interfaces: 140 pivot pitch; 5 thick; 14 horn PCD; 3.3 horn holes.
Fasteners / retention: 8 nominal M3x10 horn; 4 M3x60 cross bolts.
Print orientation: broad face down. Layers 0.2 mm; 6 walls; 40% infill baseline.
Feature locations: Proximal horn4x3.3 at(0,+/-7),(+/-7,0). Cross bolts3.4 at(35,0),(88,0). Cassette bolts at(115,+/-18.5). Distal shaft140 from proximal.
General fit: inspect and deburr holes; check the fit coupon and actual hardware. No unlisted machining tolerances are implied.

120 mm forearm side rail

forearm_rail.stl | PRINT QUANTITY 2 | PETG

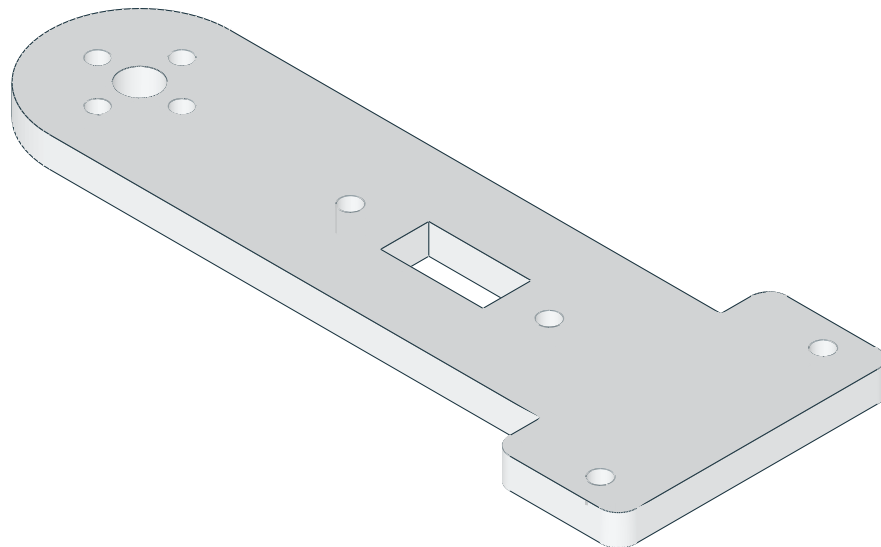
TOP / XY



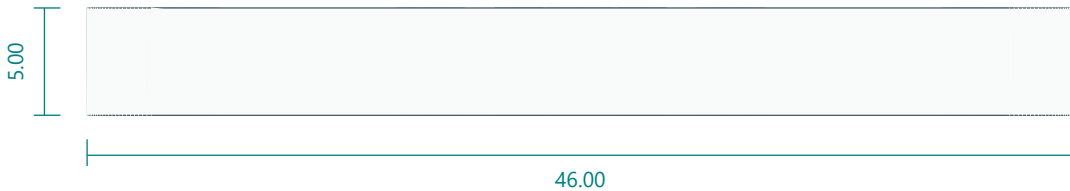
FRONT / XZ



ISOMETRIC / REFERENCE



RIGHT / YZ

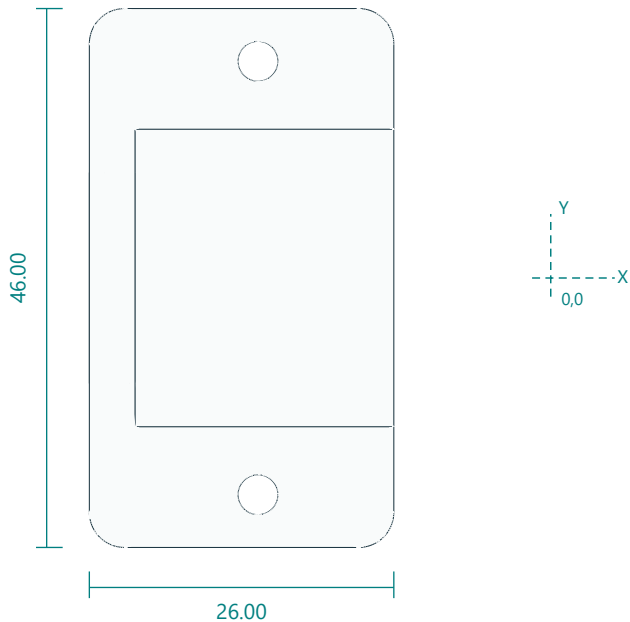


Interfaces: 120 pivot pitch; 5 thick; 14 horn PCD; 3.3 horn holes.
Fasteners / retention: 8 nominal M3x10 horn; 4 M3x60 cross bolts.
Print orientation: broad face down. Layers 0.2 mm; 6 walls; 40% infill baseline.
Feature locations: Proximal horn4x3.3 at(0,+/-7),(+/-7,0). Cross bolts3.4 at(35,0),(68,0). Cassette bolts at(95,+/-18.5). Distal shaft120 from proximal.
General fit: inspect and deburr holes; check the fit coupon and actual hardware. No unlisted machining tolerances are implied.

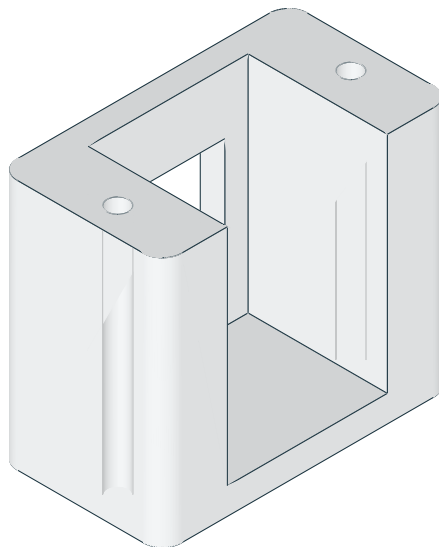
Elbow/wrist rear servo body cassette

servo_cassette.stl | PRINT QUANTITY 2 | PETG

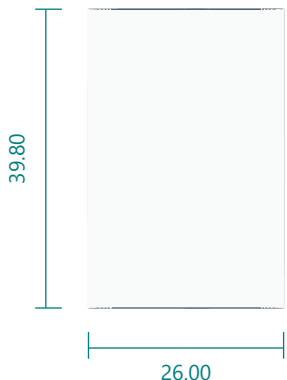
TOP / XY



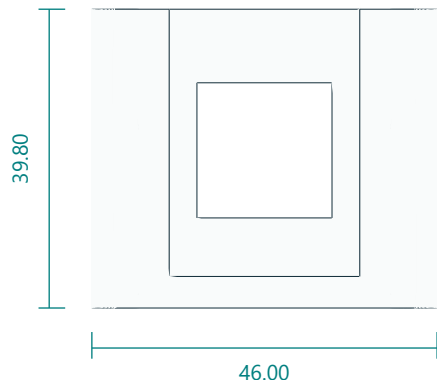
ISOMETRIC / REFERENCE



FRONT / XZ



RIGHT / YZ

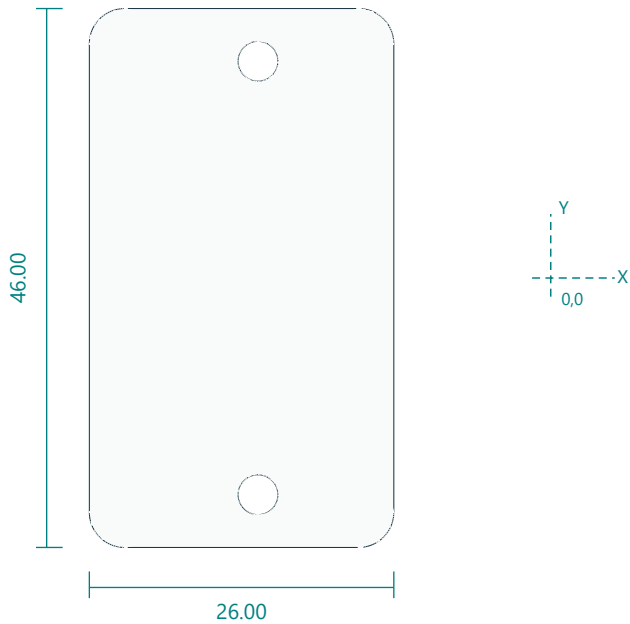


Interfaces: 26 x46 x39.8; 25.4 wide; 35.6 clamped axis width.
Fasteners / retention: 2 M3x60 each shared with rails.
Print orientation: closed side down. Layers 0.2 mm; 4 walls; 25% infill baseline.
Feature locations: Bolts3.4 at(-25,+/-18.5). Body pocketY+/-12.7; floor4.2; assembledcaseaxiswidth35.6.
General fit: inspect and deburr holes; check the fit coupon and actual hardware. No unlisted machining tolerances are implied.

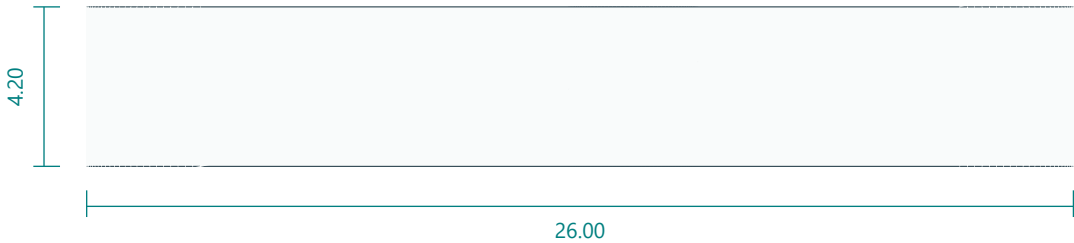
Servo body cassette keeper

cassette_cap.stl | PRINT QUANTITY 2 | PETG

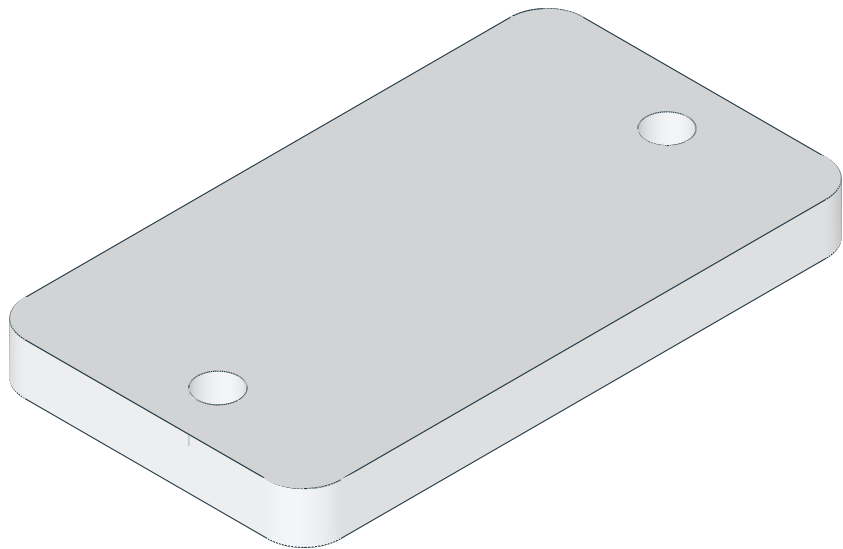
TOP / XY



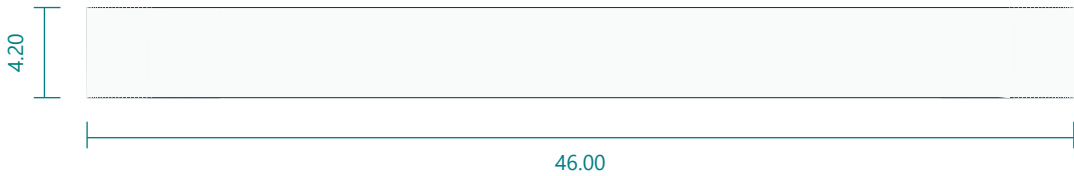
FRONT / XZ



ISOMETRIC / REFERENCE



RIGHT / YZ

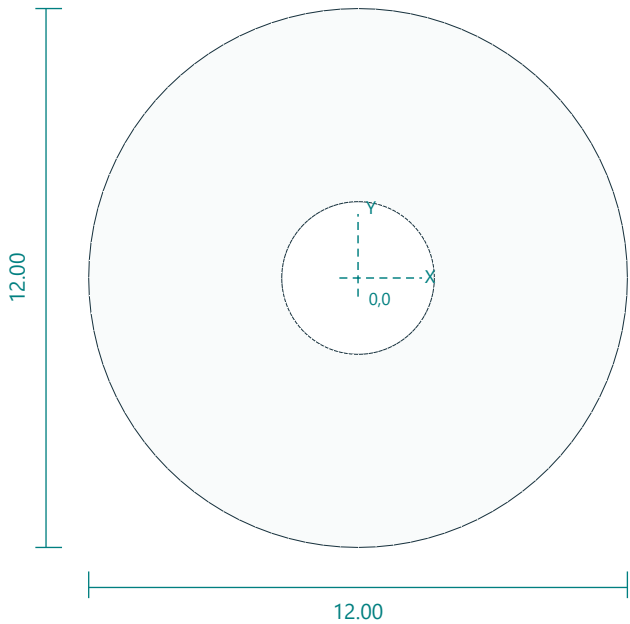


Interfaces: 26 x46 x4.2.
Fasteners / retention: shared M3x60.
Print orientation: flat down. Layers 0.2 mm; 4 walls; 25% infill baseline.
Feature locations: Bolts3.4 at(-25,+/-18.5); axialthickness4.2.
General fit: inspect and deburr holes; check the fit coupon and actual hardware. No unlisted machining tolerances are implied.

Arm through bolt spacer

cross_spacer.stl | PRINT QUANTITY 4 | PETG

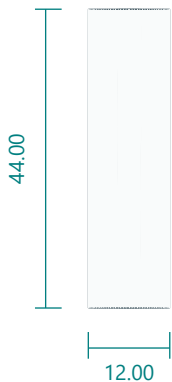
TOP / XY



ISOMETRIC / REFERENCE



FRONT / XZ



RIGHT / YZ

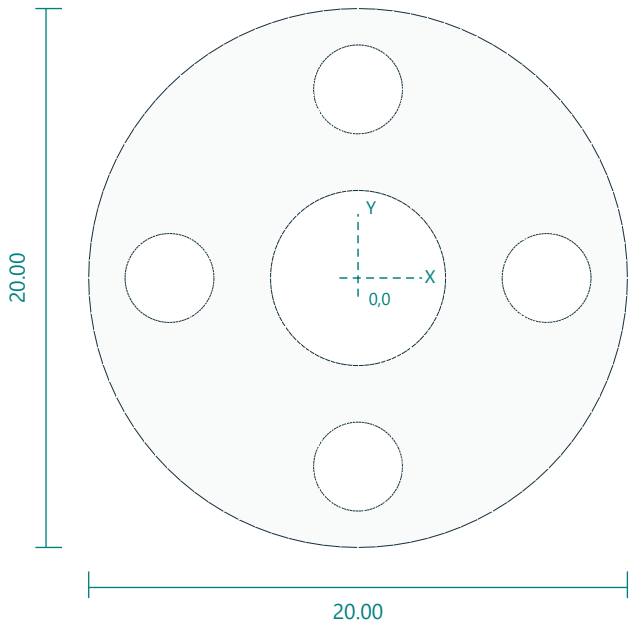


Interfaces: 12 diameter x44; bore3.4.
Fasteners / retention: 1 M3x60 each.
Print orientation: cylinder upright. Layers 0.2 mm; 4 walls; 25% infill baseline.
Feature locations: Axialbore3.4; length44; OD12.
General fit: inspect and deburr holes; check the fit coupon and actual hardware. No unlisted machining tolerances are implied.

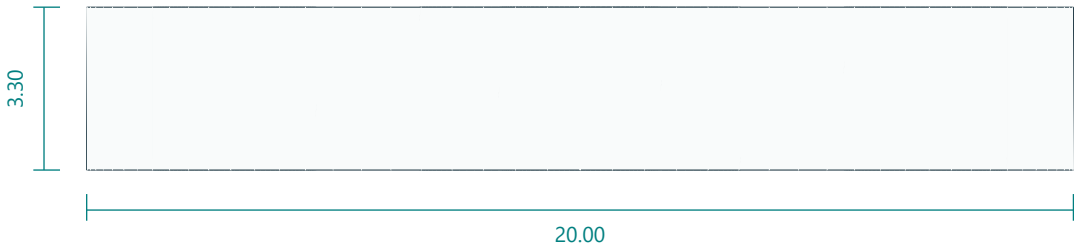
Metal horn to rail stand off

horn_spacer.stl | PRINT QUANTITY 6 | PETG

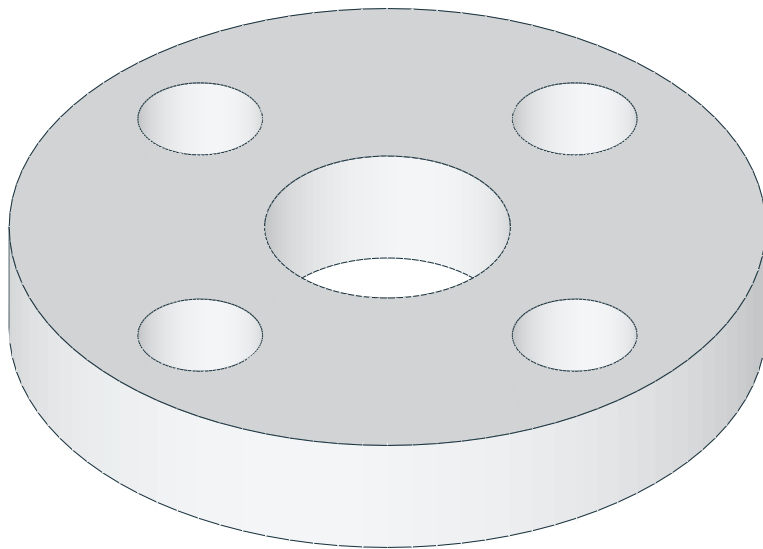
TOP / XY



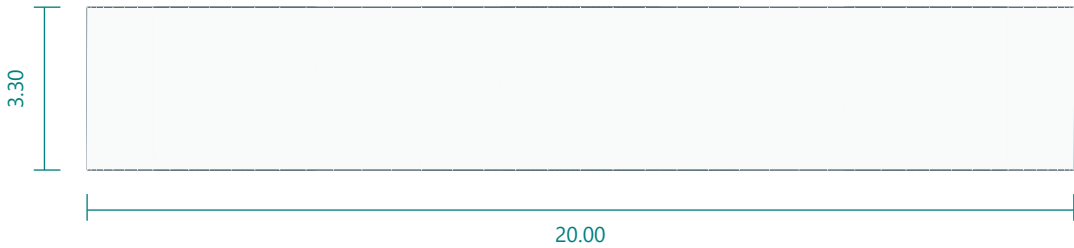
FRONT / XZ



ISOMETRIC / REFERENCE



RIGHT / YZ

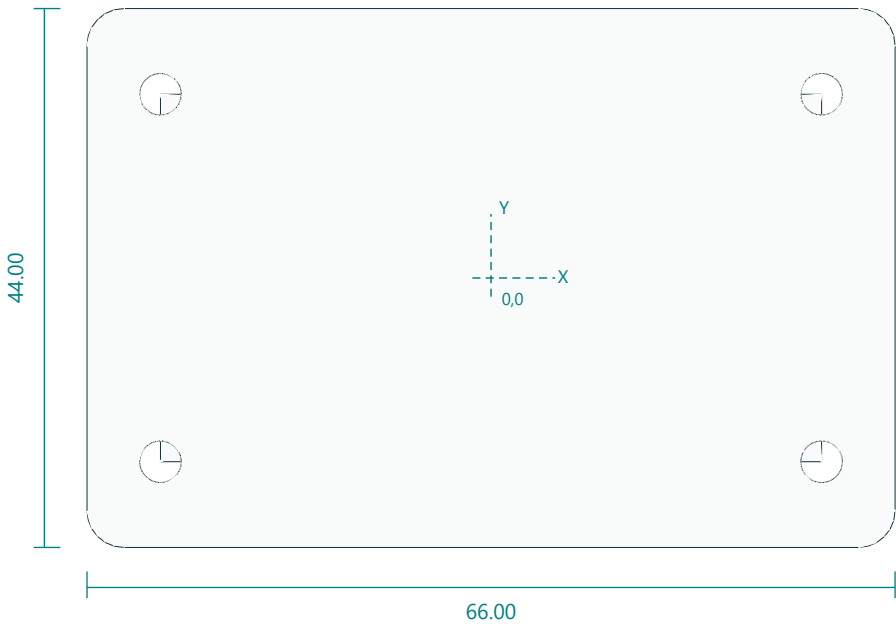


Interfaces: 20 diameter x3.3; PCD14; center6.5.
Fasteners / retention: shared nominal M3x10 horn screws.
Print orientation: flat down. Layers 0.2 mm; 4 walls; 25% infill baseline.
Feature locations: PCD14 four3.3 holes; center6.5; thickness3.3.
General fit: inspect and deburr holes; check the fit coupon and actual hardware. No unlisted machining tolerances are implied.

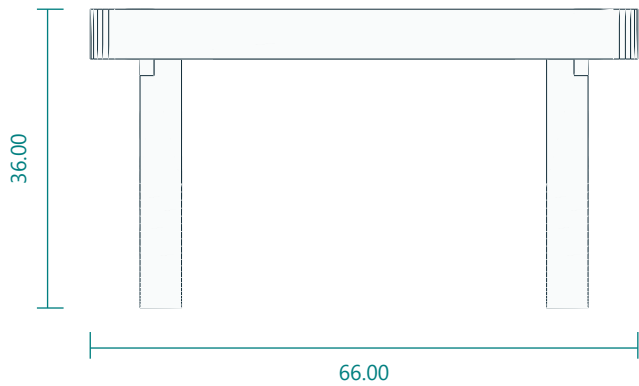
Head tilt fork and rear mount

head_yoke.stl | PRINT QUANTITY 1 | PETG

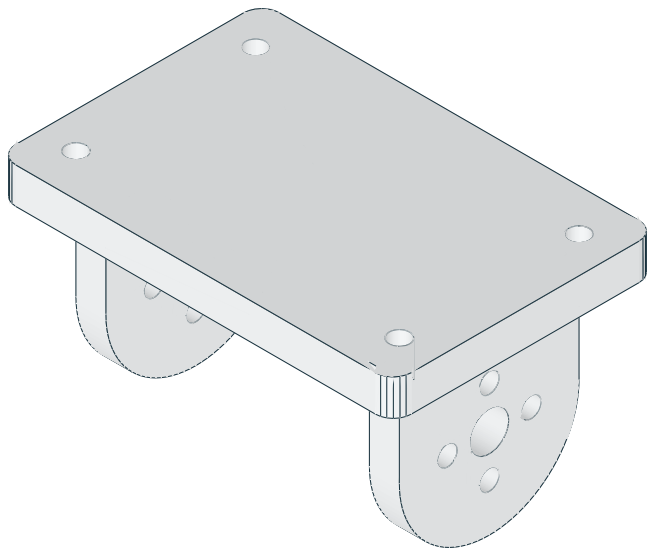
TOP / XY



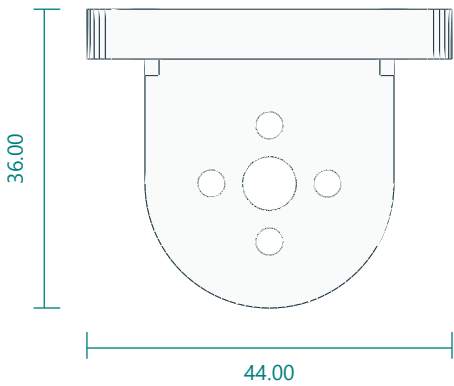
FRONT / XZ



ISOMETRIC / REFERENCE



RIGHT / YZ

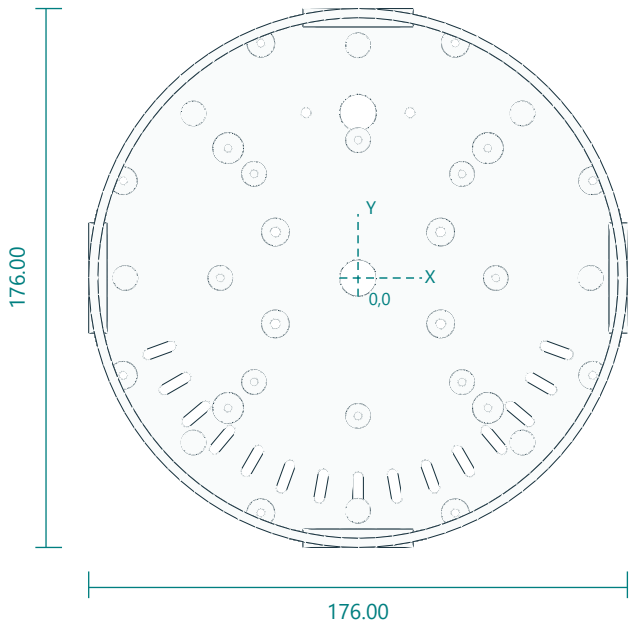


Interfaces: 54 fork span; axis21 behind head rear; rear holes54 x30.
Fasteners / retention: 8 nominal M3 horn screws; 4 M3x12 rear screws.
Print orientation: rear mounting plate on bed; rotate180X. Layers 0.2 mm; 4 walls; 25% infill baseline.
Feature locations: Back bolts3.4 XY(+/-27, +/-15); attachmentplaneZ36. HornaxesX atY0,Z15, PCD14. Forkinsidewidth44.
General fit: inspect and deburr holes; check the fit coupon and actual hardware. No unlisted machining tolerances are implied.

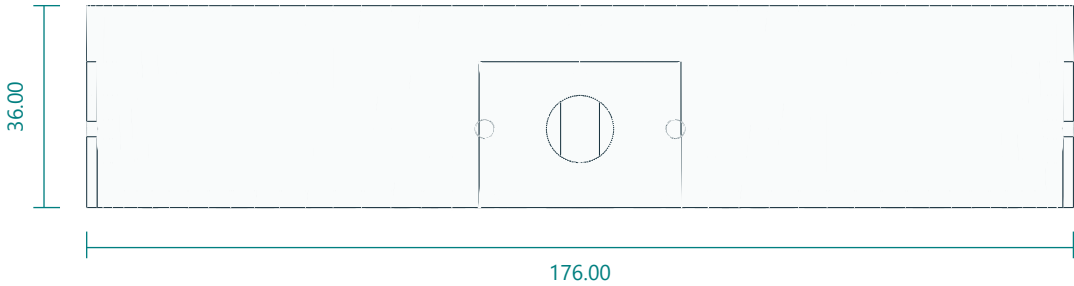
Optical head rear enclosure

head_shell.stl | PRINT QUANTITY 1 | PETG

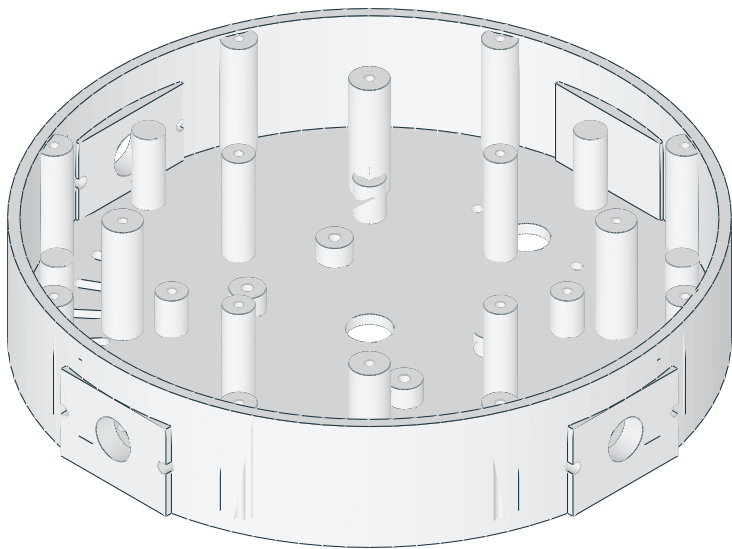
TOP / XY



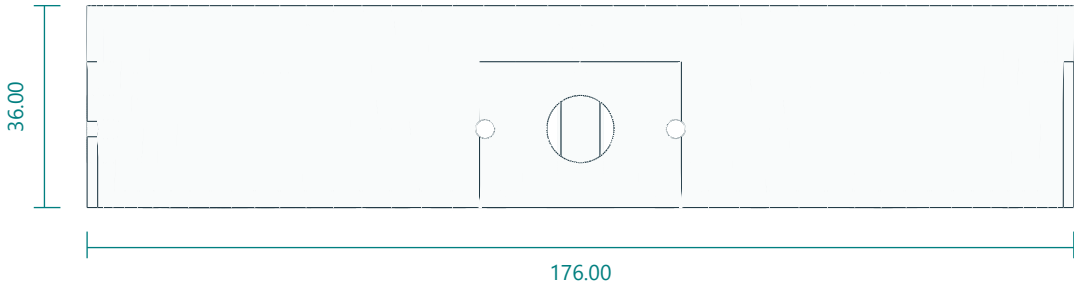
FRONT / XZ



ISOMETRIC / REFERENCE



RIGHT / YZ

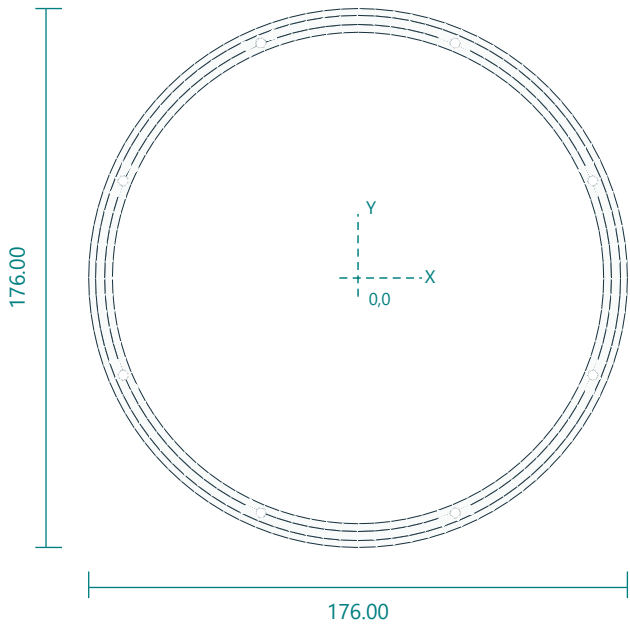


Interfaces: 176 diameter x36; rear thickness3; white carrier bosses radius45 top32.5.
Fasteners / retention: 8 M3x10 bezel; 4 M3x12 yoke.
Print orientation: rear disc down. Layers 0.2 mm; 4 walls; 25% infill baseline.
Feature locations: Rear bolts(+/-27,+/-15); LCDpilots(+/-34,+/-34),top14. Whitecarrierpilotsradius45 cardinal,top32.5. Facebossradius60 angles45+90n,top35. Bezelradius83 angles22.5+45n,top35.
General fit: inspect and deburr holes; check the fit coupon and actual hardware. No unlisted machining tolerances are implied.

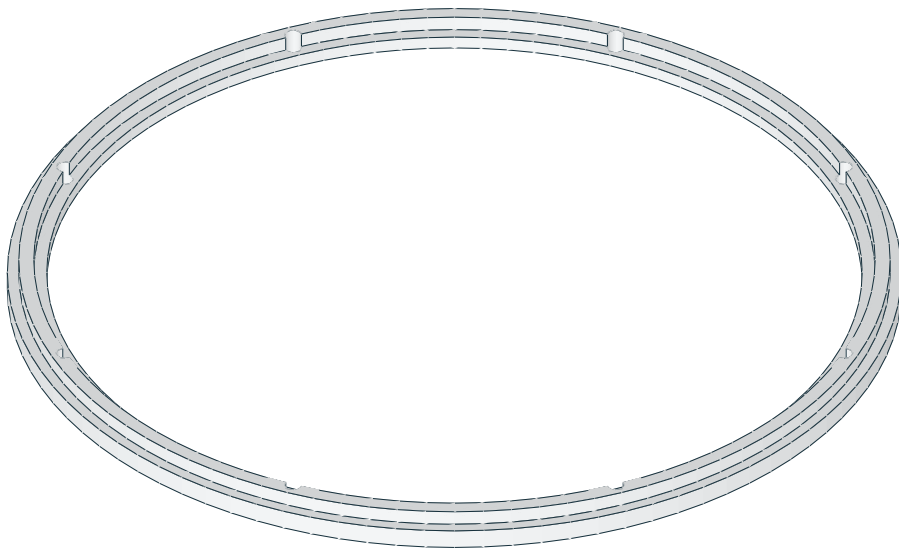
Outer RGB diffuser retaining bezel

outer_bezel.stl | PRINT QUANTITY 1 | PETG

TOP / XY



ISOMETRIC / REFERENCE



FRONT / XZ



RIGHT / YZ

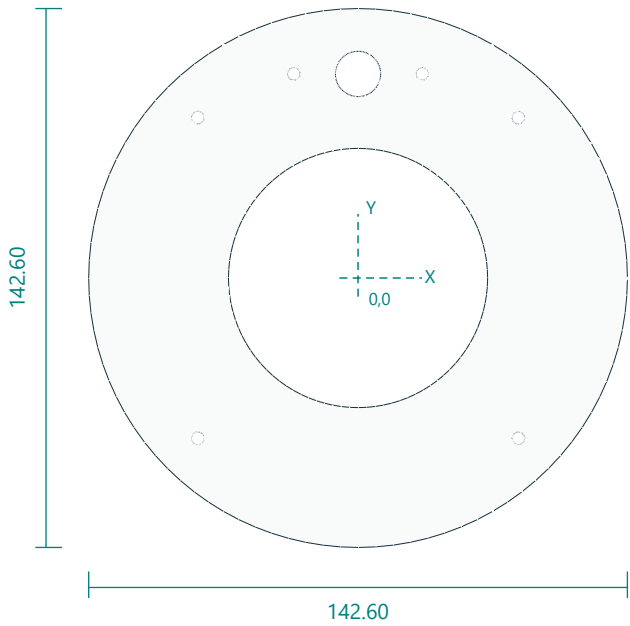


Interfaces: 176 diameter; window160.5; 8 holes radius83.
Fasteners / retention: 8 M3x10.
Print orientation: front lip down. Layers 0.2 mm; 4 walls; 25% infill baseline.
Feature locations: Eight3.4 holes radius83 angles22.5+45n. Rearflangerecessdiameter165.5 by1.4 deep; facewindow160.5.
General fit: inspect and deburr holes; check the fit coupon and actual hardware. No unlisted machining tolerances are implied.

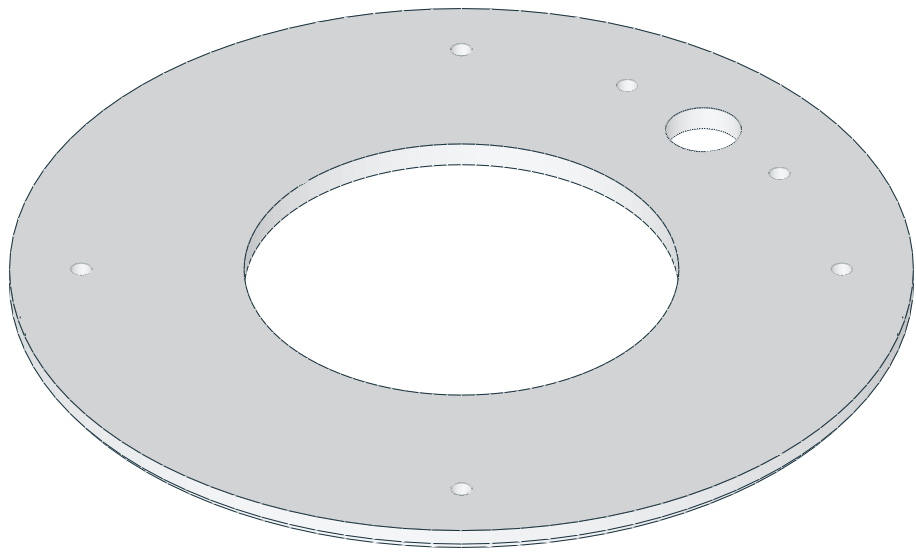
Central optical baffle plate

face_center.stl | PRINT QUANTITY 1 | black PETG

TOP / XY



ISOMETRIC / REFERENCE



FRONT / XZ



RIGHT / YZ

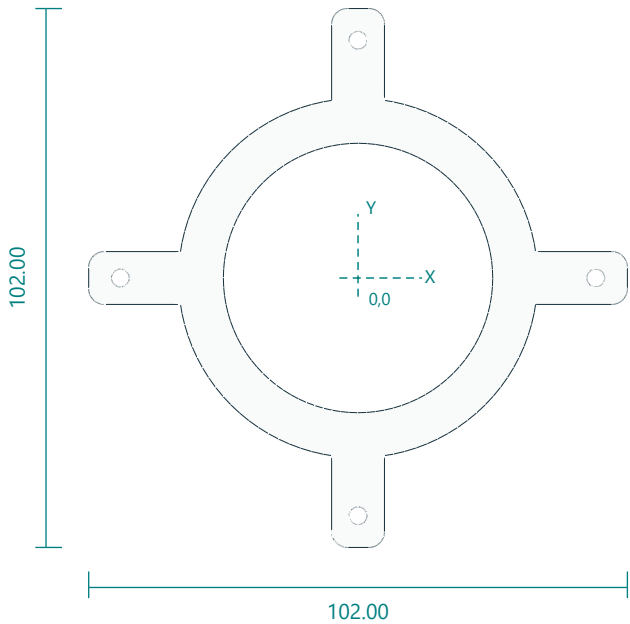


Interfaces: 142.6 diameter x4; center68.6; screws radius60.
Fasteners / retention: 4 M3x12 into shell bosses.
Print orientation: flat front down. Layers 0.2 mm; 4 walls; 25% infill baseline.
Feature locations: Four3.4 holesradius60 angles45+90n. Frontpodholes(+/-17,54); cablehole(0,54)diameter12.
General fit: inspect and deburr holes; check the fit coupon and actual hardware. No unlisted machining tolerances are implied.

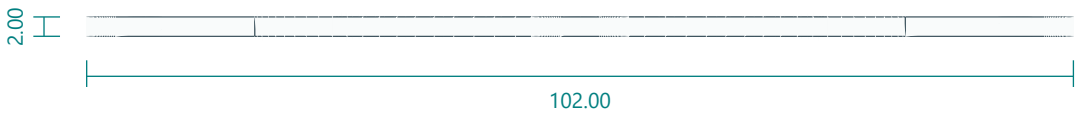
White ring carrier and mounting arms

white_carrier.stl | PRINT QUANTITY 1 | PETG

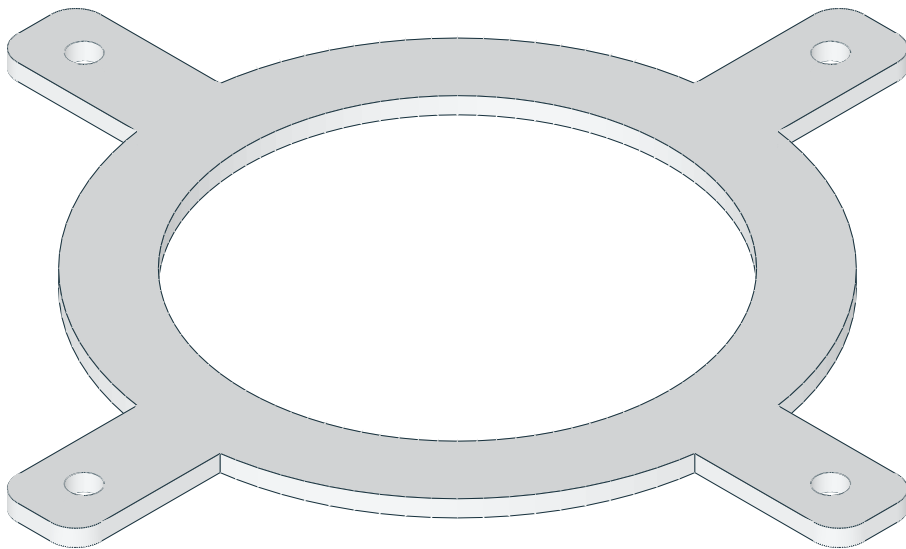
TOP / XY



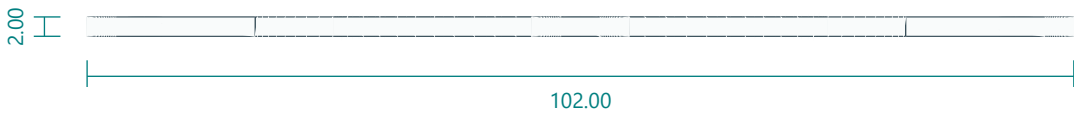
FRONT / XZ



ISOMETRIC / REFERENCE



RIGHT / YZ

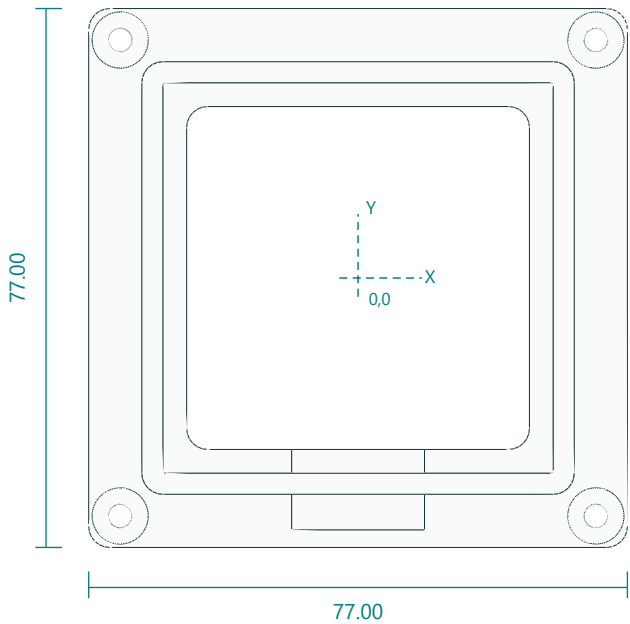


Interfaces: 68 OD51 ID; 4 cardinal mounting holes radius45; 2 thick.
Fasteners / retention: 4 M3x10.
Print orientation: flat down. Layers 0.2 mm; 4 walls; 25% infill baseline.
Feature locations: Four3.4 holes at(+/-45,0),(0,+/-45). Allarmsandring2mmthick.
General fit: inspect and deburr holes; check the fit coupon and actual hardware. No unlisted machining tolerances are implied.

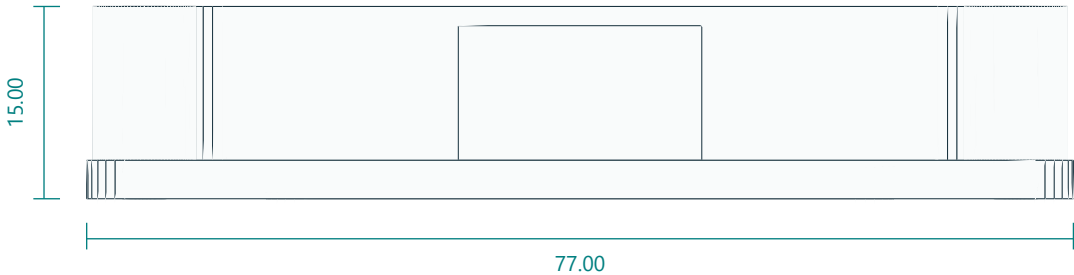
55 mm LCD edge cradle

lcd_cradle.stl | PRINT QUANTITY 1 | PETG

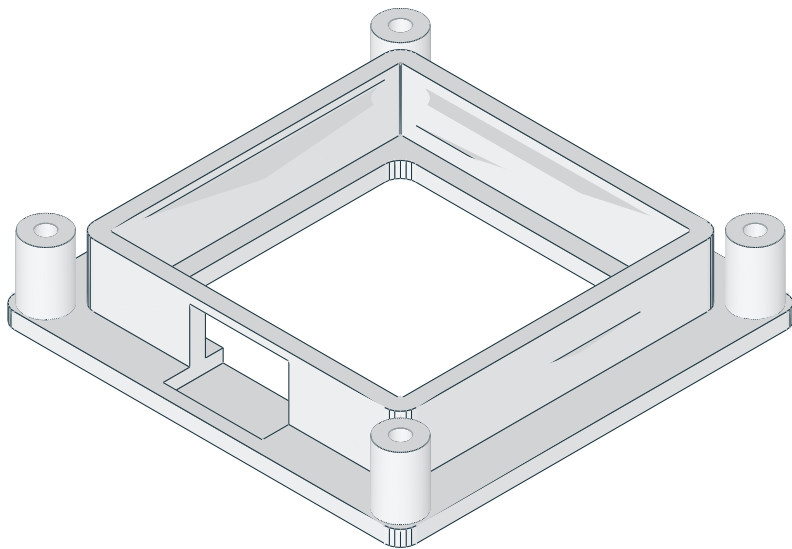
TOP / XY



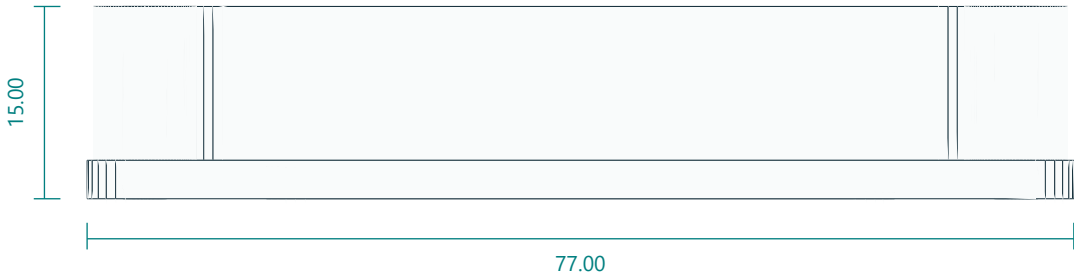
FRONT / XZ



ISOMETRIC / REFERENCE



RIGHT / YZ

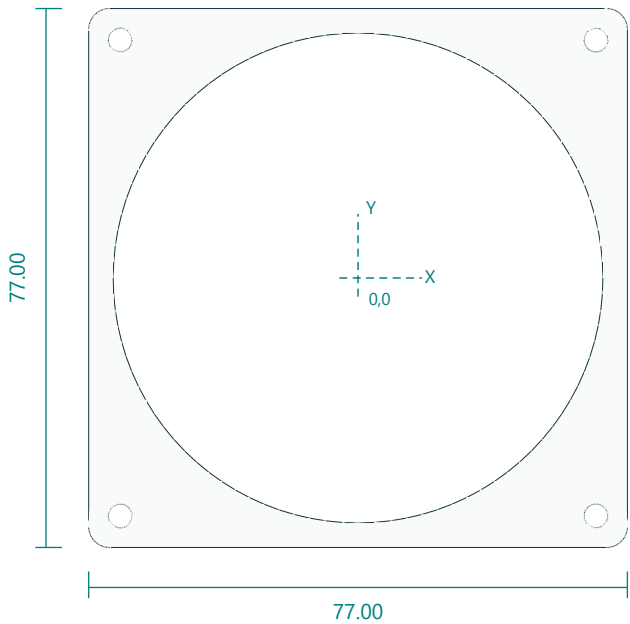


Interfaces: 55.8 square pocket; 15 depth; USB19 x13 relief.
Fasteners / retention: 4 M3x25; thin closed cell foam shims.
Print orientation: flat base down. Layers 0.2 mm; 4 walls; 25% infill baseline.
Feature locations: Four3.4 holes(+/-34,+/-34). Inner55.8square cavity. Rear49squareaccess. USBopening19wide at-Y.
General fit: inspect and deburr holes; check the fit coupon and actual hardware. No unlisted machining tolerances are implied.

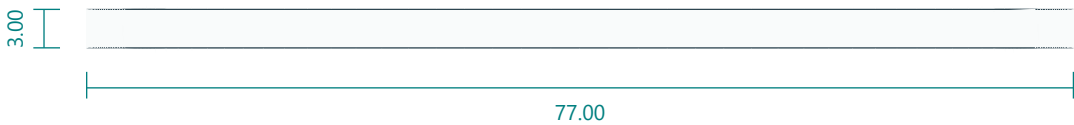
LCD PCB corner retaining plate

lcd_retainer.stl | PRINT QUANTITY 1 | black PETG

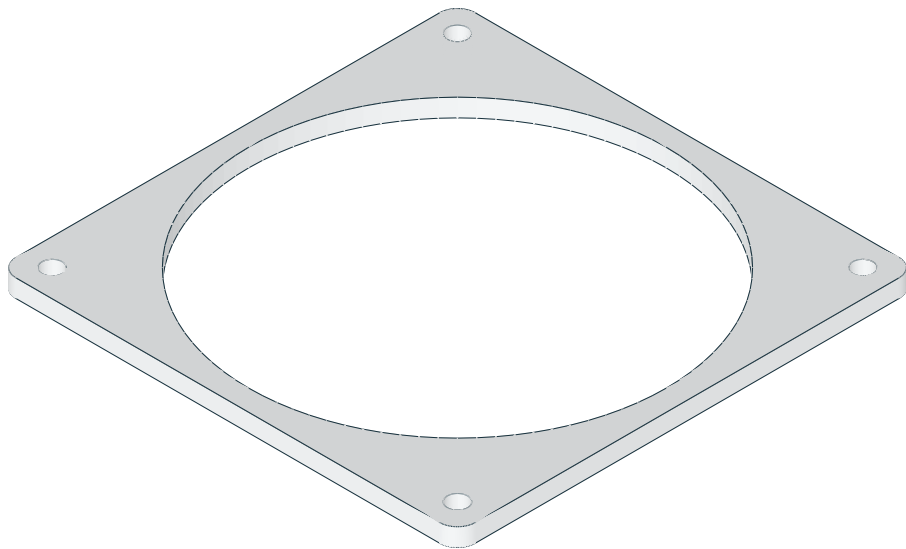
TOP / XY



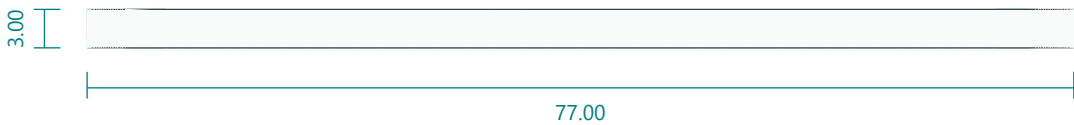
FRONT / XZ



ISOMETRIC / REFERENCE



RIGHT / YZ

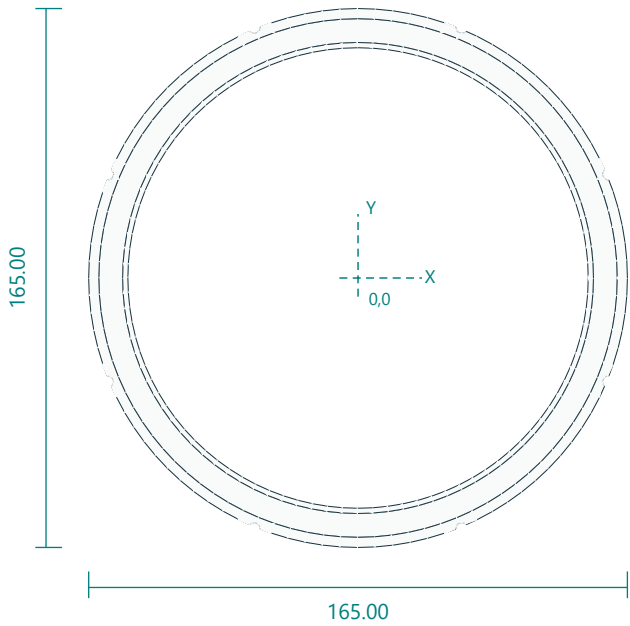


Interfaces: 77 square x3; opening70; mounting pitch68.
Fasteners / retention: 4 M3x25 shared.
Print orientation: flat down. Layers 0.2 mm; 4 walls; 25% infill baseline.
Feature locations: Four3.4 holes(+/-34,+/-34). Diameter70centeropening retainsPCBcorners, not roundglass.
General fit: inspect and deburr holes; check the fit coupon and actual hardware. No unlisted machining tolerances are implied.

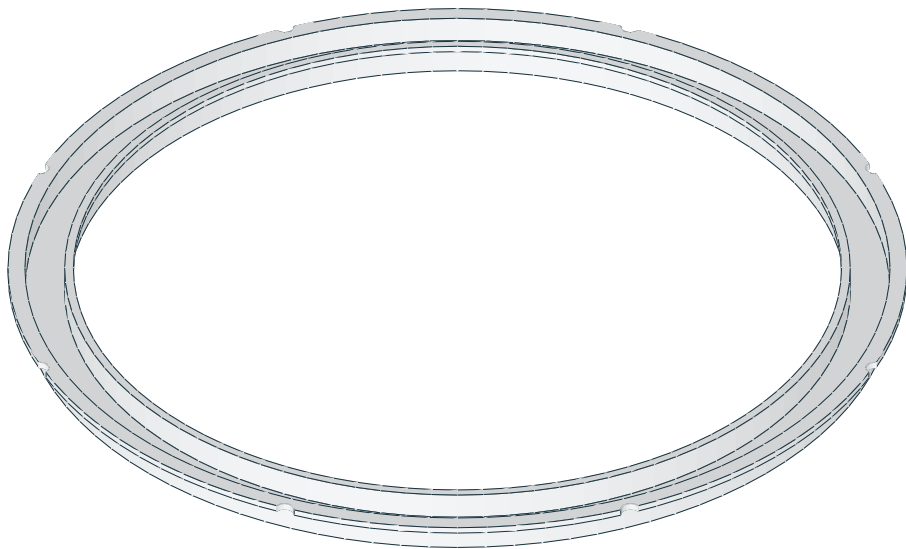
Frosted outer RGB annular diffuser

outer_diffuser.stl | PRINT QUANTITY 1 | natural/translucent PETG

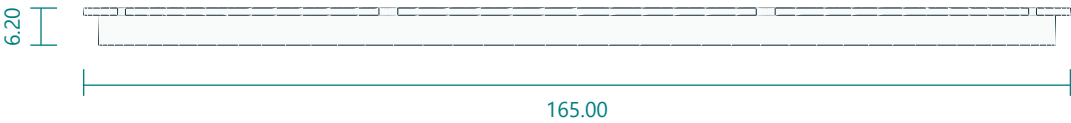
TOP / XY



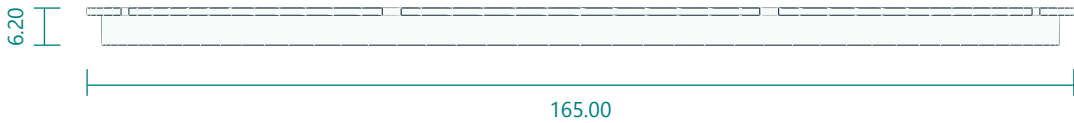
ISOMETRIC / REFERENCE



FRONT / XZ



RIGHT / YZ

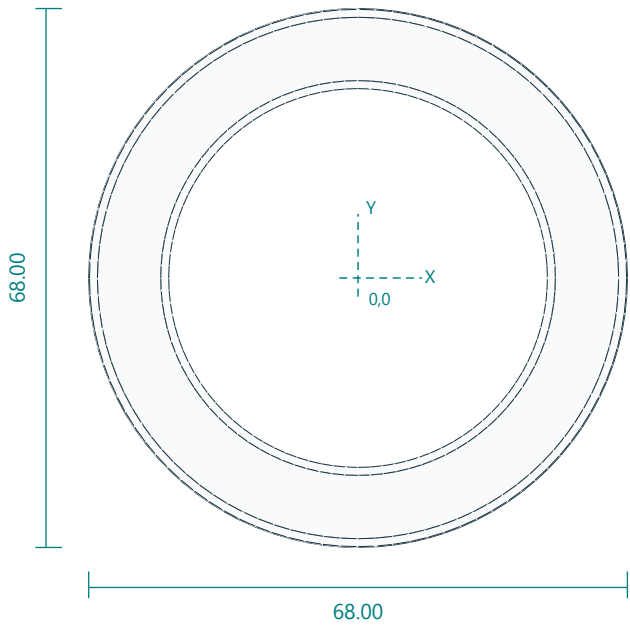


Interfaces: 160 face OD143 ID; 165 flange; 1.2 face; 6.2 depth.
Fasteners / retention: captured by outer bezel/center plate.
Print orientation: smooth optical face on bed. Layers 0.15 mm; 3 walls; 100% infill baseline.
Feature locations: FaceOD160 ID143. RearflangesOD165/ID158.8 andOD144.2/ID141. M3edgeclearancenotchesradius83.
General fit: inspect and deburr holes; check the fit coupon and actual hardware. No unlisted machining tolerances are implied.

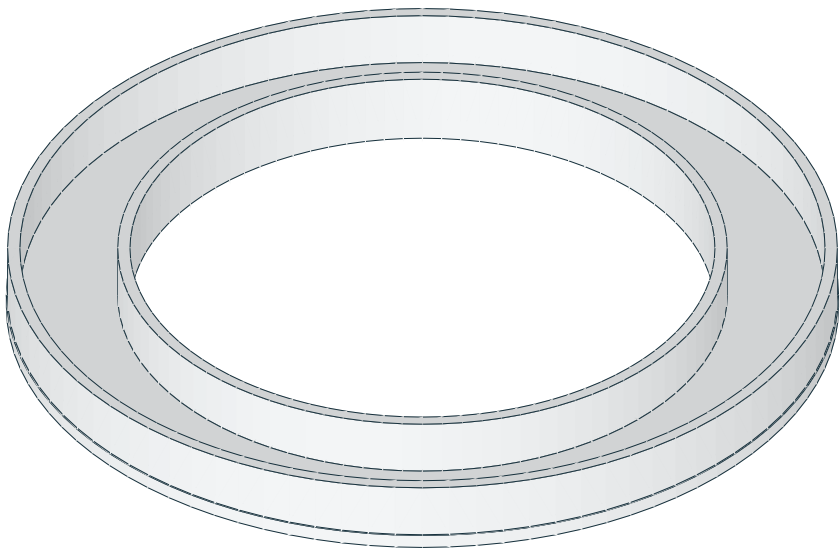
Frosted inner RGBW light diffuser

inner_diffuser.stl | PRINT QUANTITY 1 | natural/translucent PETG

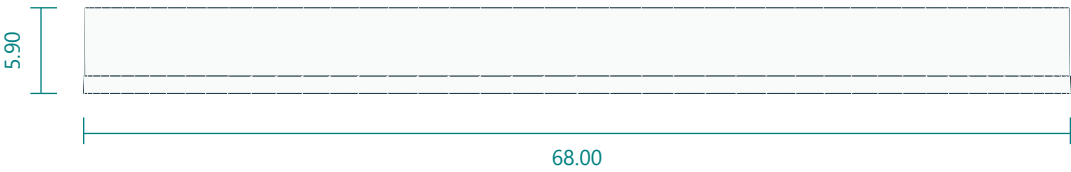
TOP / XY



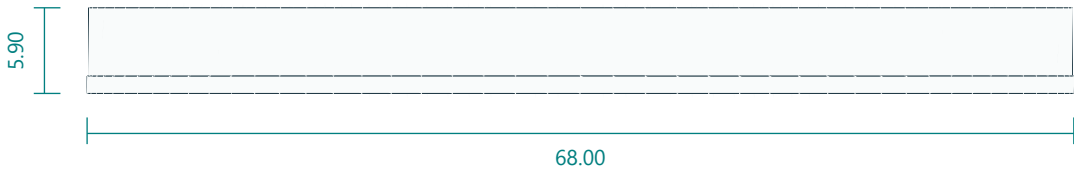
ISOMETRIC / REFERENCE



FRONT / XZ



RIGHT / YZ

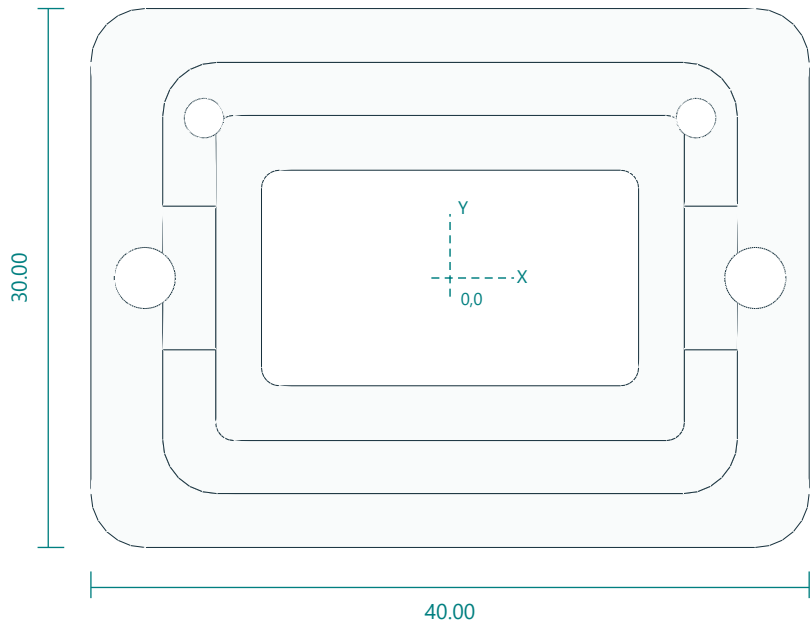


Interfaces: 68 OD47.8 ID; 1.2 face; 5.9 depth.
Fasteners / retention: 3 tiny neutral cure silicone retention dots.
Print orientation: smooth optical face on bed. Layers 0.15 mm; 3 walls; 100% infill baseline.
Feature locations: OD68 ID47.8; skin1.2; depth5.9. No sensorwindowmaterial.
General fit: inspect and deburr holes; check the fit coupon and actual hardware. No unlisted machining tolerances are implied.

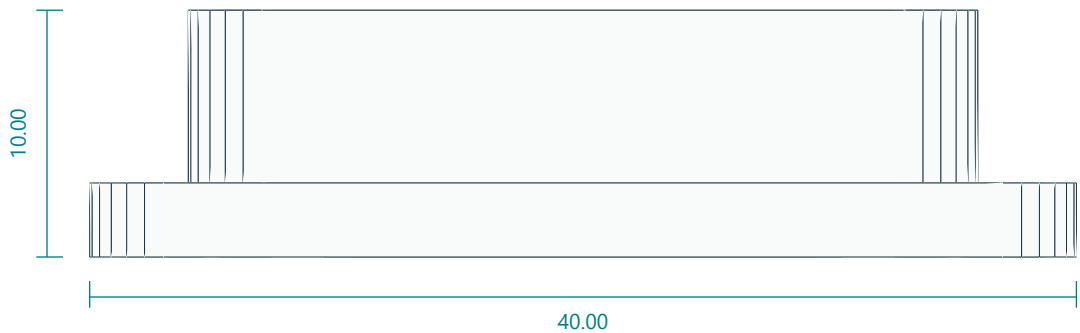
Open optical ToF sensor body

sensor_pod.stl | PRINT QUANTITY 5 | PETG

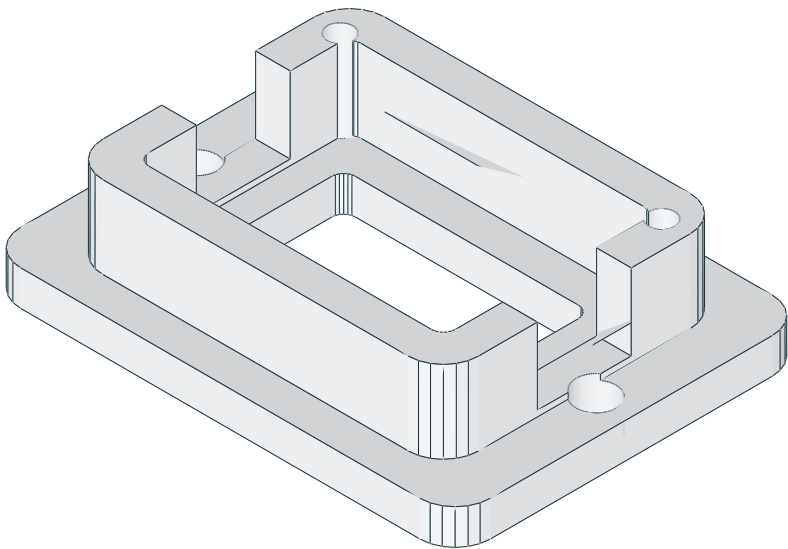
TOP / XY



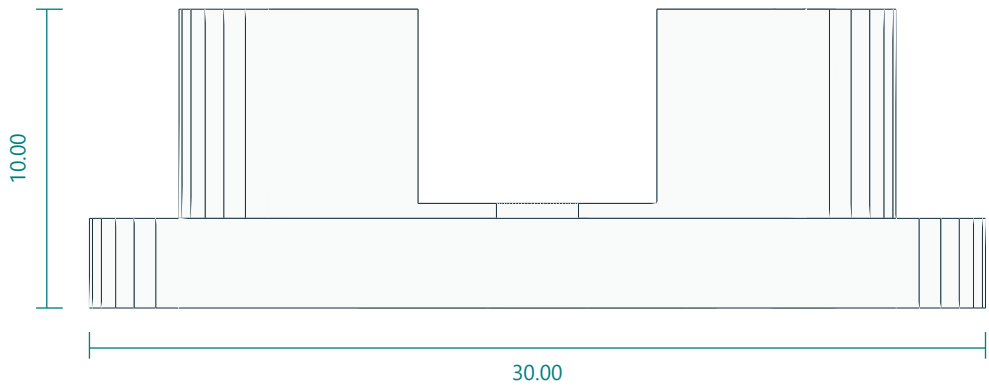
FRONT / XZ



ISOMETRIC / REFERENCE



RIGHT / YZ

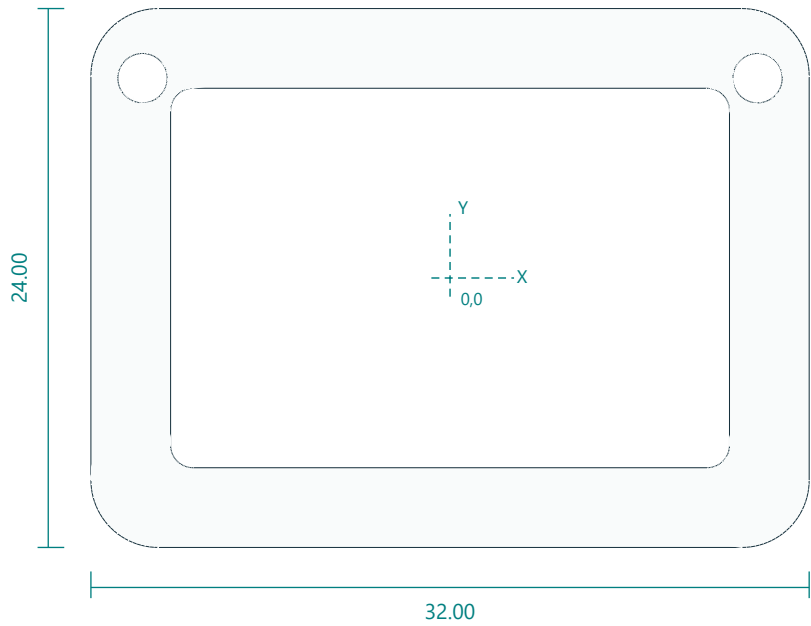


Interfaces: 40 x30 x10; PCB26.1 x18.1; mount pitch34; two side QT plug windows.
Fasteners / retention: 2 M3x10; 2 M2x14 each; two 18x2x6mm foam edge strips compressed to5.4mm.
Print orientation: rear flange down. Layers 0.2 mm; 4 walls; 25% infill baseline.
Feature locations: Mount3.4 at(+/-17,0). M2holes2.2 at(+/-13.7,8.9). SideQTplugreliefs on +/-X, width8inY, Z3.5to10.
General fit: inspect and deburr holes; check the fit coupon and actual hardware. No unlisted machining tolerances are implied.

ToF board edge retaining frame

sensor_retainer.stl | PRINT QUANTITY 5 | PETG

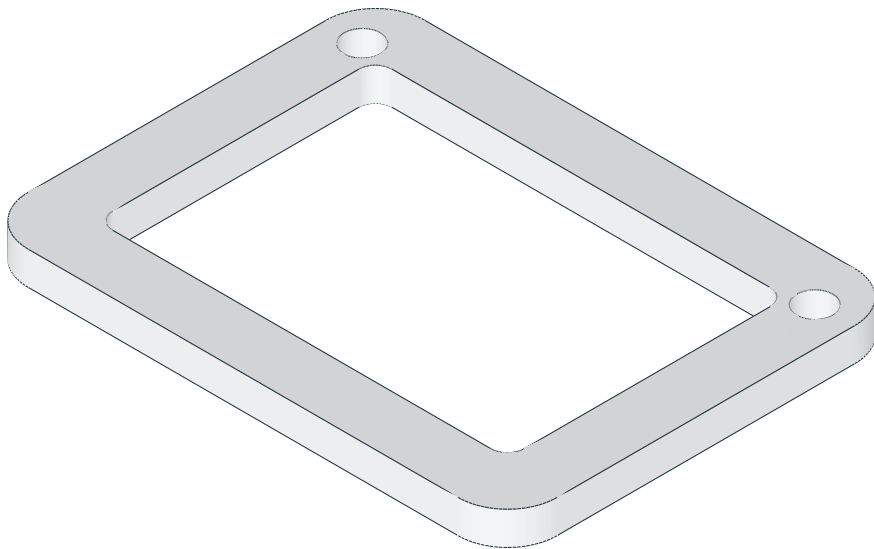
TOP / XY



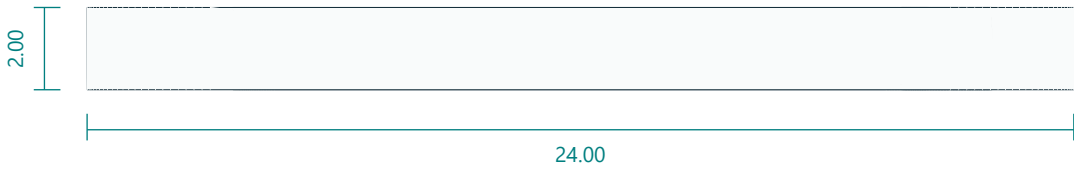
FRONT / XZ



ISOMETRIC / REFERENCE



RIGHT / YZ

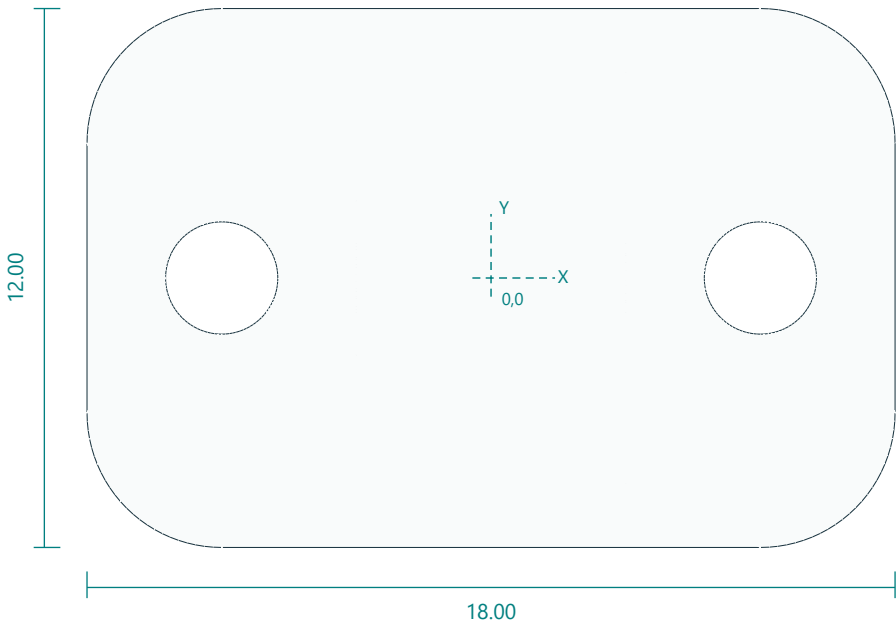


Interfaces: 32 x24 x2; unobscured aperture24.9 x16.9.
Fasteners / retention: 2 M2x14 each.
Print orientation: flat down. Layers 0.2 mm; 4 walls; 25% infill baseline.
Feature locations: M2holes2.2 at(+/-13.7,8.9). Opticalopening24.9x16.9.
General fit: inspect and deburr holes; check the fit coupon and actual hardware. No unlisted machining tolerances are implied.

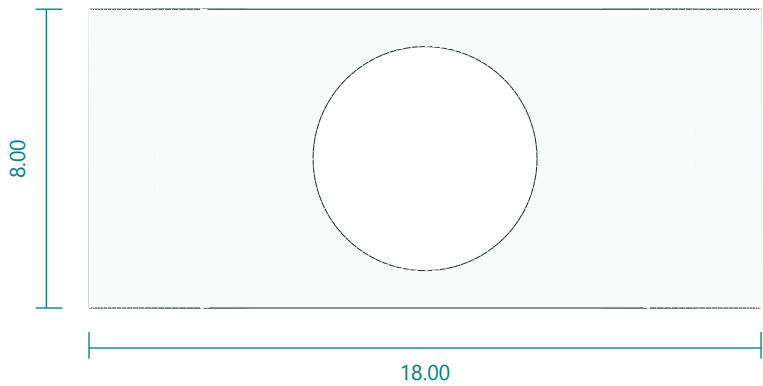
Harness saddle clip

cable_clip.stl | PRINT QUANTITY 6 | PETG

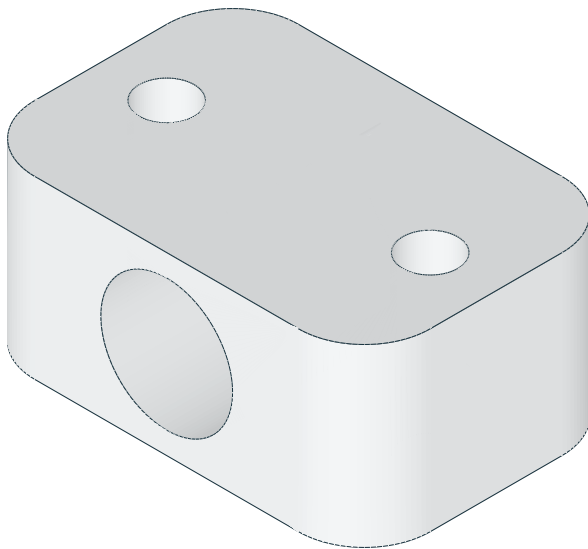
TOP / XY



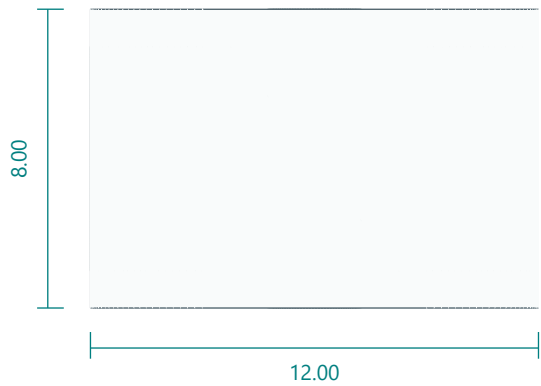
FRONT / XZ



ISOMETRIC / REFERENCE



RIGHT / YZ

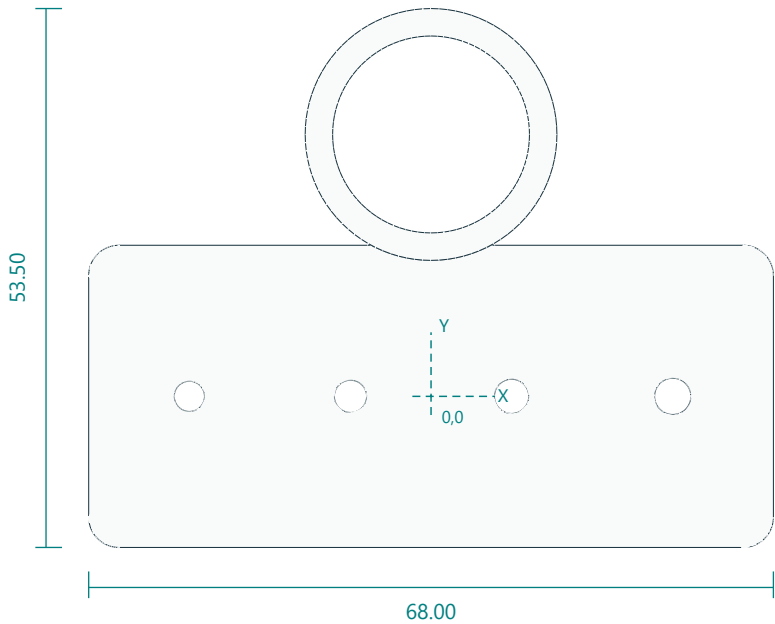


Interfaces: 18 x12 x8; cable6; screws12 pitch.
Fasteners / retention: 2 M2.5x12 each optional.
Print orientation: flat down. Layers 0.2 mm; 4 walls; 25% infill baseline.
Feature locations: Two2.5holes(+/-6,0); cablebore6 alongY centeredZ4.
General fit: inspect and deburr holes; check the fit coupon and actual hardware. No unlisted machining tolerances are implied.

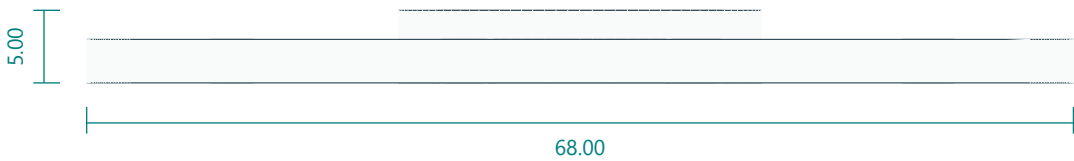
Horn and bolt clearance coupons

fit_coupon.stl | PRINT QUANTITY 1 | PETG

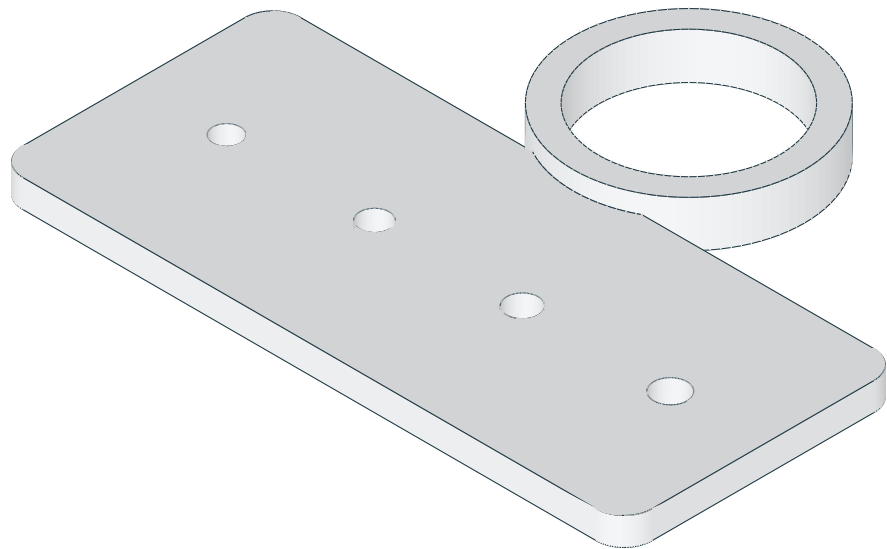
TOP / XY



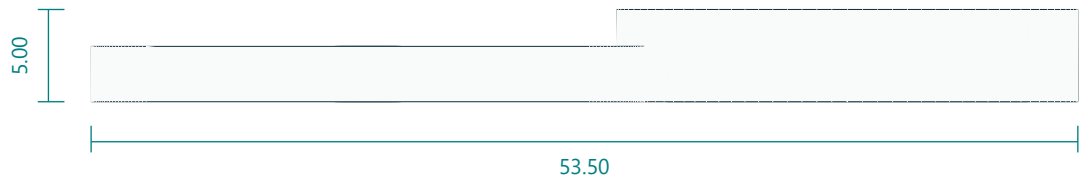
FRONT / XZ



ISOMETRIC / REFERENCE



RIGHT / YZ



Interfaces: 3.0 /3.2 /3.4 /3.6 bores; horn socket19.55.
Fasteners / retention: none.
Print orientation: flat down. Layers 0.2 mm; 4 walls; 25% infill baseline.
Feature locations: Bolt bores3.0,3.2,3.4,3.6 atX=-24,-8,8,24; hornsocket19.55 centeredY26.
General fit: inspect and deburr holes; check the fit coupon and actual hardware. No unlisted machining tolerances are implied.